

EHBEA 2023 Abstract booklet

The speaker is highlighted in **bold** in all author lists.

Plenaries

Beyond Nature vs. Nurture: An Interdisciplinary Exploration of Sexual Politics and Ideological Divides

Blake, K.

For too long, our approach to human sexual politics has quarantined the biological from the sociocultural, as if one has nothing to do with the other. Yet a closer look of the drivers of traditional gender ideology, intimate partner violence, and female beauty practices shows that nature and nurture often intertwine. In this talk I review a growing body of work implicating mating market conditions in some of the most ideologically divisive issues societies face. By incorporating insights from behavioral ecology, social psychology, economics, and international security, I aim to provide a functional account of sexual politics, one that highlights the need for cross-disciplinary collaboration. With this account I offer a new approach to understanding how and why ideologies concerning sex manifest, and what this means for the future of gender equality.

Kinship, Monotocyt and the Evolution of Cooperative Breeding

Clutton-Brock, T.

To put the evolution of primate societies in perspective, it is necessary to look beyond primates themselves to other groups of social vertebrates. Here I describe how contrasts in patterns of life histories affect the kinship structure of social groups and how contrasts interspecific contrasts in kinship structure among mammals are commonly associated with contrasts in the frequency of competitive and cooperative behaviour. I describe how monotocyt (the production of single young) is commonly associated with low average levels of kinship and high variance in kinship between group members and how specialised cooperative breeding in mammals is usually associated with polytocyt and with high average levels of kinship and relatively low variance in kinship between group members.

Children as agents of culture change in hunter-gatherer societies and beyond

Lew-Levy, S.

Studying how contemporary hunter-gatherer children learn can shed light on the evolution of human cognition, life history, and social organization. Further, cross-cultural comparisons can help us understand the ways in which humans flexibly adapt to our diverse socioecological niches. Drawing upon systematic literature reviews and empirical observational, experimental, and interview research with Tanzanian Hadza and Congolese BaYaka, I outline cross-cultural similarities and differences in contemporary hunter-gatherer children's learning. I first show how play, teaching, participation, and imitation biases contribute to children's skill acquisition. Second, I demonstrate that hunter-gatherer children are eager innovators of subsistence practices and social norms. Third, I show how cultural beliefs, ecology, settlement structure, and subsistence opportunities contribute to cross-cultural variation in hunter-gatherer children's economic work and learning. Despite considerable diversity in ecological and social environments, children across a range of hunter-gatherer societies spend much of their time learning with peers in the mixed-gender multi-age playgroup. I argue that this peer learning may uniquely contribute to dynamics of culture change by encouraging less costly knowledge transfers in humans on the one hand, and innovation discovery on the other. I support these points using examples from the ethnographic canon. These findings compel us to rethink children's agency in cultural evolution in the present and past.

Eugenics and the Misuse of Mendel

Rutherford, A.

With the invention of race in the 18th century, and the birth of eugenics in the 19th, the roots of biology, anthropology and genetics are inextricably intertwined with the political ideologies of colonialism and White Supremacy. In particular, the re-emergence of Mendel's discoveries about genetics fuelled the rise of eugenics, and provided its most devoted apostles with a biological mechanism to justify their bigotry. An ideological commitment to mendelian pedigrees and scientific racism formed the pseudoscientific basis for mass sterilisation and genocide. We now have a much more sophisticated understanding of human genetics and evolution, but this eugenic spectre limps on in our culture, reinforcing a view of biology that is simplistic, deterministic and wrong.

The Cuckoldry Conundrum

Scelza, B.

The idea that men's reproductive strategies are driven in large part by a desire to maintain paternity certainty underlies major tenets in the evolutionary psychology of mating. When they fail, men are said to be the victims of "cuckoldry," unwittingly investing in children they did not father. In this talk I will use more than ten years of ethnographic, demographic and genetic data from Himba pastoralists to present and challenge two assumptions that are built into standard models of cuckoldry: (1) men are being tricked into investing in non-biological offspring; and (2) investment in non-biological offspring is wasted. First, I will show that extra-pair paternity (EPP) is common among Himba (and variable across cultures) and briefly explain why normative concurrency is a viable strategy for Himba men and women. Next, I will show that the concepts of pater and genitor are complex and locally constructed ideas that often include explicit knowledge of extra-marital relationships and extra-pair paternity. When this context is considered, we see that paternity losses can be associated with important gains to offset them. Finally, I will propose that cuckoldry as a concept has limited use in understanding paternity in humans. Future studies should pay close attention to the cultural milieu surrounding fatherhood, rather than assuming men everywhere will be equally driven by the same concerns.

Dairying, diseases and the evolution of lactase persistence in Europe

Thomas, M.

In European and many African, Middle Eastern and southern Asian populations, lactase persistence (LP) is the most advantageous monogenic trait to have evolved over the past 10,000 years. Although natural selection on LP and the consumption of prehistoric milk must be linked, considerable uncertainty remains concerning their spatiotemporal configuration and specific interactions. Here we provide detailed distributions of milk exploitation across Europe over the past 9,000 years using around 7,000 pottery fat residues from more than 550 archaeological sites. European milk use was widespread from the Neolithic period onwards but varied spatially and temporally in intensity. Notably, LP selection varying with levels of prehistoric milk exploitation is no better at explaining LP allele frequency trajectories than uniform selection since the Neolithic period. In the UK Biobank cohort of 500,000 contemporary Europeans, LP genotype was only weakly associated with milk consumption and did not show consistent associations with improved fitness or health indicators. This suggests that other reasons for the beneficial effects of LP should be considered for its rapid frequency increase. We propose that lactase non-persistent individuals consumed milk when it became available but, under conditions of famine and/or increased pathogen exposure, this was disadvantageous, driving LP selection in prehistoric Europe. Comparison of model likelihoods indicates that population fluctuations, settlement density and wild animal exploitation—proxies for these drivers—provide better explanations of LP selection than the extent of milk exploitation. These findings offer new perspectives on prehistoric milk exploitation and LP evolution.

Wednesday, April 19th**Inequality****10.45-1.15pm; Lecture Theatre 1****A cross-cultural exploration of the association between material and relational wealth***Power, E., Redhead, D., Koster, J., the ENDOW Team*

It is now quite well established that people's social relationships are crucial to their wellbeing, livelihood, and life prospects. We may therefore expect to see a broadly positive association between the social support and social position that a person or household has—what we could term their “relational wealth”—and their material wealth. Or, to use a different set of closely related terms, if social capital is productive of economic capital (and vice versa) then we should expect the two to co-vary. As yet, however, we have little understanding of how different kinds of “wealth” are associated, nor do we know how such associations may vary across different communities and societies. While relational and material wealth may often go hand-in-hand, there may also be contexts in which they function instead as substitutes, such that investing in one may make the other less essential. To explore this, we draw on data gathered as part of the “ENDOW” project (Economic Networks and the Dynamics of Wealth inequality, a large NSF-funded collaboration) from over forty communities around the world. These communities show immense diversity in their institutional, cultural, and economic arrangements, ranging from farming villages in India to fishing settlements in Ecuador to pastoral groups in Namibia. Anthropologists working with each community gathered extensive demographic, economic, and social support network data that form the basis of our analysis. The social network data comprises multiple name generators getting at a range of different types of support that sharing units (i.e., households) can call upon. Measures derived from these networks (e.g., basic centrality and clustering metrics) serve as our proxies for relational wealth, while detailed enumerations of sharing units' property and assets serve as our proxies for material wealth. Here, we present preliminary analyses looking at the associations between a sharing unit's material and relational wealth, looking both within and between communities. While we broadly find a positive association within communities, there is substantial variation in the strength of that association, the patterning of which we discuss further. We conclude by highlighting some of the methodological challenges of this work and our efforts to resolve them.

Gender equality and childhood in pre-industrial societies*Pákozy, C., Brown, G. R.*

Objective(s): Across societies, there is variation in the extent to which adults' roles and rights are differentiated along gender lines. This study investigated whether this extends to childhood in cultures beyond the WEIRD societies normally focused on within the literature. Thus, this study looked for a correlation between structural gender (in)equality and the differential treatment of male and female children in pre-industrial societies.

Methods: Data on the traits adults try to inculcate in children (40 variables), the activities children do (58 variables) and a structural gender equality measure (5 subscales relating to property ownership, inheritance and control of goods and property) were obtained from ethnographic literature within the Standard Cross-Cultural Sample in the electronic Human Relations Area Files database. The researcher also coded 12 new societies for the gender equality measure, bringing the total used to 105 societies. Differences between the rating of the importance of various tasks, activities and traits for boys and girls were calculated.

Results: The dataset is richly varied and showcased the full range of diversity across the societies studied, both in terms of gender equality and in the degree of childhood differentiation along gender lines. A multiple regression model revealed no significant relationship of gender equality to differences in children's activities or inculcation of traits, although the activities data showed a weak positive trend.

Conclusion(s): Unlike industrialised nations, pre-industrial societies do not exhibit a correlation between gender equality and differences in how children are treated. The eHRAF database is a promising avenue for evolutionary human sciences research, and allows for expanding scopes into non-WEIRD and pre-industrial societies.

Modelling the role of environmental circumscription in the evolution of inequality*Perret, C., Currie, T. E.*

Circumscription theory proposes that complex hierarchical societies emerged in areas surrounded by barriers to dispersal e.g. mountains or seas. This theory has been widely influential but the lack of formal modelling has resulted in theoretical and empirical challenges. For instance, it has not always been clear what aspects of the idea are most important and which key factors should be measured in the real world. Here, we use a model to investigate (i) under which conditions circumscription does affect inequality, and if yes, (ii) which component of circumscription matters e.g. size versus connectivity of an area. This theory shares parallels with reproductive skew models from evolutionary ecology where inequality depends of the capacity of subordinates to escape from despotic individuals. Building on these similarities, we extend reproductive skew models to simulate the concurrent evolution of inequality in many connected groups. We disambiguate between cost of migration, number of polities and connectivity by running simulations on different network structures. Our results show that a higher cost of migration does not directly limit inequality in the long term but it controls the rate of increase in inequality. On the other side, the size of an area and its connectivity affect inequality on the long term because they control how much a random decrease in inequality in a given polity propagates through others. Running the model on networks that approximate different geographic scenarios we find support for the idea that inequality tends to be lower when there are larger open plains than when groups are found in valleys with relatively few links to other groups. Overall, our model helps clarify some issues about how the presence of outside opportunities may affect inequality.

Gender and Artificial Intelligence: The Zero-Sum Game of Thing-People Interests*Borau, S.*

Objective(s): Women are under-represented in the fields of robotics and AI (Artificial Intelligence). Does this career gender imbalance extend to AI acceptance, i.e. are women more likely to resist adopting AI than men, and if so, why? Here I argue that because women are more likely to endorse communal goals and exhibit stronger people orientations (i.e. interest in persons rather than objects), they may be more likely to resist AI, as AI might be perceived as incompatible with communality, or an orientation to care about others.

Methods: I tested these predictions in seven studies: one qualitative study and 6 preregistered quantitative studies (N = 2,143 Americans) to explore gender differences in acceptance of AI, the mediating role of orientation toward things and people, and interventions to tackle this gender gap.

Results: I show that men express greater acceptance of AI than women, and that thing orientation (but not people orientation) mediates the relationship between gender and acceptance of AI. Furthermore, participants regard AI as more closely associated with the concepts of things (vs. people), agency (vs. communality), competence (vs. warmth), and masculinity (vs. femininity); such perceptions also extend to people who like AI, and are rooted in the belief that liking AI and humans is a zero-sum game.

Conclusion(s): These findings illuminate a tension regarding the role of AI in our lives, by suggesting that people, and especially women, believe that AI might decrease rather than expand our humanity. Several proposed interventions are tested to overcome such beliefs.

Explaining social inequalities in health behaviors: the role of family mortality history and perceived uncontrollable mortality risk*Job, M., Heisig, J.P., Nettle, D.*

Objective(s): Within affluent societies, socioeconomic disparities in health are large, and substantially driven by differences in behaviours. Disadvantaged people may perceive a large part of their mortality risk to be beyond their control, making them less willing to look after their health. The present study aims to examine the role of perceived uncontrollable mortality risk (PUMR) in shaping those socioeconomic differences in behaviors. Additionally, it explores the environmental cues that may influence perceptions of control over mortality risk, with a focus on family mortality history.

Methods: We surveyed a socioeconomically balanced age-homogeneous sample of 1,000 UK residents for their time orientation, reported health effort, exposure to familial death, PUMR, locus of control, and health effort. Data were analyzed using linear regression and structural equation modeling.

Results: PUMR fully mediated the relationship between subjective socioeconomic status and health effort. PUMR also mediated the association between exposure to familial death

and both reported health effort and time orientation, although these associations were not as strong.

Conclusion(s): The present study provides evidence for the Uncontrollable Mortality Risk Hypothesis, which suggests that individuals with higher perceived risk of death from uncontrollable factors will be less motivated to engage in positive health behaviors. Additionally, we found that environmental cues, such as exposure to familial death, may influence perceptions of control over mortality risk and contribute to socioeconomic inequalities in health behaviors. These findings may have important implications for interventions aimed at reducing socioeconomic disparities in health outcomes.

How gender inequality affects attitudes concerning sex and gender

Harmon-Jones, S.K., Blake, K.R., Brooks, R.

Objective(s): Economic inequalities shape the environment within which individuals seek out romantic/sexual partners, influencing how individuals view and behave towards potential mates. Across cultures, heterosexual women often show a preference for men of higher wealth and status than their own (i.e., hypergyny). Simulation models demonstrate that, assuming a hypergynous preference, the pool of suitable mates available to poorer heterosexual men and richer heterosexual women decreases as economic gender gaps shrink. Given that experiences of romantic/sexual success and rejection can shape the way individuals view themselves as well as potential mates, an individual's income may interact with the overall degree of gender equality within their society to predict how they view themselves (e.g., their self-perceived masculinity/femininity), others (e.g., their endorsement of sexist attitudes), as well as how appropriate they believe it is to use particular mate retention strategies (e.g., their perceived acceptability of intimate partner coercion).

Methods: In the present experiment, we used a role-playing paradigm to simultaneously manipulate the mean gender gap within a hypothetical virtual society, and the participant's income relative to others in the society. Experiments manipulating economic inequalities in this way have assisted in determining the causal effects of inequality on attitudes and behaviour. However, we do not yet know whether heterosexual individuals' response to gender inequality differs depending on their own income within the distribution, as theory predicts.

Results: We used linear mixed models to examine how overall gender equality, the percentage of men, and the percentage of women that the participant out-earned interacted with the participant's sex and gender to predict their degree of self-perceived masculinity/femininity, implicit and explicit sexism, and perceived acceptability of intimate partner coercion.

Conclusion(s): We compare these results to theoretical accounts and simulation models of mating behaviour under different economic circumstances.

The cost of change: social and reproductive ramifications of supporting women's empowerment

Kilgallen, J.A., Sebarua, E., Charles, P., John, A., Chizi, T., Nicholans, I., Urassa, M., Lawson, D.W.

Objective(s): Gender egalitarian men are vital for the advancement of women's empowerment globally, yet anthropological scholarship assessing the ramifications for men who challenge inequitable gender norms is limited. This research addresses this gap, utilizing the theoretical frameworks of cultural evolution and human behavioral ecology to elucidate the social and fitness ramifications men experience when violating patriarchal norms.

Methods: All data was sourced from a single urbanizing Tanzanian community, selected because it combines patriarchal values, with recent and dramatic shifts in gender roles accompanying market integration, providing an ideal case study to examine emergent diversity in men's gender ideology. Focus groups with community members (i.e. young men, young women, and elders) first captured local understandings of what it means to be an outlier in support for women's empowerment across multiple dimensions (i.e., economic, social, sexual, and reproductive autonomy), as well as perceived social and fitness ramifications associated with performing these behaviors. In-depth interviews provided situated knowledge of lived experiences of young men who challenge the status quo.

Results: Preliminary findings suggest that support for women may lead to coinciding decreases in fertility for more gender equitable men as women gain autonomy in decision-making and greater reproductive and sexual autonomy, but young women may favor more gender equitable men for marriageability purposes. Highly supportive men face costs of social ostracism or even violence from lesser educated male peers and elders but gain social prestige via their association with relatively urban values or the 'cultural scripts' of global health agencies.

Conclusion(s): While going against prevailing gender inequitable norms may come at some cost, we find evidence that there may also be novel incentives to emulate support for women's empowerment via improvements in men's "cultural fitness"; suggesting pathways as to how support for women's empowerment may spread even if counter to men's reproductive interests.

Why is violence high and persistent in deprived communities? A formal model

De Courson, B., Frankenbuis, W.E., Nettle, D., van Gelder, J.-L.

There is massive variation in rates of violence across time and space. These rates are positively associated with economic deprivation and inequality. They also tend to display a degree of local persistence, or 'enduring neighbourhood effects'. Here, we identify a single mechanism that can produce all three observations. We formalise it in a mathematical model, which specifies how individual-level processes generate the population-level patterns. Our model assumes that agents try to keep their level of resources above a 'desperation threshold', to reflect the intuitive notion that one of people's priorities is to always meet their basic needs. As shown in previous work, being below the threshold makes risky actions, such as property crime, beneficial. We simulate populations with heterogeneous levels of resources. When deprivation or inequality is high, there are more desperate individuals, hence a higher risk of exploitation. It then becomes advantageous to use violence, to send a "toughness signal" to exploiters. For intermediate levels of poverty, the system is bistable and we observe hysteresis: populations can be violent because they were deprived or unequal in the past, even after conditions improve. We discuss implications of our findings for policy and interventions aimed at reducing violence.

Personal relative deprivation negatively predicts engagement in group decision-making

Rotella, A., Mishra, S., Neville, L., Chaudhuri, S.

In social circumstances, people spontaneously assess their status relative to others through social comparisons. In conditions of inequality, social comparisons result in perceptions of relative advantage or disadvantage, compared to interaction partners. In these instances, perceived disadvantage can lead to the experience of personal relative deprivation (i.e., subjective feelings of anger and resentment), which in turn causes psychosocial stress. To date, little empirical work has investigated how individual differences in personal relative deprivation influences group dynamics. Across two studies investigating real group dynamics, we sought to determine how feelings of personal relative deprivation and group characteristics influenced group engagement. In Study 1 (n = 150), first-year business students engaged in a high-pressure scenario that was regularly used as a teaching exercise. In small groups (4-6), participants with competing motives were tasked to reach a consensus decision. Personal relative deprivation explained 9% of variance in group exercise engagement over and above demographic and situation-related variables. In Study 2 (N = 356; pre-registered), we collected data participants who were employed full-time via Prolific. Participants reported recent team experiences in real work scenarios. Feelings of relative deprivation accounted for 13% of variance of group engagement, over and above demographic variables. Group size and relative seniority moderated the relationship between relative deprivation and group participation. Additionally, out of 8 possible points of comparison, only status, organizational seniority, and wealth predicted feelings of relative deprivation. These findings suggest that social comparisons resulting in feelings of relative deprivation are related to reduced engagement in small groups. The implications for small group dynamics, including both individual and group outcomes, will be discussed.

Predictors and fitness-relevant outcomes of women's status in Himba pastoralists

Prall, S.P., Saelz, B.A.

Objective(s): Gender inequalities in status are a common feature of many populations, but while the drivers of men's status have been well-studied, women's status receives little attention. To understand the causes and consequences of women's status in a small-scale population, we explore the determinants of women's status, how status determinants differ for men and women, and how women's status impacts fitness-related outcomes.

Methods: Demographic data collected as part of the Kunene Rural Health and Demography Project was combined with dyadic rating data on traits and preferences in Himba pastoralists of northern Namibia. Multi-level models were used to predict status as well as preferences and child anthropometrics.

Results: Perceptions of generosity, intelligence, and respectfulness best predicted women's status. Surprisingly, although men tend to reach higher status at older ages, men were not more likely than women to be rated influential by their peers. Women's status had several important effects on outcomes related to reproductive success. Women's status predicted residuals for children's height, particularly for younger children, but not weight or BMI. Status also predicted estimates of women's desirability as a partner. In marital partnerships, women's and men's status was positively correlated, and in a multi-level model, women's status predicted the status of their partner.

Conclusion(s): Results from this study indicate underlying similarities in the drivers of men's and women's status, and in line with previous research, suggest that women primarily use status to benefit their children. This work suggests that women's status is an important mediator for fitness-relevant outcomes, and should be considered alongside men's status in evolutionary studies.

Social learning

10.45-1.15pm; Lecture Theatre 2

Analogy as a critical component for communicating complex cultural information

Brand, C.O., Holding T.H., Smaldino P.E.

Complex behaviours are said to be transmitted from one person to another via high-fidelity social learning mechanisms. However, many behaviours are too opaque to be learnt via imitation alone, and require teaching with verbal communication. Although human language facilitates the explanation of complex behaviours in precise detail, we argue that direct, literal communication is overly costly and inefficient in many cases. Instead, humans use shortcuts such as analogies that enable compression of meaning (Brand et al. 2021). In their model of tacit knowledge, Miton & Dedeo (2022) find that "a small amount of guidance, well-presented, allows students to lock-in a culturally-widespread practice". We predict that the "well-presented guidance" relies on analogies that relate novel behaviour to familiar behaviour in the learner's repertoire. We present a formal model in which behavioural components are represented by bit strings in an NK landscape. In our model, individuals go through a developmental period in which they learn a range of behavioural solutions. Teachers use analogies to aid the learning of novel solutions by transmitting the closest solution in their shared repertoire with the student. In this way we do not assume high-fidelity transmission because the learner still has to employ substantial individual learning to reach the new solution. Preliminary results suggest that teaching with analogy allows learners to both reach the optimal solution more often, and more efficiently, than non-analogical teaching. We present further results from the model and their implications for the origins of language and cognition.

How culture affects fitness: causal evidence at the Swiss language border

Faessler, L., Lalive, R., Efferson, C.

Cultural evolution theory implies that social learning can lead cultural groups to differ in the same environment and drive genetic selection. Yet, distinguishing the effect of culture from other confounds is a difficult empirical challenge. To meet this challenge, we exploit a Swiss language border that divides Switzerland without any institutional boundary and can potentially act as a cultural border. Using a regression discontinuity design, we estimate discontinuities at the border in health and fertility-related attitudes and behaviors. Preliminary results about voting behaviors confirm multiple discontinuities related to health and fertility. German speakers are more likely to adopt healthy behaviors and vote against initiatives to share health care costs, while French speakers visit the doctor more often. German speakers report significantly better health but surprisingly do not live longer. Our results provide new evidence that social learning generates cultural variation without environmental variation in health and fertility domains, thus demonstrating the potential for cultural processes to create variation in domains directly related to genetic fitness.

The Forgotten Social Benefits of Social Learning

Harrison, R., Dongre, P., van Schaik, C., van de Waal, E.

Introduction: Theoretical and empirical scholars of cultural evolution have traditionally studied social learning strategies, such as conformity, as adaptive strategies to obtain accurate information about the environment, whilst within social psychology there has been a greater focus upon the social function of such strategies. Although these two approaches are often used in concert when studying human social learning, we believe the potential social functions of conformity, and social learning more broadly, have been overlooked in studies of non-humans.

Methods: We review evidence from studies of homophily, imitation, and rapid facial mimicry in both humans and non-human animals, as well as cases of conformity in dispersers.

Results: The literature reviewed suggests that behaving like others affords social benefits to non-human animals and that behaviour matching may be deployed strategically to increase affiliation. Studies of conformity in dispersers suggest that forgoing personal information or preferences in favour of those of the new group during immigration may be a strategy to facilitate social integration. Conclusions: We therefore propose that the informational and social functions of conformity apply to humans and non-human animals alike. We use this perspective to generate several interesting research questions to inspire work in this field. For example, under what conditions do animals use informational or social conformity? What role does uncertainty play in social learning in immigrant individuals?

The Monty Hall Dilemma: An Evolutionary Perspective

Kendal, J.

Objective(s): In the Monty Hall (MH) dilemma, a contestant chooses between three doors, one of which conceals a valuable prize. Before opening their chosen door, the gameshow host opens one of the other doors that does not contain the prize, before giving the contestant an opportunity to switch their choice between the two unopened doors. Most people, at least from Western samples, incorrectly choose to stick with their original choice, believing there's a 50:50 chance that each of the two unopened doors contain the prize. In reality, the chance their prior choice is correct remains at one third, so it's better to switch their choice. The MH dilemma may arise in real-world scenarios where: individuals are faced with a choice from a small set of behaviours, one of which returns a higher payoff than the others; payoffs are uncertain; and individuals can learn from others' mistakes. These conditions are fairly common, so why do many humans struggle to solve the MH dilemma? This study brings an evolutionary perspective, identifying conditions under which choice-switching evolves.

Methods: Analysis of a mathematical model. In a variable environment, individuals have a limited choice of behaviours, one of which is more adaptive than the others. Prior behaviour preference is culturally inherited but its adaptive value may be outdated by environmental change. Individuals can socially learn from payoff cues provided by asocial learners. The MH-dilemma arises when, like the gameshow host, asocial learners reveal a non-adaptive behaviour, distinct from the social learner's prior preference.

Results: Choice-switching evolves in variable environments when the set of alternative behaviours is small.

Conclusion(s): Humans living in stable environments, or environments with a lot of choice, may be less likely to respond adaptively to the MH-dilemma than those living in unstable environments with only a few behaviour choices.

Social Learning on Social Networks

Smolla, M., Kandler, A., Fogarty, L.

Objective(s): Cultural information is transmitted via social learning along the connections of a social network. While there is extensive research, for example, on the diffusion speed on social networks, less is understood about the transmission of cultural information, which can be subject to repeated innovation, transmission biases, and cultural drift. Here, we draw together and synthesise insights from diverse fields such as animal social learning, economics, and network science to better understand the interactions between network structure and cultural evolutionary processes.

Methods: We use existing cultural evolutionary as well as population genetics models, to generate (analytic and simulation-based) expectations for the behaviour of cultural

information on fully connected networks. We then compare these with simulations on regular, random, and small-world networks. In addition to unbiased (neutral) transmission, we also model novelty and conformity biased cultural transmission.

Results: We find that more socially clustered networks (where friends of a friend are friends) show more cultural clustering (where friends share the same trait) and have on average more unique cultural traits under unbiased transmission. The number of cultural variants is an important measure for standard tests of neutrality. Networks with social clustering comparable to real-world networks deviate in their cultural composition under unbiased transmission in such a way that these tests fail to detect unbiased transmission. Additionally, we find that these networks deviate from our expectations for trait count under novelty (fewer than expected) and conformity (more than expected) biased transmission.

Conclusion(s): Adding to our basic mechanistic understanding of the interaction between cultural evolution and typical human social structures is a vital step towards an accurate and complete understanding, and interpretation, of cultural data. Our study shows how important it is to take information about the social network structure into account when making inferences of real-world data.

The evolution of developmental flexibility in reliance on social learning

Molleman, L., Van den Berg, P., Vu, T.V.

Objective(s): Human ecological success is often attributed to our unrivalled capacity for social learning, which facilitates the spread of adaptive behaviours. All humans rely on social learning to acquire language, skills, and cultural norms, but mounting evidence reveals substantial individual variation: people differ in their reliance on social learning, and this reliance changes throughout development. In this study, we provide a formal explanation for the existence of these differences.

Methods: We present an evolutionary simulation model that shows how variation in reliance on social learning can emerge, both between individuals and over individuals' lifetimes.

Results: Our model indicates that it can be adaptive to flexibly adjust reliance on social learning if payoffs associated with cultural traits vary between individuals in a population; traits that benefit some individuals may be detrimental for others. Under such circumstances, evolution leads to developmental programmes that enable individuals to adjust their reliance on social learning based on their learning histories, causing some to end up relying more on social learning than others.

Conclusion(s): Our results show that individual variation – often disregarded in models of cultural evolution – can shape how individuals learn to learn from others, with potentially considerable downstream effects on the pace and direction of cultural evolution.

Two-step social learning strategy: from social ties to information use

Pykälä, M.L.E., Efferson, C.M.

Objective(s): Network formation and social learning on a given network both drive information transmission, yet little is known about how they interact. We use a laboratory experiment to investigate how individuals choose ties to form networks, and how the resulting social ties and networks shape information use. To do so, we introduce new methods in which social learning happens in two steps. First, social learners choose ties, and social networks result. Second, social learners make choices based on the information distributed on these networks.

Methods: Participants were separated into two types to complete a learning task. Demonstrators were in self-contained networks and learned both individually and socially. Social learners first observed demonstrator networks and purchased ties to specific demonstrators. We then used the strategy method to elicit each social learner's complete strategy given her ties.

Results: Social learners prefer well-connected demonstrators, and they vary the number of ties they purchase depending on demonstrator network structure. Social learners use both success and frequency information from demonstrators, but how they do so depends on the number of ties purchased.

Conclusion(s): People use multiple, interacting forms of social information. They form social ties strategically, which in turn shapes social networks and how people use social information about choice frequencies and success on a network. These kinds of interactions should be critical to social transmission and cultural evolution.

The wisdom and madness of interactive crowds with a structured network

Xi, J.G., Toyokawa, W.

Previous studies showed that conformist social learning could be beneficial even though individuals potentially exhibit adverse bias. In a collective learning situation with a fully connected social network structure, the self-organizing group dynamics can rescue individuals from the suboptimal bias (collective behavioral rescue). However, social information is not always equally accessible to all individuals during social interaction, and the amount and quality of social interaction are determined by how information travels through the structured social network. Varying information efficiency of different networks might impact group performance in social learning. In order to understand how diverse social network structures may alter the dynamics of collective decision making by social learners, we used an agent-based simulation in a repeated two-armed risky bandit task with varying social network structures from the ring topology (the least density, locally connected network) to the perfect graph (the fully connected network; a replication of the previous model). The results showed that (1) collective rescue emerged even in the ring network, and that (2) increased social network density did not qualitatively change the trend of collective behavioral rescue, although (3) they amplified the existing biases. Interestingly, the magnitude of the amplification of risk aversion was larger than that of risk seeking. The results suggest that having an efficient network topology is a double-edged sword: it boosts both possibilities of collective intelligence and suboptimal informational herding. These results shed light on how social network structure influences the evolution of social learning strategies.

Cultural variation in first- versus third-party attention and imitation

Wen, N.J., Clegg, J.M., Cicardo, C.G., Jacome, A.P., Legare, C.H.

Objective(s): We examined variation in children's observational learning behaviors and imitation in the U.S. and Vanuatu (N=128 8-11 years-old). We selected these populations due to distinct differences in children's experiences with observational learning and time spent with peers, rather than in direct interaction with adults.

Methods: An experimenter demonstrated a novel necklace-making task to one child (Child 1) while another child (Child 2) observed. In a between-subjects design, the demonstration was prefaced with conventional or instrumental language regarding the task goal. Following the demonstration, Child 1 could interact with the materials before leaving the room, followed by Child 2.

Results: We examined three measures: Child 1's imitative fidelity, Child 2's attention during the demonstration, and Child 2's imitative fidelity. First, we examined whether being observed impacted children's responses to instrumental versus conventional behavior. We found that U.S. children imitated conventional behavior with higher fidelity than instrumental behavior, $t(20)=-0.30$, $p=.054$. Ni-Vanuatu children imitated with high fidelity across conditions, $t(39)=0.30$, $p=.763$. Second, we examined whether children acting as observers paid more attention to demonstrations of conventional than instrumental behavior and whether Ni-Vanuatu children paid more attention overall than U.S. children. U.S. children paid greater attention when observing a conventional than instrumental behavior, $t(20)=-2.07$, $p=.052$, whereas Ni-Vanuatu children paid the same attention across conditions, $t(40)=-.107$, $p=.915$. Finally, we examined whether the onlooker child's imitative fidelity differed depending on the type of task observed. U.S. children imitated with high fidelity overall, $t(20)=-.301$, $p=.767$. Ni-Vanuatu children also imitated with high fidelity, but demonstrated higher imitative fidelity in conventional than instrumental behavior, $t(38)=-2.05$, $p=.048$.

Conclusion(s): Our data demonstrate sensitivity to distinct behavioral goals in children's attention and imitative fidelity. We also document high levels of imitative fidelity across highly diverse populations of children, which provides evidence for keen attention to the behavior of others in childhood.

Heterogeneity in frequency-dependent social learning under cognitive load

Von Flöe, L., Vogt, S., Efferson, C.

Social learning can be radically heterogeneous, both within, as well as across individuals. However, we do not yet fully understand how, when, and why individuals vary in their use of social learning strategies and how this heterogeneity precisely affects evolutionary processes. In our behavioural experiment we examine different factors influencing frequency-dependent social learning. We focus on cognitive load, a factor that, to our knowledge, has not been studied before. Participants conducted a binary choice task. One group, which we

call social learners, observed the choice distribution of another group, which we call demonstrators, before making their choices. Our study manipulated whether social learners were similar or dissimilar to the demonstrators of their ingroup and whether they observed ingroup or outgroup demonstrators. These two treatment variations created four conditions that were informationally equivalent. In other words, if social learning of participants was completely flexible, they could perform equally well in all conditions. In addition, we implemented a cognitive load treatment that required half of the social learners to memorise a multi-digit number during the social learning task. We find that social learning was not flexible. Social learners performed worse when being dissimilar and they performed worse when observing outgroup demonstrators. Surprisingly, cognitive load did not affect social learners' performance. Our study indicates that other factors outweigh cognitive load as determinants of social learning. Crucially, certain social learning biases might be so robust that not even cognitive distractions affect them.

Teaching by leaders drives the evolution of opaque culture and egalitarian cooperation

Garfield, Z., Peña, J., Lew-Lery, S.

Despite a sizable literature on teaching in the evolution of instrumental culture (e.g., subsistence technology, manufacturing) there has been relatively little attention on the role of teachers in transmitting opaque culture, including social values, kinship, norms, and cultural history – key cultural features facilitating our species' unique cooperative propensities. Furthermore, the transmission of opaque culture is more likely to require teaching, as these cultural features are complex and not easily observable. Socially influential individuals, i.e., leaders, are best positioned to transmit such cultural information and will be incentivized to do so to benefit themselves and their social partners and kin via the gains of cooperation. We leverage cross-cultural data and introduce an empirically driven theory suggesting leader-biased teaching is a key cultural evolutionary process facilitating cooperation within more egalitarian contexts. We draw on the hunter-gatherer social learning data, a cross-cultural ethnographic database of 23 relatively egalitarian foraging societies, to compare evidence for teaching to other forms of social learning using Bayesian multilevel modeling. We find teaching is more associated with the transmission of cultural values, kinship knowledge, and religious beliefs and less associated with subsistence skills or manufacturing knowledge. Teaching is also closely linked to leadership. Drawing on comparative results, we developed a mathematical two-population replicator dynamic model to explore the costliness of teaching by informed leaders and learning by naive followers in facilitating the evolution of cooperative norms. We suggest that leader-biased teaching is a critical mechanism in the evolution of cooperation, particularly within more egalitarian socio-cultural contexts.

Life history theory

4.15-6.00pm; Lecture Theatre 1

How wealth influences age at first reproduction among Pimbwe women from Tanzania.

Varas Enríquez, P.J., Borgerhoff Mulder, M., Collieran, H., Lukas, D.

Objective(s): Age at first reproduction is a crucial component of fitness. When a woman has her first descendant influences the length and pace of her reproductive career, and lifetime reproductive output. Evidence suggests that women postpone their reproductive onset because they either have large amounts of wealth (i.e. absolute) or have a stable amount through time (i.e. variability). Regarding absolute wealth, research has focused on the allocation of resources towards survival, reproduction, and parental investment (e.g. life history trade-offs), whereas for wealth variability the focus has been towards uncertainty and early life conditions (e.g. adaptive developmental plasticity). However, studies focus on the relationship of either absolute or variability of wealth to explain individual differences of age at first reproduction among women. Here, we use longitudinal data to study how both aspects of wealth, absolute and variability, influence the timing of reproductive onset among Pimbwe women from Tanzania to explore the extent to which both might explain different patterns of age at first reproduction.

Methods: We build a multivariate Bayesian logistic regression model to analyse how the absolute (i.e. average) and variability (i.e. coefficient of variation) of household material wealth throughout pre-reproductive years may shape the timing of reproductive onset among Pimbwe women (n=335). We use a Gaussian process model to account for autocorrelation of wealth among individuals due to age.

Results: The results will show the extent to which the average and coefficient of variation of material wealth that a woman experiences throughout her pre-reproductive years can account for her age at first reproduction.

Conclusion(s): Knowing how the absolute amount of wealth or its variations through life influence the timing of women to start their reproduction might shed light on the mechanisms behind reproductive decisions, allowing researchers to disentangle the mixed evidence regarding the link between wealth and the reproductive strategies of women.

Exploring Subjective Financial Strain, Present-Orientation, and Material Context Using a Large, Representative UK Panel Dataset

Buzan, J., Sheehy-Skeffington, J.

Objective(s): Previous research has explored the relationship between present-orientation and socioeconomic conditions, pointing to the role of financial strain in prompting individuals to act in present-oriented ways. However, it remains unclear if present-oriented behaviour in adverse environments should be interpreted as a product of financial strain or as an adaptive recalibration to the socioeconomic context. This paper uses the large, representative UK panel dataset Understanding Society to examine the relationships between subjective financial strain, present-orientation, and material environmental situation.

Methods: We first test if the relationship between subjective financial strain and present-orientation is mediated by general strain and sense of control. Next we consider if this relationship maps onto real-world behaviour, using longitudinal analysis to build evidence towards the causal impact of subjective financial strain on smoking. Last, we use a dominance analysis to explore the underlying drivers of subjective financial strain, and a latent factor analysis to assess the relationship between material environmental situation, present-orientation, and subjective financial strain.

Results: We find the relationship between subjective financial strain and present-orientation (first as a survey-based measure of present-orientation and next as reported smoking behaviour) is mediated by sense of control but not by general strain. Tentative evidence also suggests that subjective financial strain predicts smoking behaviour over time. Next, we find material deprivation ranks most important in predicting subjective appraisals of financial strain, while monthly household income ranks least important. Last we find the relationship between material environmental situation and present-orientation is mediated by sense of control but not by general strain or by financial strain.

Conclusion(s): While more research is needed to bolster causal claims, our findings provide tentative evidence that present-oriented behaviour in environments of adversity may be better understood as a recalibration to specific, concrete experiences of material adversity rather than as a product of financial strain.

Exploring existential risk from a life-history perspective

Gordon, D.S., Birney, M.E.

Objective(s): Behavioural responses to intrinsic and extrinsic risks have been heavily studied from a life-history perspective. However, human intelligence means we also face an evolutionarily novel type of risk: existential risk, the risk of human extinction as the result of a seemingly far-removed phenomenon (e.g., climate change). However, how existential risk affects general human behaviour is poorly studied. Using priming, the current study investigates how responses to existential risk compare to responses to extrinsic risk by measuring family planning intentions.

Methods: Study 1 piloted the priming material. Participants (N=400) were exposed to either a control group or one of two types of risk: local crime (extrinsic risk) or the threat of asteroids (existential risk). Using these primes, Study 2 (N = 900) will ask participants to state their ideal number of children, their ideal age of first child, and about their childhood environment.

Results: Participants in S1 did perceive local crime to be a greater risk to themselves than asteroids, and vice versa for risk to "humanity". In S2, we expect both risk-type primes to affect family planning decisions compared to control but make no prediction as to their relationship to one another.

Conclusion(s): Given the increased intensity of existential risks we now face it is important to understand how these impact on human behaviour. No matter now participants in S2 respond, we hope the pattern of results will help us further understand the impact of existential risks on behaviour, and whether communicating about such risks might produce unintended consequences.

Effects of social support on energy expenditure, perceived fatigue, and performance outputs during aerobic physical activity

Davis, A.J., Greenhouse-Tucknott, A., Beedie, C., Cohen, E.

Objective(s): This research investigates how cues to social support affect energy expenditure, perceived fatigue and effort, and performance during endurance exercise, and how these effects vary according to individuals' ontogenetic adaptation to early life social environments. Social support is associated with safety, greater access to resources, and positive health outcomes. As a result, fitness-relevant homeostatic functions, such pain, stress, and fatigue, are tied to perceptions of social support, and ontogenetic adaptations to early life social environments tune how individuals respond to social cues. In the context of endurance exercise in humans, previous laboratory-based research suggests that social support can reduce perceptions of fatigue while enhancing performance. This field study further investigates whether socially-modulated energy regulation (rather than motivation or efficiency) underlies these effects.

Methods: We used a within-subjects design to investigate how young rowers (14-18 years old; $n = 25$) respond to a cue to social support during an ecologically valid rowing trial.

Results: Our analyses test whether participants produce greater physical outputs (total Watts) while expending more energy (total oxygen consumption) in the social support condition, while reporting similar levels of effort and fatigue as in the control condition. Analyses also test whether positive social support effects were strongest in individuals who reliably experienced social support during childhood.

Conclusion(s): Social support alters homeostatic processes governing energy expenditure, facilitating greater physical outputs without corresponding increases in perceived effort and fatigue. Responses to social cues may be ontogenetically tuned to match individuals' histories of benefiting from social resources in early life.

The Economic Origins of Ascetic Values and Self-Discipline: Evidence from Medieval Europe

Baumard, N., Huillery, E., Zabrocki, L.

Max Weber famously argued that ascetic values and self-discipline, the so-called 'protestant ethics' played a key role in the economic development of Western societies. However, this association between ascetic values and economic development is not specific to Europe, and has now been observed in other ancient societies (e.g. China, Japan, Greece). Moreover, the causality could be the other way around, from economic development to ascetic values. Evolutionary human sciences have shown that high levels of resources have important effects on human preferences: when resources are abundant, individuals tend to be more future-oriented and to postpone immediate gratification in order to obtain later but greater rewards, hence favoring self-discipline over indulgence. In this paper, we use Medieval Europe as a test case for the hypothesis that the higher levels of resources explain the rise of self-discipline values in Eurasian societies. To do so, we construct a unique database of Christian saints to measure the historical importance of self-discipline over 1200 years (300 - 1500 CE). We use soil suitability to the introduction of the heavy plow as an instrument for increasing levels of resources, as measured by population density and urbanization, to identify the causal impact of increasing levels of resources on ascetic values. Our results show that a rise in resources led to a rise in the proportion of saints described as ascetic. This paper provides the first longitudinal and causal evidence that economic development cause changes in social preferences and religious values in history.

Twinning rate increases with improved living standards across 68 low-to-middle income countries

Lee, D.S., Barclay, K.

Objective(s): The 'insurance ova' hypothesis states that twinning is a by-product of selection for polyovulation in older age against increased risk of fetal loss. From this view, twinning rate is expected to increase with i) age, as well as with ii) improvement in living conditions that would generally reduce the risk of fetal loss. We aimed to test these ecological processes shaping the variation in twinning rate.

Methods: We used the Demographic and Health Surveys from 68 low-to-middle income countries, where medically assisted reproduction (MAR) is not yet prevalent and thus does not confound the aging effect (i.e., more MAR uptake with older age). First, we estimated generalized linear models for per-birth twinning probability for each country. Using a within-mother comparison, we adjusted for individual variation in twinning propensity beyond that due to the fixed-effects of i) age and ii) a country's Human Development Index (HDI) at the time of birth. Testing the net effect of maternal age is important, given that the general increase in HDI within countries over time means that women, as they age, also conceive in a more favorable living condition. Second, we conducted meta-analyses to summarize findings across countries.

Results: The probability of twinning peaked at late thirties, replicating findings from high-income countries before the spread of MAR. An increase in HDI by 0.1 within a country was associated with the probability of twinning higher in average by 7 %. The findings were consistent across countries, withstanding the between-country variation in twinning rate.

Conclusion(s): Given the rising mean age at childbearing in low-to-middle income countries, our findings imply greater proportion of twins in future generations even if the spread of MAR remains low. The present study demonstrates how the evolved mechanisms of reproductive functioning may interact with ecological changes.

Genetic clustering of reproductive traits with other components of the pace-of-life syndrome

Hörak, P., Valge, M., Meitern, R.

Objective(s): Research in evolutionary psychology has a tradition of explaining individual variation in behavioural and physiological traits related to reproduction as adaptive plastic responses to psychosocial stress in the growth environment. Such an approach often ignores the alternative explanation that parents and offspring may share a genetic liability for a spectrum of correlated life-history, behavioural and physiological traits. Under this scenario, traits characteristic of a fast pace of life, such as early maturation, sexual debut and reproduction, cluster genetically with a propensity to live in a stressful environment, short lifespan and associated physiological, behavioural and health-related traits. We evaluate the evidence for this scenario.

Methods: First, we review the published evidence of genome-wide association (GWA) studies about genetic correlations between traits related to the timing of reproduction, growth environment and lifespan. Second, we present the original data on inter-generational phenotypic correlations between the rate of daughters' sexual maturation and parental lifespan/longevity among the participants of the Estonian Biobank and schoolchildren sampled in the middle of the 20th century. In the latter dataset, we also test whether children's age of first reproduction predicts parental longevity.

Results: GWA studies generally demonstrate moderate to strong genetic correlations between life-history, behavioural and physiological traits, consistent with the much-debated concepts of the fast-slow continuum of life-history speed and the pace-of-life syndrome. The menarcheal age of the Estonian Biobank participants did not predict their mothers' survival but did predict fathers' survival in well-educated families. In the schoolchildren's dataset, the rate of breast development robustly predicted the longevity of both mothers and fathers. Daughters' (but not sons') age of first reproduction only predicted fathers' longevity in unskilled manual professions.

Conclusion(s): GWA studies provided robust and inter-generational studies provided partial evidence for the genetic clustering of reproductive traits with other components of the pace-of-life syndrome.

Joint evolution of life-history and the deleterious mutation rate

Avila, P., Lehmann, L.

The cost of germline maintenance gives rise to a trade-off between lowering the deleterious mutation rate and investing into life-history functions. Life-history and the mutation rate therefore coevolve, but this joint evolutionary process is not well understood. Here, we develop a mathematical model to analyse the long-term evolution of individual resource allocation traits affecting life-history and the deleterious mutation rate. We show that the invasion fitness of alleles controlling allocation to life-history functions and mutation rate reduction can be approximated by the basic reproductive number of the least-loaded class, which is the expected lifetime production of offspring without deleterious mutations born to individuals with no deleterious mutations. Second, we analyse two specific biological scenarios: (i) coevolution between reproductive effort and germline maintenance and (ii) coevolution between age at maturity and germline maintenance. This provides two broad biological predictions. First, resource allocation to germline maintenance, at the expense of

allocation to survival and reproduction, tends to increase as external causes of mutation rate increase (e.g. environmental stress, oxygen levels) and to tilt allocation towards reproduction instead of survival. Second, when such external causes increase, life-histories tend to be faster with individuals having shorter life spans and smaller body sizes at maturity.

An Ecological Energetics Approach to the Industrial Revolution, 1800-2023

Beheim, B.

Since roughly the year 1800, the human species has begun occupying a strange new niche among mammals, evident in three related phenomena: the systematic decrease in fertility and increase in longevity called the Demographic Transition, the appearance of the WEIRD phenotype, and, most alarmingly, the shocks to Earth's climate and biosphere that define the Anthropocene. Here, I approach these three novel phenomena with the tools of ecological energetics, the study of energy flows through the living ecosystem. Drawing on ecological models of population growth, I show how the Industrial Revolution can be understood as an unprecedented and sustained increase in our species' access to energy, primarily through the sudden and intense exploitation of calories derived from fossil fuels. Beginning with coal mining in England the early 19th century, and still ongoing today with the arrival of electricity and gasoline motors to subsistence populations in the Global South, this transition has resulted in enormous increases standard of living even as global population size has repeatedly doubled. This process is comparable to other examples of ecological release, in which a species rapidly breaks out of its existing niches. The negative consequences of these events to Earth's biosphere, however, now appear catastrophic: mass extinctions of global wildlife and increasingly severe weather irregularities. Situating recent human history in a larger ecological and evolutionary framework is essential for understanding how humanity got to our current ecological position, and navigating the worst long-term consequences of our unusual mammalian lifeways.

Attractiveness and mate choice

4.15-6.00pm; Lecture Theatre 2

What can sex toys tell us about preferences for male genital morphology?

Johns, S., Bushnell, N.

There is limited research into the morphology of sex toys, and specifically into (the often phallic-shaped) vibrators and dildos and what they may represent in terms of user preferences for male genital morphology. This study provides insight into consumer preference around vaginally insertable sex toys, their features, and what contributes to their popularity. Using a data set compiling information from the world's largest online sexual wellness retailer Lovehoney, we examined the dimensions, price, and morphological features of 265 sex toys designed for vaginal insertion to determine what contributes to item popularity. Using regression models, we found that realistic features did not predict item popularity, whereas price ($p < 0.001$) and circumference ($p = 0.01$) significantly predicted the overall popularity of a toy. It appears that consumers show a preference for insertable sex toys that are not direct replicas of the male penis, which suggests they are not seeking a realistic partner substitute. Further, we found that the length of the toy did not significantly predict popularity which is consistent with other work showing that women do not place considerable emphasis on large phallus size. Our results can contribute to future product design and marketing, as well as reveal preferences towards particular characteristics of the phallus (whether real or toy).

The Autumn Years: Age Differences in Preferences for Sexually Dimorphic Faces

Han, C., Li, X., Chen, X., Lei, X., Liao, C., Zhang, L., Li, P., Peng, X., Morrison, E.R.

Objective(s): Life history theory proposes that it is adaptive for older people to shift investment away from reproductive effort (such as mating) to survivorship. However, it remains unclear whether the shift is also present at the psychological level. We investigated this question by comparing preferences for mate choice-relevant cues, sexually dimorphic facial images.

Methods: 92 older (60 years+, $n = 92$) and 86 younger adults (18–40 years, $n = 86$) viewed 30 pairs of faces whose shape had been masculinised or feminised using computer-graphics techniques. Responses were analysed using mixed binary logistic regression.

Results: Older adults had significantly smaller preferences for sexually dimorphic faces of both sexes than young adults. Specifically, both older men and women showed no significant preferences for sexually dimorphic traits when judging opposite-sex faces, and smaller preferences for masculine male faces and feminine female faces when judging same-sex faces. Young adults generally showed strong preferences for masculine male faces and feminine female faces. In a further study, we confirmed that the absent/reduced preferences in older adults for sexually dimorphic faces did not result from poor visual ability.

Conclusion(s): The smaller preferences for sexually dimorphic facial cues in older adults compared to young adults suggest that older adults may shift away from mating-oriented psychology as they become less fertile.

Save the Last Dance for Me: Dynamic Cues and Social Perception

Abdul Kader, M.I.

Evolutionary psychological theories suggest that social perception plays a key role in making decisions on mate choices, relationships, etc. Previous studies have explored social perception using static cues like facial symmetry, skin colour, and body shape from static stimuli such as photographs. However, real-life judgements are made on moving bodies and less is known about the dynamic cues that may have an influence on our perceptions. This pilot study chose two contrasting forms of movements—walking and dancing—to study the perception of three basic factors: attractiveness, competence and warmth. While walking is a simple and common form of movement, dancing is a complex form of movement which also plays a role in wooing and courtship. To separate dynamic from static cues, motion-capture technology was used. Normal videos and motion-captured videos of walking and dancing were collected from 30 individuals along with their photographs which were rated for attractiveness, competence and warmth by 12 raters. Multiple regression analysis showed static cues were most important when judging attractiveness and warmth, and dynamic cues were only important in the perception of competence with dance as the stimulus. These findings suggest that movements play a role in social perception and should be considered in related studies. This pilot shows this is an effective way to study the role of movements in the perception of attractiveness, competence and warmth, paving the way for further studies with a larger stimulus set and more raters. Future research will explore how dynamic sexual dimorphism influences social perceptions in dance.

Internalisation of the cultural ideal predicts judgments of physical attractiveness

Ridley, B., Cornelissen, P., Maalin, N., Mohamed, S., Kramer, R. S., McCarty, K., & Tóré, M. J.

We investigated perceptions of female and male body attractiveness, and how individual participant differences influence these perceptions. In study one, male and female participants rated the attractiveness of 242 biometrically valid male and female computer-generated bodies, which varied independently in muscle and adipose. Across the whole-body composition space, therefore, we were able to assess changes in attractiveness and calculate peak attractiveness for both men and women. In the second study, participants used a method of adjustment task and were required to choose the most attractive male and female bodies. The task allowed them to independently vary muscle and adipose content to create the most attractive body. Individual differences in internalization of cultural ideals, drive for muscularity, eating disorder symptomatology and depressive symptoms were also assessed, to investigate if they could systematically shift the location of peak attractiveness in body composition space. A clear preference was found by both genders for a male body with high muscle and low adipose, and a female body that was toned and had low adipose. The large individual differences in the composition of the most attractive bodies were predicted by the degree to which an individual internalised the cultural ideals.

Halo effect of faces and bodies: Cross-cultural similarities and differences between German and Japanese observers

Russ, J., Kordsmeyer, T., Freund, D., Ueshima, A., Kuroda, K., Kameda, T., Penke, L.

Objective(s): According to the halo effect, person perceptions are globally biased by specific traits or characteristics. Attractive people are attributed positive traits like prosociality, health, and dominance. However, due to a strong focus on facial stimuli it remains unclear whether this effect can also be found for bodies. Furthermore, most studies involved observers from individualistic cultures. This preregistered study explored the consistency of halo effects for men's faces and bodies for observers from individualistic and collectivistic cultures.

Methods: Facial photos and 3D body scans of 165 German men were judged separately for attractiveness, prosociality, health, and dominance by 123 German and 100 Japanese observers.

Results: Results were mostly consistent between both observer groups and revealed strong attractiveness halo effects for faces and bodies, and a dominance halo effect for bodies, but not faces. Further predictions of the one ornament hypothesis were supported.

Conclusion(s): This study provides new insights on halo effects as consistent cognitive biases in person perception for faces and bodies for observers with different cultural backgrounds.

The effect of humor styles on mate value and preferences in an online experiment

Grundstroem, H., Fredriksson, A., Nyman-Kurkiala, P., Hemberg, J., Madison, G.

Humor is likely to serve as signals of fitness in potential partners. Less is known about how different styles of humor affect partner attractiveness. This study aimed to test the attractiveness of the four different humor styles proposed by Martin et al. (2003) categorized according to being benign (affiliate, self-enhancing) or detrimental (aggressive, self-defeating). Participants were presented with a series of potential partners, much like on a dating site. Each partner was described by a portrait picture and a vignette, which included examples of one of the four humor styles. The participants' task was to rate a number of items about partner preference (date, intercourse, short-and long-term relationships) and mate value (intelligence, health, social status and parenting skill). A total of 170 women and 81 men between 18-40 years of age completed the experiment. The results showed significant effects on all measurements of partner interest and mate value for women with the aggressive humor style being rated as less attractive and lower in mate value than the other humor styles. For men there was a significant effect on two measurements on mate value (social status, parenting skill), showing that the self-defeating style was rated less attractive. The results support the notion that humor is used as a fitness signal, that this is used to a substantially greater extent by women, and that women find the aggressive humor style to be particularly unattractive in potential partners.

Testing ovulatory cycle shifts in motivational priorities and dual sexuality in a large diary study of romantic couples

Penke, L., Schleifenbaum, L., Stern, J., Driebe, J.C., Wieczorek, L.L., Gerlach, T.M., Arslan, R. C.

Objective(s): Human female ovulatory cycles have been compared to other primates' estrus. Based on this, earlier studies proposed a dual sexuality, with ovulatory cycle shifts in female preferences for genetic quality and male partner responsiveness to resulting threats of sexual infidelity. However, recent large-scale replication attempts were mostly unresponsive. An alternative hypothesis is that female motivational priorities for mating vs. foraging and somatic investment shift over the ovulatory cycle, adaptively regulating effort allocation.

Methods: In a preregistered 40-day online diary study, 390 heterosexual women and 384 of their romantic partners reported daily on their motivations and behaviours in the domains of sexuality, mate retention, and eating, as well as their perceptions of their partner's motivations and attractiveness. Women also reported on their menstrual cycle and hormonal contraception usage.

Results: We compared 209 naturally cycling women to 181 women taking hormonal contraceptives (HC) to infer ovulatory shifts. We found robust ovulatory decreases in food intake and increases in general sexual desire, in-pair sexual desire, and initiation of dyadic sexual behaviour, but not extra-pair sexual desire. However, we found no compelling evidence that male partners notice women's fertility status (as potentially reflected in women's attractiveness, sexual desire, or wish for contact with others) or display mid-cycle increases in mate retention tactics (jealousy, attention, wish for contact or sexual desire towards female partner).

Conclusion(s): Results support the motivational priorities shift hypothesis, but contradict the good genes ovulatory cycle shift hypothesis. The state of these hypotheses in the literature will be critically discussed.

Menstrual cycle shifts in female competitiveness: an evolutionary account

Arthur, L.C., Casto, K.V., Blake, K.R.

Objective(s): An evolutionary account of the endocrine system suggests that hormones mediate the allocation of limited energetic resources towards particular behavioural endpoints. A growing body of research has begun investigating the relationship between hormones and female competition for resources that enhance survival and reproduction. The aim of the current research was to use a rigorous and interdisciplinary perspective to investigate the relationship between menstrual cycle phase, hormonal contraceptive use and female competitiveness.

Methods: This talk will outline the methods and results of a citizen science project and a lab-based behavioural quasi-experiment. The citizen science project used a longitudinal diary study design (3,900+ observations from 21 countries) to explore cycle-phase variations in competitiveness among hormonal contraceptive users and non-users. In a quasi-experiment, participants in different cycle-phases (mid-follicular and mid-luteal) and hormonal contraceptive users, participated in a series of competitive tasks. Salivary progesterone, cortisol and testosterone were assayed.

Results: In our longitudinal study, fertility was associated with an increase in achievement-based competitiveness amongst naturally cycling females but not hormonal contraceptive users. In our behavioural design, participants in the follicular phase had elevated competitive performance compared to the luteal phase and hormonal contraceptive users.

Conclusion(s): Results suggest that naturally cycling females alternate between periods of high and low competitive motivation, and that hormonal contraceptives disrupt this effect. Given the significance of competition for survival and reproduction, understanding how female hormones are associated with competition is crucial. Failing to investigate the way behaviour changes across the menstrual cycle risks the perpetuation of masculinity as the dominant biological framework for humans.

Re-examining hormonal effects on sexual behavior and desire across the menstrual cycle

Arslan, R., Botzet, L. M., Jones, B. C., DeBruine, L. M., Hahn, A., Mumford, S., Schistermann, E., Penke, L., Grebe, N. M., Blake, K., Roney, J., Fiers, T., Marcinkowska, U., Stern, J.

Objective(s): To compare the informativeness of several approaches to investigate hormonal effects on sexual behaviour and desire across the menstrual cycle.

Methods: I draw on eight published datasets (total N > 1,300) with repeated measures of progesterone and estradiol, a large-scale online diary study (total N > 900) and data from a popular cycle tracking app (N > 1,000,000) which allow imputation of serum sex steroids. I investigate the effects of cycle hormones and the time lag at which they might act.

Results: Salivary, but not serum, estradiol enzyme-linked immunoassays have surprisingly low validity for estimating cycle phase and may suffer from substantial bias. Results are not much better for progesterone. In addition, many previous studies underestimated the time lag of hormonal effects. Estradiol and progesterone consistently show lagged effects on sexual desire, leading to a small peri-ovulatory peak. Effects on sexual behaviour are less clear.

Conclusion(s): Greater transparency through preregistration, code and data sharing have moved the field forward, in part by making problems more obvious. Now, further methodological improvements are necessary to characterise replicable phenomena. We need to walk before we can run, that is, agree on improved methodological standards, before we can answer theoretical questions such as whether ovulatory changes in humans are vestiges or evolutionary adaptations. Individual differences may be the telltale cue of selection or its absence.

Thursday, April 20th**Reproduction****10.45-12.30pm; Lecture Theatre 1****Increased birth rank of homosexual males: disentangling the older brother effect and sexual antagonism hypothesis***Raymond, M.*

Male homosexual orientation remains a Darwinian paradox, as there is no consensus on its evolutionary (ultimate) determinants. One intriguing feature of homosexual men is their higher male birth rank compared to heterosexual men. This can be explained by two non-exclusive mechanisms: an antagonistic effect (AE), implying that more fertile women have a higher chance of having a homosexual son and producing children with a higher mean birth rank, or a fraternal birth effect (FBOE), where each additional older brother increases the chances for a male embryo to develop a homosexual orientation due to an immunoreactivity process. However, there is no consensus on whether both FBOE and AE are present in human populations, or if only one of these mechanisms is at play with its effect mimicking the signature of the other mechanism. An additional sororal birth order effect (SBOE) has also recently been proposed. To clarify this situation, we developed theoretical and statistical tools to study FBOE and AE independently or in combination, taking into account all known sampling biases. These tools were applied on new individual data, and on various available published data (two individual datasets, and all relevant aggregated data). Support for FBOE was apparent in aggregated data, with the FBOE increasing linearly with fertility. The FBOE was also supported in two individual datasets. An SBOE is generated when sampling in presence of FBOE, suggesting that controlling for FBOE is required to avoid artefactual SBOE. AE was not supported in individual datasets, including the analysis of the extended maternal family. The implications of these findings are discussed, in particular the evolutionary significance of the FBOE and its origin in the human lineage.

A matter of time: Bateman's principles and mating success as count and duration in contemporary Finland*Andersson, L., Jalovaara, M., Saarela, J., Uggle, C.*

Objective(s): Despite enduring criticisms, Bateman's principles continue to influence the understanding of human reproductive strategies. However, few rigorous examinations of the Bateman gradient in industrialized populations exist, and previous studies have suffered from small sample sizes and the exclusion of non-marital unions.

Methods: Here we address these shortcomings by assessing mating success and reproductive success using population-wide Finnish register data of both marital and non-marital partners and children ever born. We compare two different operationalizations of mating success, the number of partners and cumulated union duration (in years), for men and women 18-46 years (n= 218,748).

Results: These data support Bateman's first and second principles, but the association between mating success and reproductive success is less clear. The number of partners is somewhat more positively associated with reproductive success for men than women, but the mating success-reproductive success association turns negative when adjusting for years with a partner. Having had more partners is associated with lower reproductive success, than having had one partner. Only among men in the lowest income quartile, are more partners positively associated with higher reproductive success.

Conclusion(s): Union duration (controlling for number of partners) is associated with higher male reproductive success and should be incorporated as an important dimension of mating success. For men in contemporary Finland, fitness payoffs to multiple mating appear to vary by social status.

Personality Concepts in Conambo, Ecuador*Lukaszewski, A., Durkee, P.K., Zerbe, J., Patton, J.Q.*

Objective(s): We present a study testing the existence and fitness-linked correlates of personality concepts in the Amazonian village of Conambo, Ecuador, which is home to Achuar and Zapara forager-horticulturalists.

Methods: Lexical terms to describe three focal personality concepts—Sociability, Immodesty, and Un-emotionality—were translated from Spanish into the indigenous languages of Achuar and Quichua. These terms were employed in a photo ranking task wherein 76 adult community members ranked the relative standing of same-sex others on each personality concept. Inter-ranker agreement was high for Sociability and Immodesty, but low for Un-emotionality. Using Bayesian models and psychological networks including age controls, we tested the associations among individual differences in (i) Sociability and Immodesty, (ii) hierarchical status and fertility, which are hypothesized fitness-linked benefits of high Sociability and Immodesty, and (iii) physical strength, which is a hypothesized calibrator of status-oriented personality strategies.

Results: We found good evidence that men's (but not women's) physical strength associates positively with Sociability, Immodesty, and status. Evidence that status associates with fertility was strong for men, but weaker among women. Among both sexes, Sociability and Immodesty exhibited strong positive correlations with status, but evidence was weaker that the personality traits associated with fertility.

Conclusion(s): We conclude that two personality concepts imported from the HEXACO and Big Five taxonomies, Sociability and Immodesty, exist with common meaning in the minds of Conambo villagers and appear adaptively patterned in relation to physical strength and fitness-linked outcomes.

A proposal for measuring costs of reproduction in preindustrial women*Young, E.A., Lummaa, V., Postma, E., Dugdale, H.L.*

Objective(s): Why people age at different rates is poorly understood. However, life-history theory states trade-offs in energy allocation between reproduction and somatic maintenance can help shape this variation. Thus, it is predicted that individuals who invest more resources in reproduction will suffer reduced longevity as a consequence. However, even in pre-industrialized societies, where high reproduction results in substantial reproductive effort, the evidence is inconsistent. There are two key factors that potentially explain this: (1) reproductive effort has largely only been estimated as number of children born; and (2) there is unaccounted variation in resource availability among individuals which is known to mask trade-offs

Methods: Structural equation modelling with latent variables, using an exceptional 19th century dataset from 11,252 reproductive Finnish women, provides an opportunity to resolve this. Here, total reproductive effort can be estimated through combining information from multiple proxies (including children birthed, years spent raising children, and reproductive tenure). Data on individual resource availability (proxied through crop yields and social class) can then be incorporated to examine how it moderates the effect of reproductive effort on longevity.

Results: We predict that the longevity costs of reproductive effort will be highest in individuals with the least resources.

Conclusion(s): In this study we aim to advance our fundamental understanding of ageing while providing a robust method for testing the costs of reproduction in populations with variation amongst individuals in resources.

Subsistence strategy does not predict fertility at the group level, but women who farm have more children*Page, A.E., Ringen, E.J., Koster, J., Borgerhoff Mulder, M., Kramer, K., Shenk, M., Stieglitz, J., Starkweather, K., Ziker, J., Boyette, A. H., Collier, H., Moya, C., Du, J., Mattison, S., Greaves, R., Sum, C., Lai, R., Lew-Lew, S., Kiabiya Ntamboudila, F., Prall, S., Townner, M., Blumenfeld, T., Migliano, A.B., Major-Smith, D., Dyble, M., Salali, G.D., Chaudhary, N., Derks, I., Ross, C., Snela, B., Gurven, M., Winterhalder, B., Cortez, C., Pacheco-Cobos, L., Schacht, R., Macfarlan, S.J., Leonetti, D., French, J.C., Alam, N., tug Zohora, F., Sear, R*

While it is commonly assumed that agriculturalists have higher fertility than foragers, we lack robust supportive evidence. Previous studies have been limited by 1) use of population averages, limiting statistical power and discarding intra-population variation and 2) reliance on subsistence typologies. We test the hypothesis that agriculturalists have higher fertility in 10,250 women from 27 small-scale societies using Bayesian multi-level modelling, finding substantial variation in fertility. This variation did not correspond to subsistence typology given substantial diversity in subsistence strategies themselves: average fertility in societies dependent largely on agriculture was not clearly differentiated from others. We re-tested this

relationship across a range of individual-level subsistence measures, leveraging the intrapopulation diversity in subsistence activities. We found that fertility was positively associated with farming, and negatively associated with foraging and market integration. Overall, however, (traditional) livelihood activities were not strong predictors of fertility, likely because the determinants of fertility, and their associations with subsistence activities, are complex.

Fertility Outcomes and the Division of Household Labor in Australia

Snopkowski, K.

Objective(s): Low fertility appears evolutionarily puzzling and requires examination of conscious decisions to have few children. Gender revolution theories of fertility posit that if women who work in the labor market face a second shift of household responsibilities, they opt to reduce fertility. This study examines how gender equity in the household and gender ideology influence couple's decisions to have additional children among dual-income couples in Australia. Several possible mediators, including mental health, partner relationship quality, and pre-reproductive ideal family size, are examined to elucidate psychological mechanisms that influence parents' decision to have additional children.

Methods: The Household, Income and Labour Dynamics in Australia (HILDA) dataset, a nationally representative household-based longitudinal study covering family life, well-being, and employment patterns of 17,000 Australians is used to test hypotheses. Married and cohabitating heterosexual couples are included in the sample. Models examining progression to each birth are analyzed as discrete-time event history analyses.

Results: Gender equity and gender ideology both influence fertility outcomes, particularly for couples progressing to their third birth. Those women who report greater gender equal ideology are significantly less likely to progress to their third child. In families where husbands contribute more to household chores, women are more likely to progress to their third child if they are working part-time, but less likely if working full time. An examination of mediators shows that only pre-reproductive ideal family size mediates this relationship. When couples report higher relationship satisfaction, they are more likely to progress to both their second and third child when women work part-time.

Conclusion(s): This research demonstrates that division of household labor is one factor that contributes to fertility decision making, providing evidence that women are cognizant of the support they need to raise children.

Using discrete choice modelling to understand the drivers of reproductive delay in the UK.

Sheppard, P.

Small family sizes, as is common across Europe, remains an evolutionary puzzle. However, although many people want small families, on average they get fewer children than they had hoped or planned for. This is known as the fertility gap and is seen in every European country. The fertility gap is largely driven by prolonged postponement of childbearing. Demographers have broad-stroke ideas on what drives this delay but our understanding of what these things mean to aspirant parents or how they interact together is still limited. Following on from qualitative results that were presented last year, I now present the findings from a Discrete Choice Experiment (DCE) conducted on a nationally-representative sample from the UK. One of the most striking findings was that university-educated women value hands-on dads for partners much more than any other factor, including lost career progression, throwing doubt on the male-breadwinner ideology. An estimation of reproductive delay to have a hands-on dad was around eight years which may explain the very lengthy delays observed for this demographic. University men prioritised their partners' readiness for babies suggesting that results for women drive the macro-level trends. I discuss these, and more results from the non-university educated sample, in line with policy implications.

Cooperation and competition

10.45-12.30pm; Lecture Theatre 2

Global characteristics of witchcraft beliefs and accusations

Peacey, S., Mace, R.

Objective(s): Accusations of witchcraft practice, using harmful magic against others, are a widely distributed and harmful cultural trait. To date there has been little cross-cultural quantitative investigation of global patterns of witchcraft accusations. The type of relationships leading to accusations may associate with features of socio-ecology: if accusations are a mechanism for targeting competitors, socio-ecology could determine where competition is highest. So accusations between brothers may occur in societies where they are forced to compete for inheritance or witchcraft beliefs may be less common in nomadic societies than sedentary ones, as individuals can avoid those they are in conflict with. We will 1) describe the cross-cultural distribution of witchcraft beliefs and common accusation patterns and 2) examine their association with socio-ecology.

Methods: We will code variables on witchcraft beliefs and accusation types, using a global sample of small-scale societies from the Ethnographic Atlas and eHRAF (large online databases of historic ethnographic materials). We will test several hypotheses relating to the evolution and socio-ecology of these traits.

Results: We predict patterns of witchcraft beliefs and accusations will broadly correlate with socio-ecology. For example, witchcraft beliefs may be more common in sedentary societies than mobile ones. But relationships may not be universal.

Conclusion(s): Identifying global patterns in witchcraft beliefs and accusations in a new cross-cultural dataset may provide further insight into how and why they evolve.

The Norm Enforcing Role of Witchcraft Beliefs in Mauritius

Willard, A., Rosun, N., de Barra, M.

Objective(s): This project examined if witchcraft beliefs can enforce normative behaviour in Mauritius, and if these beliefs enforce different types of behaviours than religion. Specifically, we are interested in if witchcraft beliefs enforce norms around not being the object of envy.

Methods: We ran two pre-registered studies examining perceived causes of misfortune. In study 1, participants (N=453) read four vignettes about an ill person who was described as having either good (normative) or bad behaviour. Participants were then asked how likely it was that the illness was caused by natural causes, another person (witchcraft), or God. In study 2 participants (N=273) were given a set of 3 short vignettes that described a character neutrally, as making others envious, or being selfish, and then experiencing a misfortune. Participants were asked if the misfortune was caused by God and magic, and if others acting similarly would also experience a misfortune.

Results: Participants in study 1 rated God and another person (witchcraft) as more likely causes of the illness when the character had behaved badly rather than well. In study 2, participants gave higher ratings for God as a potential cause of misfortune when the person behaved selfishly, rather than neutrally, and for witchcraft when a person behaved in a way that could make other envious. Acting selfishly and evoking envy were also rated as more likely to cause misfortune to someone acting similarly than neutral.

Conclusion(s): Across two studies, witchcraft beliefs are seen as a potential cause of bad things happening when people behave in a non-normative way. This suggests that these beliefs can enforce normative behaviour. In study two we showed that witchcraft, but not God, is believed to be more likely to causing misfortune when a person has evoked envy, suggesting witchcraft beliefs may specifically enforce norms around not being the object of envy.

Punitive justice serves to restore reciprocal cooperation in three small-scale societies

Fitouchi, L., Singh, M.

Objective(s): Fines, corporal punishments, and other procedures of punitive justice recur across small-scale societies. While they are often assumed to enforce group norms, we propose the relation-restoration hypothesis of punitive justice, according to which punitive procedures function to restore dyadic cooperation and curtail conflict between offender and victim following violations of reciprocal obligations. We test this hypothesis's predictions against the justice systems of three small-scale societies.

Methods: We code ethnographic reports of 97 transgressions among Kiowa equestrian foragers (North America); analyze a sample of 302 transgressions among Mentawai horticulturalists (Indonesia); and review retributive procedures documented among Nuer pastoralists (South Sudan).

Results: Consistent with the relation-restoration hypothesis, we find that third-party punishment is rare; that most third-party involvement aims at resolving conflicts; that costs paid

by offenders serve to achieve forgiveness by repairing victims; that punitive justice is accompanied by ceremonial procedures aimed at limiting conflict and restoring goodwill; and that failures to impose costs contribute to a decline in reciprocal cooperation. Although much rarer, we find some instances of third-party punishment among the Kiowa (6.6% of offenses), consistent with a norm-enforcement function.

Conclusion(s): Most often, punitive justice serves to appease victims' urge for revenge while not overly harming offenders' interests to ensure reconciliation.

Reputation Management In Everyday Life

Engeler, N.C., Raihani, N.J.

As people monitor the reputations of others to better decide whether and how to interact with them, it is valuable for those individuals to influence how they are viewed, i.e., to manage their reputation. Since reputation has usually been studied in isolation, focusing on single, context-dependent behaviours at a time, we aimed to investigate how reputation management varies across social relationships and situational contexts in everyday life. To collect naturally occurring instances of reputation management, 160 participants used a phone app to complete daily assessments for a week. Herein, they reported on a recent instance in which they managed their reputation, or if they did not recall doing so, answered question on a recent social interaction instead. Findings were largely comparable to previous reports from the laboratory and field. We found that people experienced more reputational concern during reputation management events than during other social interactions, likely because heightened reputational concern motivates us to engage in reputation management. How important an interaction partner was to the participant most strongly influenced levels of reputational concern. Moreover, participants were most concerned about their reputation when engaging with their superiors or strangers rather than close others. Interestingly, social relationship factors influenced reputational concern more than social visibility cues (i.e., factors that increase the 'audience' that could learn from an individual's behaviour, such as social connectedness and likelihood of future interactions). Our findings highlight the need to further study social relationships, particularly the value we place on a relationship, in relation to reputational concern.

The repeated punishment game explains why, and when, we seek revenge

Lie-Panis, J., Burum, B., Hilbe, C., Hoffman, M.

Objective(s): Revenge is often understood in terms of deterrence. When we seek revenge on someone who has wronged us, we pay an immediate cost, which may be offset by the long-term benefit of a retaliatory reputation. Yet, revenge does not always seem designed for optimal deterrence. We forgive others' dangerous behavior based on variables out of their control, such as a lucky positive outcome. Minuscule transgressions can lead to full-blown conflict, rather than a proportionate response. In addition, it is unclear when apologies will mitigate the risk of retaliation.

Methods: To understand revenge and its apparent quirks, we rely first on a mathematical model. The repeated punishment game involves two players. In each round, player 1 may first transgress on player 2, and player 2 may second pay a cost to punish player 1. We look at the conditions under which player 2 is expected to punish transgression, when players are in a subgame perfect equilibrium that sustains cooperation. We rely second on psychological vignettes to test the qualitative predictions arising from our model.

Results: Revenge functions as a deterrent in our model. By punishing transgressions, player 2 reduces the chance of future transgressions against her. In addition, our model makes several qualitative predictions concerning the conditions under which retaliation is expected to occur. For instance, we predict that player 2 will punish even following minuscule transgressions, and that apologies will mitigate the risk of retaliation only when their cost exceeds the benefit of transgression. Our vignette experiments confirm the qualitative predictions arising from our model, for a diversity of realistic scenarios.

Conclusion(s): A dyadic model helps us better understand the deterrent function of revenge, and its apparent quirks. While revenge may ideally ignore luck and take into account the magnitude of transgression, it is constrained by the dyadic context in which it evolved.

Misrepresentation of group contributions undermines conditional cooperation in a human decision making experiment

Van Den Berg, P., Liu, S., Wenseleers, T., Zhang, J.

Cooperative behaviour can evolve through conditional strategies that direct cooperation towards interaction partners who have themselves been cooperative in the past. Such strategies are common in human cooperation, but they can be vulnerable to manipulation: individuals may try to exaggerate their past cooperation to elicit reciprocal contributions or improve their reputation for future gains. Little is known about the prevalence and the ramifications of misrepresentation in human cooperation, neither in general nor about its cultural facets (self-sacrifice for the group is valued differently across cultures). Here, we present a large-scale interactive decision making experiment (N= 870), performed in China and the USA, in which individuals had repeated cooperative interactions in groups. Our results show that (1) most individuals from both cultures overstate their contributions to the group if given the opportunity, (2) misrepresentation of cooperation is detrimental to cooperation in future interactions, and (3) the possibility to build up a personal reputation amplifies the effects of misrepresentation on cooperation in China, but not in the USA. Our results suggest that misrepresentation of cooperation is likely to be an important factor in (the evolution of) human social behaviour, with, depending on culture, diverging impacts on cooperation outcomes.

Is violence necessary to explain the post-neolithic Y chromosome bottleneck?

Chais, R., Guyon, L., Guez, J., Heyer, E.

Objective(s): By analyzing mitochondrial and Y chromosome sequences in worldwide present-day human populations, Karmin et al. (2015) highlighted that the effective size of Y chromosome underwent a severe bottleneck around 5,000 years ago. Such bottleneck was not observed for the mitochondrial DNA. This pattern was interpreted by Zeng et al. (2018) as the result of violent intergroup competition between patrilineal descent groups. Indeed, deaths from such competition would cluster on the genealogical tree of males, increasing the rate of loss of Y-chromosomal lineages and accelerating genetic drift, inducing a severe decrease of Y chromosome diversity. In this study, we explore whether other peaceful socio-demographic scenarios may explain the Y chromosome bottleneck.

Methods: We undertook a modelling approach to evaluate the effect of different shifts in social organization on Y and mtDNA effective sizes. In particular, we simulated a population structured in 5 villages following a patrilocal residence rule. The descent rule is patrilineal (ie individuals are affiliated to their father's descent group). These descent groups follow a segmentary dynamics, meaning that can split or go extinct. Several effective size estimators were computed from the simulated data.

Results: We confirm that intergroup violence explains well the observed bottleneck on the Y chromosome, but we show also that the segmentary dynamics of patrilineal systems alone can account for most of this decrease.

Conclusion(s): A violent scenario is not necessary to explain the observed Y chromosome bottleneck. A past period of patrilineality may account for the observed bottleneck.

Gender, age and the timing of leadership across societies

Redhead, D., Power, E. A., Koster, J., the ENDOW team

Objective(s): Leadership plays an important role in sustaining collective action and cooperation in human societies. Individuals perceived as best able and willing to provide benefits for others often become leaders. Extant empirical evidence, including ethnographic descriptions, have suggested that leadership is reliably tracked by age. Alongside this, culturally-determined gender roles may constrain womens' leadership in certain societies. We examine how gender and age structure leadership across roughly 30 rural communities, and test predictions about the cultural and ecological factors that determine gender or age differences in leadership.

Methods: We collected data from at roughly 30 culturally, economically and ecologically diverse rural communities. These data included context-specific measurements of whether individuals occupied important leadership positions, reported gender and age, full genealogical information, and household wealth inventories.

Results: We examined the types of leadership positions observed across communities. We applied a Bayesian Gaussian process model that estimated the non-linear associations that age has leadership for each gender, and for each society. We find that leadership takes many forms across different societies, and that the age timing of leadership shows important similarities and difference between genders and across societies.

Conclusion(s): Our findings present some of the first empirical comparisons of the correlates of leadership across societies. These results suggest that leadership is indeed structured by gender and age in ways that are complementary to theoretical predictions, and shed new light on the cultural and ecological origins of gendered differences in leadership in human societies.

Mating strategies and sexual selection

10.45-12.30pm; Jeremy Bentham Room

Vocal masculinity, when accompanied by visual information, affects intra-sexual dominance perceptions but not inter-sexual attractiveness

Rutter, J., Boothroyd, L.G.

Objective(s): Numerous studies have found that male voices are perceived as being more attractive and dominant when their pitch and formant frequencies are lowered. However, to date, all of these studies have relied solely on audio stimuli, so we cannot say whether these findings hold in more ecologically valid contexts. For example, voice pitch may be redundant in detecting dominance when height and muscularity are visible. This study therefore tested social perceptions of vocal pitch in full audiovisual stimuli.

Methods: 17 men were recorded counting to five in the same clothing. The audio of these recordings were then manipulated in pitch and formant frequency to produce 12 variants of each video which were rated online by 20 unfamiliar participants for attractiveness, physical dominance, strength, and social dominance.

Results: In line with previous findings, men with feminised voices (i.e. raised in pitch and formant frequency) were perceived as being less attractive, less dominant (physically and socially), and less strong. Men with masculinised voices were perceived as physically stronger and more dominant (physically and socially). Attractiveness was not significantly affected by lowered pitch manipulations in these video stimuli.

Conclusion(s): These results support intrasexual selection theory of the human voice – that the male voice evolved to signal dominance to other men, rather than attractiveness to women – even when other visual signals of formidability are present.

Measuring Individual Differences in Testosterone-Induced Behaviour with a short scale

Ostermann, S., Herms, U., Penke, L.

In the evolutionary literature, testosterone is prominently discussed as an endocrine mechanism mediating trade-offs of effort allocation between competitive and nurturing behaviours. However, associations between testosterone and behaviour are often inconsistent. One potential moderator of this association is personality. Based on prior research, dominance, impulsivity and self-construal were identified as relevant personality dimensions: testosterone effects on behavior were stronger in individuals scoring high on these personality traits. A composite of self-report scales measuring this trait has been used successfully to explain heterogeneity in testosterone-aggression associations, but this composite is highly uneconomical due to a large amount of items, and the composite's psychometric properties have never been established. We aimed to rectify this by developing a short scale, measuring relevant individual differences that moderate behavioural consequences of testosterone. Psychometrically developed based on an online study (N = 520), the resulting short scale consists of 15 items, is reliable, and correlates highly ($r > .8$) with the original composite. The items indicate that the scale measures mainly self-centered and anti-communal tendencies.

Systematic review and meta-analysis of testosterone and its role in human sexual behaviour and attitudes towards sex

Ford, G.C., Morrison, E. R.

The role of testosterone in human behaviour is complex and has so far shown mixed results. This PRISMA protocol review aims to synthesize the literature, and calculate the effect sizes of testosterone and sexual behaviour, psychology (i.e. desire etc.) and attitudes (i.e. sociosexuality). The review will highlight gaps and identify the quality of the research. Prospective, cross-sectional observations of baseline and reactive endogenous testosterone levels and sexual behaviour/attitudes will be included. Exogenous testosterone administration studies will also be included. Studies are those with age related controls (if required) and are written in English, with no date of publication restrictions. Study participants are healthy men and women that are over the age of 18 years of age. Participants with disabilities, disorders or known illnesses are not included. All immunoassay measures of testosterone will be included (i.e. salivary and plasma). Studies must also control for diurnal testosterone fluctuations by collecting samples at a similar time of day, or statistically adjust for this accordingly. Pregnant women/couples will be excluded. This review searched PubMed, Web of Science, PsycInfo, PsychArticles from January to November 2022 and resulted in 37,741 studies. Risk of bias within the results will be assessed. Estimated effects (and confidence intervals) will be reported for each main outcome, depending on heterogeneity. For example, testosterone and number of sexual partners; one-night stands, extra-pair copulations, sexual debut, and frequency of masturbation. An estimated effect will also be calculated for testosterone and sexual desire, frequency preferences and sociosexuality.

Cheating Up: Testing Two Explanations for Female Infidelity in Humans

Murphy, M., Philips, C.A., Blake, K.R.

Objective(s): Evolutionary scholars argue that infidelity benefits men by increasing their offspring quantity, but the benefits to women are debated. Two leading theories of female infidelity both focus on quality: The “good genes” hypothesis posits female infidelity evolved so women could couple the (in some way) “better” genes of an affair partner with the paternal investment of a primary partner, whereas the mate-switching hypothesis argues women's affairs mainly help them trade to a more desirable primary partner. Minimal empirical work tests these theories against each other, and both share theoretical challenges. Our study tests these hypotheses against each other and their alternatives.

Methods: Our study examines how individuals rate their in-pair and extra-pair mates' physical, personal, parental, and overall desirability by administering an online questionnaire to 772 individuals recently engaged in infidelity from multiple countries. This is among the largest and most diverse samples in the evolutionary infidelity literature.

Results: If the focus of the mainstream debate is justified, women, more than men, should find their extra-pair mate more desirable than their in-pair mate on at least one scale.

Different scales should exhibit this preference depending on whether women's affairs are to obtain genes or a new mate. For example, women rating extra-pair mates more attractive as co-parents would match the mate-switching hypothesis but not the good genes hypothesis. Other results would lend credence to the importance of the many alternatives to these two hypotheses, such as Brooke Scelza's multiple-investors hypothesis.

Conclusion(s): Our results will substantially contribute to the female infidelity debate either by challenging the shared assumption of frontrunners (that women cheat for quality, not quantity) or by distinguishing them. My presentation will contrast theories and explain the implications of our study, which will be complete by EHBEA 2023.

Parental interference in their offspring's mate choice: not as disruptive as one may think?

Fišerová, A., Hanlíček, J., Štěrbová, Z.

From the evolutionary perspective, mate choice in humans is specific by the involvement of kin both in mate choice and in the offspring's relationship functioning. The various reasons of parental interference are rather complex ranging from genetic relatedness to their offspring, resources' protection and distribution, to establishing functioning family coalitions. The main aim of the current study was to map sets of parental actions from both parental and their offspring's perspectives. In particular, we focused on parental actions toward the offspring and their partners. Further, we mapped whether and to what extent parental actions differ regarding the target person. We conducted semi-structured interviews exploring parental actions applied within current and previous offspring's relationships. The sample consisted of 20 offspring (mean age = 31.4y; SD = 5.6) and their 20 parents (mean age = 58.05y; SD = 7.82). We identified 32 parental actions classified as disruptive, ambivalent, and supportive; however, the most frequently reported were ambivalent and supportive ones by both parents and offspring. Interestingly, thematic analysis revealed that parents often interfered toward both offspring and their partners at once, i.e., the couple.

Parental actions differed regarding the target person: toward the offspring were the most numbered and most disruptive, whereas toward the couple were the least numbered and least disruptive. Our findings show that real-life interactions might not give rise to extreme actions contrary to prior findings based on hypothetical scenarios. Moreover, the necessity to attain extremely disruptive behavioural patterns might slowly decrease with the offspring's age as parental influence weakens with offspring's independence.

Antioxidative potential and masculinity – testing oxidative handicap hypothesis in men

Żelazniewicz, A., Nowak-Kornicka, J., Studzińska, I., Pawłowski, B.

Objective(s): Testosterone-dependent morphological traits are perceived as cues of a man's biological condition. However, so far the evidence for the relationship between morphological masculinity and health is scarce and the results are not conclusive. According to the oxidative handicap hypothesis (OHH) the reliability of masculine traits as cues of a man's biological condition is assured by the cost of pro-oxidative properties of testosterone. Oxidative stress (OS) is a state of imbalance between reactive oxygen species and antioxidants, that negatively affects health, fertility, and longevity. An individual's ability to resist oxidative imbalance (antioxidant potential) is supposed to be a component of biological condition that may be conveyed by masculine traits. Thus, this study aimed to verify if masculinity is related to a man's antioxidant potential.

Methods: The study sample included 166 healthy, non-smoking men aged 28.9-44.3 years ($M=35.1$, $SD=3.44$). Serum levels of antioxidants (catalase, superoxide dismutase, vitamin C), total antioxidant capacity, and OS markers (8-isoprostaglandin, RNA/DNA, PCs) were determined. Testosterone, cortisol, C-reactive protein levels, age, and body adiposity were controlled in the analyses. Body masculinity was assessed based on shoulder-to-hip ratio, 2nd to 4th digit length, and muscle mass. Facial masculinity was assessed in the photos, based on sexually dimorphic traits.

Results: There was no relationship between masculinity and antioxidant potential or oxidative stress markers, also when controlled for age, testosterone and cortisol levels, inflammation or body adiposity.

Conclusion(s): The results indicate that in the studied western population masculinity is not a cue of a man's antioxidant capacity.

Pathways to online infidelity: the roles of perceived online dating success, perceived availability of alternative partners, and mate value discrepancy

Nascimento, B.S., Adair, L., Vione, K.C.

Dating apps have become increasingly popular in recent years. The use of dating apps may pose several consequences for close relationships. Indeed, previous research has found that users of dating apps tend to have a higher short-term relationship orientation than non-users (Barrada et al., 2021). Similarly, relationships initiated online may be more vulnerable to infidelity and relationship termination (Alexopoulos et al., 2020). One potential explanation is that dating apps may give users the perception that there is an unlimited pool of potential partners available. A second explanation is that the use of dating apps may boost individuals' self-perceived desirability among those who perceive themselves to be more successful at online dating. As a result, these may motivate individuals to keep on searching for better alternatives and engage in infidelity. This study aims to explore both hypotheses to best understand the consequences of dating apps for close relationships. Specifically, this study aimed to examine the association between perceived online dating success and online infidelity-related behaviours. We explored two possible indirect paths through perceived number of alternative partners or mate value relative to one's partner and attention to alternatives. A total of 338 individuals ($M_{age} = 28.99$, $SD_{age} = 8.31$) that were in an exclusive long-term relationship participated in this study. A serial mediation analysis with two parallel paths revealed that perceived online dating success contributes to higher perceived availability of alternative partners and higher mate value relative to one's partner, both of which are associated with attention to alternatives that, in turn, increases engagement in online infidelity-related behaviours. No direct association between perceived online dating success and online infidelity-related behaviours was found. Our findings offer important insights on the implications of online dating services for relationships.

Does flawed perception of consent facilitate a sexually coercive mating strategy? Relationships between the Dark Triad of personality and perceptions of sexual consent

Lyons, M., Blinkhorn, V.

Objective(s): Sexual coercion could have evolved as an alternative mating strategy alongside intra and inter-sexual selection. Sexually coercive strategies can be facilitated by personality traits, such as (in the case of humans), the Dark Triad of personality (i.e., primary and secondary psychopathy, Machiavellianism, and narcissism). Research has established that these traits relate to sexually coercive behaviours, intentions, and cognitions. However, less is known about perceptions of consent, which could be a mechanism facilitating coercion. The objective of this pilot study is to investigate associations between the Dark Triad and perceptions of sexual consent when alcohol is involved in the situation.

Methods: In an online study ($N = 228$), participants filled in questionnaires assessing the Dark Triad, and alcohol and perceptions of consent. In addition, they watched a video where rape of an intoxicated woman was implied, and answered questions on perceptions of consent.

Results: Cross-correlational analyses indicated that primary and secondary psychopathy and Machiavellianism in men, and narcissism and primary psychopathy in women were significantly, positively correlated with the perception that alcohol does not influence the ability to consent to sex. Machiavellianism and primary psychopathy in women, and narcissism, Machiavellianism, and primary psychopathy in men related to the perception that the woman in the video was not raped and that she did give her consent to sex. Out of all the predictor variables, in both sexes, primary psychopathy was the most significant predictor of flawed consent.

Conclusion(s): Flawed perceptions of sexual consent could promote coercive mating strategies. Future research should investigate whether perceptions of consent are malleable in individuals with callous and uncaring personality dispositions.

Digital Lovers and Jealousy: Anticipated Emotional Responses to Sexual Technologies, and Implications for Theories of Jealousy

Prochazka, A., Brooks, R.C.

Objective(s): We consider how people anticipate reacting to technologies of the near future that are expected to stimulate human social and sexual impulses. Further, we use these anticipated reactions to test evolutionary and non-evolutionary theories regarding the origins of sexual and romantic jealousy.

Methods: We conducted two concurrent experiments in which participants anticipate their reactions to (1) 'virtual friend' chatbots of varying emotional sophistication, and (2) 'digital lover' sex toys, sex robots, or virtual reality entities of varying physical sophistication. Participants considered descriptive vignettes, and then answered whether, if a partner were to use the described technology, they would anticipate feeling jealous or angry, and whether they would prefer to see the technology banned.

Results: Both the jealous-angry response to chatbots and the inclination to ban them increased with greater emotional sophistication, especially for female participants. By contrast, both sexes anticipated greater jealous-angry responses and inclination to ban digital lovers of higher levels of physical sophistication, but female participants expressed higher levels of both responses. There was no interaction, suggesting the physical sophistication itself did not elicit a sex difference in response.

Conclusion(s): Our results show limited consistency with evolutionary theories concerning sex differences in jealousy. They also indicate that most participants do not anticipate strong jealous or angry responses, and, at least at this stage, few participants favour banning any of the technologies.

Sex ratios

2.45-4.00pm; Lecture Theatre 1

Theoretical and empirical evidence that polygyny does not necessarily exclude significant numbers of men from the marriage market

Gaddy, H., Sear, R., Fortunato, L.

Objective(s): Polygyny has been the subject of much interest in the evolutionary behavioural sciences. Often this interest focuses around the consequences of polygyny, in particular, the assumption that polygyny will necessarily result in significant numbers of men being permanently excluded from the marriage market: “For every man who takes two wives, another man must remain single” is a statement often included in work on polygyny. Secondary consequences are typically then assumed to follow from this male ‘marriagelessness’, notably that polygynous societies have high levels of violence, caused by unmarried men. These discussions, however, neglect demography. The assumption that every polygynous man must exclude at least one other man from the marriage market only holds true if there are equal numbers of marriageable men and women. Yet sex ratios in human populations are often not equal; they are influenced by, for example, mortality rates which are typically higher for men than women. Differential mortality rates then interact with factors such as population growth and differences in marriageable ages for men and women (typically higher for men than women) to produce unbalanced sex ratios.

Methods: Here we build theoretical models which determine likely sex ratios for marriageable men and women, under a range of realistic demographic scenarios. We then use census data from multiple populations to empirically examine whether polygyny is associated with a high proportion of unmarried men.

Results: both theoretical and empirical evidence suggest that polygyny does not necessarily produce an excess of unmarried men, because sex ratios of marriageable men and women are often female-biased. On the contrary, the empirical data show that polygyny is associated with higher proportions of men being married.

Conclusion(s): Incorporating demography into the analysis of polygyny requires a re-examination of the widespread belief that polygyny leads to high levels of male marriagelessness.

Decades of Trivers-Willard research on humans: what conclusions can be drawn?

Thouzeau, V., Bollée, J., Cristia, A., Chevallier, C.

Objective(s): The Trivers-Willard hypothesis predicts that parents in good condition are positively biased towards sons, while parents in poor condition are positively biased towards daughters. Moreover, theoretical work suggests that, the conditions under which the Trivers-Willard hypothesis is verified should be more restrictive in the case of post-birth investment than for sex ratio. An extensive literature testing this hypothesis has accumulated in the last five decades. The purpose of our study is to systematically review this literature to assess its validity in the human species.

Methods: We take stock of results concerning humans in a systematic review, which yielded 87 articles, reporting a total of 821 hypothesis tests. We performed a p-hacking analysis, as well as an effect size calculation.

Results: The p-curve analysis did not reveal a pattern of p-values consistent with p-hacking. Effects are consistent with the Trivers-Willard hypothesis overall. We then went on to check whether there was a difference between sex ratio and post-birth investment, and obtained mixed results.

Conclusion(s): There is robust support for the predictions of the Trivers-Willard hypothesis in terms of sex ratio, but weaker statistical support for post-birth investment. We put forward recommendations for future studies that aim to further assess the validity of the Trivers-Willard hypothesis or mechanisms subtending it, and we discuss the implications of different ways of measuring parental status and investment.

Sex inequality driven by dispersal

Chen, Y., Ge, E., Zhou, L., Du, J., Mace, R.

Inequality between the sexes is pervasive both outside and inside the home. There could be a sex-specific difference in relatedness to the group, as well as asymmetry in intersexual power, caused by the dispersal of one sex at marriage under various kinship systems. Our approach here takes advantage of both the ecological diversity and the social structures found in southwest China. Our study examines how kinship dynamics and social organization affect inequality in the sexual division of labour. We applied multilevel comparative approaches to examine the relative workload pattern between the sexes under different dispersal states and under differential operational sex ratios generated by religious practice. We show that 1) time spent on agricultural work and animal husbandry activities correlated highly with wearable activity tracker readings. Our results reveal that 2) being female and dispersing at marriage lead to an unfavorable division of workload. We also discovered that 3) under female-biased sex ratio contexts, females bear significantly heavier workloads than males, whereas, in villages of male-biased sex ratios, males' workloads significantly increase. This is consistent with the hypothesis that males have increased bargaining power when remaining in their natal home or under female-biased sex ratio contexts, leading to inequality in workload. This comparative framework provides us with new insight into the study of the evolution of labour division under different ecological contexts.

The consequence of adult sex ratio for marriage strategies across human societies

Halimubieke, N., Arnot, M., Chen, Y., Székely, T., Mace, R.

The relative frequencies of males and females in a population (termed adult sex ratio, ASR) are expected to influence traits associated with sexual selection and mating strategies. Recent comparative and experimental studies in various animal species suggest that ASR is associated with mating strategies, mate fidelity and parental investment. A male-skewed ASR facilitates males to follow long-term stable mating strategies whereas a female-skewed ASR often promotes males to behave promiscuously. The evidence from human societies, however, is contentious. Here we use a global dataset that represents 143 countries and 28 years to examine the associations between the ASR and marriage strategies and family structures across human societies. We find evidence that the divorce rate is associated with ASR, with males more likely to engage in a stable marriage when the ASR is male-skewed. We also show that male-skewed ASR predicts a lower prevalence of single-motherhood and delayed reproduction in females. To our knowledge, this is the largest cross-national, longitudinal dataset addressing ASR and marriage patterns in human societies, it supports the role of ASR in breeding system evolution. Our work also points out that the associations are more complex than predicted by the mating market theory as socio-economic factors may also mediate marriage patterns and family structures.

Increased levels of intimate partner violence in response to a relatively high sex ratio

Arnot, M.

The local sex ratio has been shown to be a strong population-level determinant of behaviours relating to mating and reproduction, such as mate guarding. In humans and other animals, a high sex ratio has been shown to predict greater levels of mate-guarding from males. However, it is not clear in humans what mate-guarding strategy is employed, with some proposing that males use violence as a mate-guarding technique, and others suggesting that the male-biased sex ratio increases female autonomy, resulting in greater relationship satisfaction. To examine how males mate-guard in humans, I look at the rates of violence and levels of relationship satisfaction relative to the local adult and operational sex ratio using longitudinal data from a UK cohort study. After controlling for relevant factors, I found that a relatively high adult and operational sex ratio predicted an increased odds of experiencing violence, but not increased relationship satisfaction. This was the case even when the objective adult/operational sex ratio was female-biased, meaning that if a region was more male-biased than other regions (but not objectively male-biased), this relatively higher sex ratio predicted increased odds of violence. This highlights that ecological cues are not objective, but contingent upon what is being cued by surrounding areas. Additionally, it highlights that even in areas where the sex ratio is not extremely skewed, if a male perceives that there are excess men in his environment, then mate-guarding behaviour could be invoked, with this mate-guarding behaviour being more likely to be violence.

Food and foraging

2.45-4.00pm; Lecture Theatre 2

Feed your mind: The evolved cognitive processes shaping human food behaviors

Rionex, C., Wertz, A., Rumiati, R., Coricelli, C.

Objective(s): Food cognition is a surprisingly under-explored aspect of human cognition. Yet, eating requires complex psychological abilities to search, evaluate and prepare food in a safe manner, and has played a crucial role in human evolution. Here we present an original framework to study food cognition. Methods and Results: Current research has overlooked that our modern food environment strongly differs from the environment in which our ancestors lived. In most cultures today, many decisions about food are made during a trip to the supermarket, where we might ask ourselves: Are these canned tomatoes healthy? but certainly not: Are they edible or toxic? In contrast, our ancestors had to navigate their natural environment to Find and Evaluate potential foods, while Excluding the costs of consuming something harmful, and finally Decide which entities to eat. We call this the FEED problem. We argue that a focus on FEED provides a window into the interplay of evolutionary, cognitive and developmental factors driving our food behaviors. Using this original FEED framework, we first present findings that show how our brains evolved to rapidly (i) Find and (ii) Evaluate candidate foods, notably by extracting inherent and useful information via visual inspection. In these sections, we also articulate the role of higher-order cognitive processes, such as memory, in evaluating foods. The paper then covers the complementary role of behavioral avoidance strategies and social learning to (iii) Exclude the cost of consuming something harmful and (iv) Decide which food to eat. Results from developmental, adult, non-human primate and neuroimaging studies are discussed. Conclusion(s): Adopting the FEED framework to study food cognition sheds a new light on the cognitive and brain mechanisms that shape our food behaviors and open promising research avenues for psychological sciences, as well as practical applications to take up the current health challenges.

Individual heterogeneity in foraging returns among Inuit harvesters

Hillemann, F., Ready, E.

Involvement with traditional subsistence activities and sharing of country foods is variable within Indigenous Arctic communities today. We previously analysed socio-economic correlates of Inuit foraging patch decisions (i.e., harvest method and target species) in the Canadian Arctic, and showed that age, gender, income, and the number of ties in food-sharing support networks affect the choice of harvest activities. Here, we extend our analyses to focus on another dimension of foraging: inter-individual variability in harvest returns (edible weight). We analyse about 250 foraging trips involving 23 Inuit harvesters from Kangiqsujuaq, Nunavik, using a Bayesian approach that simultaneously models patch choice and the harvest return, and specifically model heterogeneity among foragers in the expected harvest in different patches. We use our results on individual heterogeneity in harvest returns, which allows us to account for inter-individual differences in skill, to reconsider how socio-economic factors impact Inuit harvesting strategies. We expect these processes to interact with individual-level skill and experience. If people differ in their capacity to maintain access to harvest activities and food sharing potential depending on their income, such economic inequities may potentially exacerbate inequalities in subsistence participation, with consequences for peoples' food security and socio-cultural identity.

The Origin of the Family, Communal Property and the Community: Coevolution of Inequalities and inheritance norms in agropastoralist communities

Montalva, N., Flores-Abarado, S., Sánchez-Dinamarca, A., Varas-Enríquez, P. J., Guerra-Spencer, J., Reyes-Madrid, M., Vázquez-Estay, F., Wiesler, E.P.

Objective(s): To better understand the dynamics of wealth inheritance and inequality.

Methods: We studied wealth transmission in agropastoralist groups from Chile featuring collective land ownership and showing diverse customs of inheritance of land usage rights. Two types of phylogenetic trees were inferred for (i) surnames: using pairwise distance matrices of the distribution of surnames included on lists of the 171 registered communities, (ii) genetic markers: based on 15 autosomal STRs for a sample of 15 communities. Lists of registered commoners were used to measure four community-level traits: (i) gender-bias in inheritance, (ii) inequality in the distribution of commoner's rights using Gini's coefficients, (iii) concentration of rights in extended families measuring surnames diversity, (iv) and proportion of multigeniture based on number of registry records under the name of a whole succession rather than a single individual. These traits were treated as continuous variables to assess their ancestral states and patterns of correlated evolution along the branches of the previously described trees.

Results: Both types of trees show similar grouping patterns (entanglement = 0.16, Mantel test p-value = 0.001), which also correlate with geographic distances between communities (Mantel test genetic distances x geographic distances $p = 0.03$; surnames distances x geographic distances $p = > 0.001$). Communities in the same clades have similar Gini Coefficients, while inequality is higher between different clades. Differences in Gini coefficients between clades diminish towards the base of the trees. Ancestral estate reconstruction shows low surname diversity and high bias for male inheritors, thus is partially consistent with anecdotal and historical accounts of a very strong preference for male primogeniture in the past.

Conclusion(s): Inequalities within groups are diminished by non-partible multigeniture while unigeniture causes clustering of groups with homogeneous levels of inequality. The relevance of modes of inheritance over inequality calls for the study of the effect of other variations, such as the effect of testaments, taxation, pre-mortem inheritance and marital wealth transfer.

Socio-ecology and resource encounters shape collective human foraging dynamics in Finnish ice-fishers

Schakowski, A., Kortet, R., Niemelä, P.T., Deffner, D., Monk, C.T., Kurvers, R.H.J.M.

Objective(s): Foraging plays a central role in the ecology and evolution of all species and the diversity of human foraging styles is considered key to our evolution and ecological success. Successful foragers need to integrate resource return rates, social information, and knowledge about the foraging ecology to locate high-quality resource patches. While individual human foraging has been investigated both in laboratory experiments and observational studies, study systems investigating collective foragers in natural socio-ecological environments are largely lacking. Drawing on recent developments in bio-logging technology, we aim to close this gap by studying large groups of experienced ice-fishers competing for resources.

Methods: In February 2022, we organized 10 ice-fishing competitions on different lakes in Northern Karelia, Finland, with 39 to 51 participants per competition. All participants were equipped with smartwatches recording high resolution GPS position and heart rate, as well as head-worn cameras to identify behavioral states (i.e., drilling, angling, fish handling, relocating). Combining this data with measures of lake topology, we can use Bayesian computational models to disentangle the effects of resource return rates, social context, and ecology on foraging behavior and social dynamics.

Results: Preliminary results indicate that ice-fishers adapt their search to resource return rates (i.e., area-restricted search if return rates are high and long-range displacements if return rates are low) and gradually update their beliefs about environmental return accounting for spatial distance between angling spots. Moreover, ice-fishers use both social and ecological features to navigate potential resource locations, but mainly rely on social density when choosing angling spots. This results in spatially dispersed but locally dense clusters of ice-fishers with relatively low encounter rates of ice-fishers between clusters.

Conclusion(s): This novel study system offers unprecedented data on the socio-ecological drivers of human collective foraging and helps shed light on the cognitive and social processes facilitating flexible human adaptation.

Understanding science

2.45-4.00pm; Jeremy Bentham Room

Variance Functions for Multilevel Models

Koster, J.

Among evolutionary researchers, the increasing use of multilevel modeling approaches has been a compelling alternative for the complex data structures that typify the field. These approaches include models with random slopes, which allow the effect of a covariate to vary across higher-level units. Random slopes are used in many cases for the sake of obtaining conservative estimates of the effect of the covariate on the outcome. There is also useful information to be gained, however, from considering the ways in which higher-level variance in the outcome variable increases or decreases as a function of the covariate. In this talk, I demonstrate the calculation of variance functions and their potential applications in the evolutionary sciences.

The Politics of Female Sexuality: Use and Misuse of Evolutionary Hypotheses in Online Antifeminist Communities

Barband, L.

Objective(s): While early evolutionary accounts of female sexuality insisted on coyness and monogamy, primatological observations started challenging those assumptions in the 1970s. Nowadays, evidence suggests that female mammals enact varied and flexible reproductive strategies, dependent on life history or ecological conditions. This is important for evolutionary biology, bridging the gap between theory and reality, and for feminism, as it challenges stereotypes of female “nature”. However, evolutionary hypotheses on female extra-pair and short-term mating are also routinely invoked among the antifeminist communities collectively known as “the manosphere”. This paper analyzes the phenomenon, before suggesting ways to mitigate it.

Methods: This paper is based on the largest qualitative analysis of a manosphere discourse corpus to date: 9000 pages of mostly online material, spanning three decades and evenly divided among five communities (Men’s Rights’ Activists, pickup-artists, MGTOW, The Red Pill, and incels). Part of it was selected for its importance to antifeminist communities (70%), part for referring to evolutionary science (15%), and part was randomly sampled (15%).

Results: Evolutionary hypotheses are ubiquitous in the manosphere, where state-of-the-art research is often ideologically mobilized - usually through a moralistic and emotional frame, rooted in sexual double standards or personal grief. Antifeminist narratives also obscure the fact that female mating strategies are hypothetical, unconscious, and supposed to have evolved aggregately over time. Changing wording and emphases in scientific literature could palliate this.

Conclusion(s): Serious evolutionary research is routinely appropriated for misogynistic aims. Through analysis and suggestions, this study seeks to increase awareness and mitigate the phenomenon.

Misinformation about evolution and nonverbal behaviour: How body language experts undermine global scientific efforts

Zlotanu, M.

Objective(s): Nonverbal behaviour is the subject of thousands of scientific publications. However, on traditional and social media, “body language experts” gain unparalleled visibility, who propose that body language can be read like words on a page. These so-called experts promote many erroneous beliefs about nonverbal behaviour to the public, including to people in position of power (e.g., police officers, judges, security agents). We highlight the dangers of this misinformation, and the impact of misinformation about nonverbal behaviour on the reach of scientific research, focusing on nonverbal behaviour and body language analysis.

Methods: We present a critical analysis of the discourses of various “body language experts” who receive thousands of views on YouTube every day, including the most viewed “body language” video on YouTube, with about 45 million views since 2019, or nearly 36,000 views per day. Based on the literature on misinformation, we will also explain how, in practice, these experts undermine global scientific efforts.

Results: The analysis shows that “body language” experts disseminate unfounded and discredited claims about human behaviour and promote misinterpretations of legitimate research. Moreover, the literature on misinformation suggests that their discourses will continue to be accepted by the public, even after researchers have made their findings public.

Conclusion(s): For researchers concerned that the reach of their work extends beyond the walls of academia, given the misinformation about nonverbal behaviour, human behaviour researchers should take an active part in the fight against so-called experts.

The biological versus social dichotomy: Why giving up is the hardest thing to do

Nettle, D.

Objective(s): The unhelpful dichotomy of ‘social’ versus ‘biological’ causes (and related dichotomies like learned/innate) persists, and hinders the development of an integrative human science. I argue this duality comes from intuitive cognition, not from reality. Specifically, I hypothesize that the duality arises from people having two modes of intuitive cognition: intuitive psychology, which tracks individual variability and change and subjective utility; and intuitive biology, which classifies organisms into fixed types. Whether people think of something as ‘social’ or ‘biological’ depends on which system is most active. For many behavioural traits, the two systems compete. I investigate this hypothesis in UK non-academic adults.

Methods: Vignette experiments describing alien life forms using very minimal amounts of information.

Results: Participants readily dichotomise the traits of the life forms into those that are fixed and species-typical, which they describe as ‘biological’ and ‘in the genes’, and those that are malleable and idiosyncratic. These latter are not seen as being biological, and are the ones associated with subjective value. It is possible to cue which traits get assigned to which category. The two categories of traits are seen as mutually exclusive.

Conclusion(s): The social/biological dichotomy arises readily from everyday psychology. It is spontaneously applied to new information, and leads to unhelpful inferences and distinctions. As a field we need to understand and try to combat unhelpful folk categories that people bring to bear on the explanations we develop.

Cooperative breeding

4.45-6.15pm; Lecture Theatre 1

The influence of maternal religion on alloparenting, child health and development in the Gambia

Shaver, J., Spake, L., Sear, R., Shenk, M., Sosis, R.

Objective(s): The higher relative fertility of religious individuals in contemporary settings represents a puzzle: studies find a negative relationship between child number and developmental outcomes for each child; however, there is currently little evidence that children born to religious parents fare worse, despite higher fertility. The current work was undertaken to evaluate the hypothesis that religious behaviour, although often costly, returns benefits in the form of alloparental support among extended kin networks and unrelated co-religionists. Increased alloparental support to religious mothers would therefore contribute to enabling larger relative family sizes.

Methods: In partnership with the Medical Research Council Gambia Unit, we administered structured surveys to 705 mothers across 23 villages in rural Gambia. Surveys focused on demographics, maternal religion, social networks, and on maternal and child health. We also collected anthropometric data from mothers and focal children. Bayesian multi-level models evaluated the impact of maternal religious behaviour on alloparental support, as well as the effect of alloparental support on child health and development.

Results: In the high fertility setting of the Gambia, maternal religion is positively related to fertility. Furthermore, maternal religiosity is positively associated with temporal alloparental investments directed towards her children, but unrelated to resource-based alloparental investments. These different forms of alloparental support are found to be differentially associated with child outcomes: temporal alloparental investments are associated with child health, while resource based alloparental investments are associated with child growth.

Conclusion(s): Maternal religion is associated with higher fertility and greater alloparental support. However, alloparental support is varied, and maternal religion appears to be related to some forms of alloparental support, but not others. These findings are placed within the broader cross-cultural context of maternal support networks and child outcomes.

Exploring the role of children as allomaternal caregivers among the BaYaka hunter-gatherers in Congo

Jang, H., Boyette, A.H.

Objective(s): Humans rely on allomaternal help to raise offspring. Many studies have focused on the roles of grandmothers, fathers and kin women as caregivers. The importance of older children as caregivers has also been acknowledged. However, there is still a gap in our knowledge of how caregivers, including children, vary in different settings, and how the roles of caregivers from different age classes differ. Here, we aimed to investigate caregivers in camp versus in out-of-camp settings, and how they reduce mothers’ childcare burden.

Methods: We conducted 700 daily short interviews with 22 BaYaka nursing mothers in the Congo, asking who provided childcare in the village and in the forest when they were out for foraging. We also conducted focal follows of 140 foraging trips, and collected data on the presence of infants, group composition and their food returns. Using Bayesian

multilevel models, we examined the effects of caregivers in foraging groups on nursing mothers' food returns.

Results: We found that when mothers decided not to take infants on foraging trips, they left infants mostly with grandmothers. However, when the mothers took infants on foraging trips, it was children in early childhood who provided childcare while mothers were collecting foods. Supporting this, we found that the company of girls in early childhood increased mothers' food returns, especially when nursing children were present. This effect was stronger than that of other women or older girls. Our results suggest that young girls in foraging groups provide childcare, while adults and older girls engage in food acquisition.

Conclusion(s): Our findings highlight a critical role for children as caregivers from early childhood. This provides an insight into how the flexible pooling of childcare labour, by involving a wide range of people from different age classes, may have been an important key to the demographic success in humans.

Communal breeding by women is associated with lower investment from husbands

He, Q.Q., Rui, J.W., Zhang, L., Tao, Y., Wu, J.J., Mace, R., Ji, T.

Objective(s): According to Hamilton's rule, matrilineal-biased investment restrains men in matrilineal societies from maximising their inclusive fitness (the 'matrilineal puzzle'). A recent hypothesis argues that when women breed communally and share household resources, a man should help his sisters' household, rather than his wife's household, as investment to the later but not the former would be diluted by other unrelated members (Wu et al., 2013). According to this hypothesis, a man is less likely to help on his wife's farm when there are more women reproducing in the wife's household, because on average he would be less related to his wife's household.

Methods: We used a farm-work observational dataset, that we collected in the matrilineal Mosuo in southwest China, to test this hypothesis.

Results: As predicted, high levels of communal breeding by women in his wife's households do predict less effort spent by men on their wife's farm, and communal breeding in men's natal households do not affect whether men help on their natal farms.

Conclusion(s): Thus, communal breeding by women dilutes the inclusive fitness benefits men receive from investment to their wife and children, and may drive the evolution of matrilineal-biased investment by men. These results can help solve the 'matrilineal puzzle'.

Grandparental co-residence and grandchild survival: the role of resource competition in a pre-industrial population

Chapman, S., Danielsbacka, M., Tanskanen, A.O., Lahdenperä, M., Pettay, J., Lummaa, V.

Objective(s): Apart from being beneficial, grandparents can also compete with grandchildren over limited resources. Resource competition may exist especially if grandparents live in the same household with grandchildren and its effects can be dependent on grandchild age. By utilising demographic data collected from historic population registers in Finland between 1761 and 1895 (study sample $n = 4041$) we investigate whether grandparents living in the same household with grandchildren are detrimental or beneficial for grandchild survival.

Methods: To analyse the survival effects of co-residence on the grandchild, we used event history analysis with a discrete time-event framework, implemented as binomial generalised mixed-effects models with a logit link function.

Results: Having a living but not co-residing grandmother or grandfather were both associated with better survival whereas having a co-resident grandfather was associated with lower chance to survive for infants (age 0-1). Separating the effect between maternal and paternal grandparents and grandmothers and grandfathers revealed no differences in the effects between lineages. Negative effect of having a co-residing grandfather was not significant when grandfathers were separated for lineage specific models.

Conclusion(s): These results implicate that accounting for the co-residence status and child age, grandparents were mostly beneficial when not co-residing with very young children and that having a co-residing grandfather at that age could be associated with lower chances to survive. Predictions made by grandmother hypothesis and resource competition both received support.

Allomothering networks in cross-cultural perspective: who helps mothers and what do they help with?

Hassan, A., Spake, L., Schaffnit, S., Alam, N., Badije, J., Cerami, C., Chvaja, R., Crampin, A., Dube, A., Jagne, A.Y., Kaye, M., Kotch, R., Munthali, S., Mwalwanda, L., Prentice, A., Rai, R.K., Zohora, F.T., Shenk, M.K., Soisi, R., Shaver, J., Sear, R.

Objective(s): Considerable research supports the hypothesis that women rely on allomothers – cooperation from other individuals – to raise children. Yet much of this research has focused on specific allomothers, particularly grandparents and older siblings. Here we expand the evidence base for allomothering by presenting data on women's allomothering networks in full, detailing how many individuals women receive support from, who those individuals are and what types of support they provide. We further compare data from four countries to demonstrate whether and how allomothering networks differ between contexts.

Methods: We collected data from 3653 mothers using similar survey materials in: Matlab thana, Chandpur District, Bangladesh; West Kiang region, the Gambia; Birbhum District, West Bengal, India; Karonga District, Malawi. Our survey asked women to list all individuals who provided the following types of support: childcare, financial support, food support, help carrying out their work, support when sick or injured, emotional support. We conducted descriptive analyses to detail mothers' support networks in each context.

Results: Women receive considerable support from multiple individuals, but there is variation between contexts in their allomothering networks. The average number of cooperative partners for all types of help is 2.9 (India), 8 (Malawi), 10.3 (Gambia), and 11.1 (Bangladesh), while childcare support is received from between 1.5 (India) and 4.9 individuals (Bangladesh). Greatest variation was seen in the number of individuals providing emotional support, which was much higher in Bangladesh and the Gambia than India. Support comes from a wide range of individuals, including spouses, children and parents, but women's siblings and unrelated individuals (friends or neighbours) are also frequent cooperation partners.

Conclusion(s): We advance understanding of allomothering by demonstrating the width (how many supporters and who they are) and breadth (what they do) of women's cooperative networks, and show support is frequently received from beyond nuclear family members.

Inclusive fitness and the evolution of affinal kinship

Dyble, M.

Humans are good at recognising genetic kin and, in many contexts, preferentially cooperate with them. However, human kinship includes not only genetic kin but also kin by marriage: our affines (in-laws) and spouses. Can cooperation between these genetically unrelated kin be reconciled with inclusive fitness theory? Here, I argue that although affinal kin and spouses do not necessarily share genetic ancestry, they may have shared genetic interests in future reproduction and, as such, can derive indirect fitness benefits through cooperating. Using standard inclusive fitness theory, we can derive a coefficient of shared reproductive interest (s) that predicts altruistic investment both in genetic kin and in spouses and affines. I explore the conditions under which kin nepotism according to shared reproductive interest would be selected in humans.

Mental health

4.45-6.15pm; Lecture Theatre 2

Testing the 'gender equality paradox': cross-national gender equality indicators and gender gaps in depression

Xie, J., Brown, G.R.

Objective(s): Previous studies have reported that gender differences in some psychological traits (e.g., personality) are counter-intuitively largest in countries with higher gender equality, which is known as the 'gender equality paradox'. Evolutionary psychologists have suggested that, in gender equal societies, men and women can more fully express their evolved psychological tendencies. However, empirical support for the gender equality paradox is highly variable. This study investigated the links between gender equality, gender norms and mental wellbeing.

Methods: Cross-national datasets were used to examine the association between gender gaps in depression and gender equality indices. Data on the prevalence of depression were extracted from the international Global Burden of Disease (GBD) dataset. Gender equality measures included those based on i) economics: Gender Inequality Index (GII) and Gender Development Index (GDI), and ii) social norms: Gender Social Norm Index (GSNI). Multi-level linear regression analyses tested the relationships between gender gaps in depression and these gender equality indexes using data from 68 countries, with random intercepts for countries and random slopes for gender.

Results: While the economic indices showed no significant association with gender gaps in depression (GII: $p = .477$; GDI: $p = .272$), the social norm index (GSNI) was positively correlated with the gender gap in depression ($p < .001$) (i.e., gender gaps in depression were largest in countries with the strongest gender norms). Our results indicate that gender equality indicators based on social norms are more relevant than those based on economics for understanding gender differences in depression.

Conclusion(s): In contrast to some previous studies, our findings are not consistent with the 'gender equality paradox'. Determining gender equality in economic terms only is likely to provide an incomplete picture, and the extent of gender equality needs to be considered in terms of gender norms at the individual level.

Social Media Resignation The Only Way To Protect User Data And Restore Cognitive Balance: A Literature Review.

Motilal, R.

The birth of the Internet and the rise of social media marked an important chapter in the history of humankind. Often termed the fourth scientific revolution, the Internet has changed human lives and cognisance. The birth of Web 2.0, followed by the launch of social media and social networking sites, added another milestone to these technological advancements where connectivity and influx of information became dominant. With billions of individuals using the internet and social media sites in the 21st century, "users" became "consumers", and orthodox marketing reshaped itself to digital marketing. Furthermore, organisations started using sophisticated algorithms to predict consumer purchase behaviour and manipulate it to sustain themselves in such a competitive environment. The rampant storage and analysis of individual data became the new normal, raising many questions about data privacy. The excessive usage of the Internet among individuals brought in other problems of them becoming addicted to it, scavenging for societal approval and instant gratification, subsequently leading to a collective dualism, isolation, and finally, depression. This study aims to determine the relationship between social media usage in the modern age and the rise of psychological and cognitive imbalances in human minds. The literature review is positioned timely as an addition to the existing work at a time when the world is constantly debating on whether social media resignation is the only way to protect user data and restore the decaying cognitive balance.

An evolutionary-developmental account of sex differences and heterogeneity in Borderline Personality Disorder

Baptista, A., Farkas, B., Jacquet, P.O.

Objective(s): This study aims to investigate whether and how the degree to which individuals trade off somatic maintenance against short-term reproductive goals relates to the severity of Borderline Personality disorder (BPD) at both its general and single symptom levels, and whether these associations differ between men and women.

Methods: We fit multi-group Structural Equation Models to data drawn from wave 2 of the National Epidemiological Survey on Alcohol and Related Conditions (NESARC), a large, observational and representative US cohort [final N = 31 015 (17 928 female)]. The reproduction-maintenance trade-off and BPD severity were modelled as two continuous latent variables. Inferences about differential item functioning (DIF) were made by Chi square difference tests comparing the goodness of fit of unrestricted vs restricted models. Unlike the restricted models, the unrestricted models allowed for the reproduction-maintenance trade-off to modulate symptom expression probabilities, and for these associations to differ between the men and women subsamples.

Results: Results showed that a reproduction-oriented life history strategy – i.e. represented by the negative correlation between the pursuit of goals that provide rapid reproductive benefits and physical health functioning – linked positively with overall BPD severity. We additionally found evidence for sex-independent DIF for 5 out of the 9 DSM-IV BPD symptoms: individuals with a reproduction-oriented life strategy were more likely to express Impulsivity, Suicidal/Self-mutilation behaviour, Affective instability, and less likely to endorse Chronic feelings of emptiness and Stress-related paranoid ideation. Finally, we found some evidence for sex-dependency in Suicidal/Self-mutilation behaviour and Affective instability, such that women with a reproduction-oriented life history strategy were more likely to endorse these symptoms than men. These effects however did not survive correction for multiple comparisons.

Conclusion(s): An evolutionary-developmental framework, rooted in human life history and sexual selection theory can help explain the heterogeneous manifestation of BPD.

Early life unpredictability and the development of psychopathology

Farkas, B.C., Wjart, V., Jacquet, P.O.

Objective(s): Previous work has shown that various dimensions of early life adverse experiences, such as deprivation, threat and unpredictability are associated with both physical and mental health outcomes throughout the life course. However, the concepts and operationalisations of unpredictability vary greatly across studies. In this series of works, we tested whether distinguishing between shorter and longer timescales of unpredictability can allow for more precision in characterizing the developmental sequelae of adversity.

Methods: We utilized a Structural Equation Modelling framework, along with a dimensional model of early life adversity factors to investigate the associations between adversity dimensions and multiple important developmental outcomes, including internalising and externalising symptoms, attachment styles, linguistic-cognitive ability and reproduction-maintenance trade-offs. We made use of multiple sources and types of data, including large, longitudinal cohort datasets, nationally representative of the US population [Early Childhood Longitudinal Study, Kindergarten Class of 2010-11 (ECLS-K:2011) and the Fragile Families and Child Wellbeing Study (FFCWS)], as well as online data, collected by us.

Results: Findings generally demonstrated that dimensional models were better at capturing developmental outcomes, than simple composite models of adversity, as different dimensions of adversity were associated with different outcomes. Moreover, specifying the timescale of unpredictability also proved important in capturing interindividual differences. Deprivation was specifically associated with cognitive-linguistic ability, Short timescale unpredictability with insecure attachment and internalising symptoms, Long timescale unpredictability with 'Fast' life histories and externalising symptoms, and Threat had more broader effects. Moreover, specifying the timescale of unpredictability also proved important in capturing interindividual differences.

Conclusion(s): A dimensional model of adversity, along with a more fine-grained modelling of unpredictability allow for a more complete characterisation of the adaptive and maladaptive developmental effects of adverse childhood experiences.

Are all couples the same? Exploring the variation in dyadic depressive symptoms

Csajbók, Z., Štérbová, Z., Jonason P.K., Harlíček, J.

Objective(s): Researchers predominantly aggregate their data and study couples testing overall associations. Cluster analysis is sometimes performed, but the number of groups to identify is subjective. Mixture modelling, however, is a structural equation modelling method that can reliably guide this decision in finding meaningful classes of couples. In this research, couples' longitudinal trajectories were explored showing the diversity of depression in a large sample of Europeans. Methods. Altogether 11,136 mixed-sex couples (mean age = 60.76) were followed up for 12 years in the multicentric cohort SHARE study from all around Europe. Both men and women's depressive symptoms, and physical and mental health were measured at each time point. Couples' dyadic longitudinal trajectories were classified into meaningful latent classes with the probabilistic Dyadic Growth Mixture Modelling method.

Results: Couples were grouped into four distinct classes. In the largest Class 1, both were non-depressed (77%), couples in Class 2 both had increasing depressive symptoms (8%), in Class 3 both experienced decreasing depressive symptoms (7%), and in Class 4 only women had depression while men were non-depressed (8%). Participants experiencing any depressive symptoms had more psychological and physical health problems than the non-depressed participants. The highest rates of bereavement and relationship dissolution were observed in couples with increasing depressive symptoms. No difference was found between the reproductive success of these four classes of couples.

Conclusion(s): This large sample of couples was not homogeneous in depression trajectories, and meaningful longitudinal patterns could be observed which related to other relationship, psychological, and health outcomes. These distinct depression patterns can extend evolutionary theories on homogamy and couples' divergence or convergence over time since these varied between different classes of couples. The findings indicate that exploring the heterogeneity of data can provide a significant benefit for theorizing and interpretation of data.

Specialised minds: extending adaptive explanations of personality to the evolution of psychopathology

Hunt, A., Jaeggi, A. V.

Traditional evolutionary theory invoked natural and sexual selection to explain species- and sex-typical traits. However, some heritable inter-individual variability in behaviour and psychology – personality – is probably adaptive. Here we extend this insight to common psychopathological traits. Reviewing key findings from three background areas of importance – theoretical models, non-human personality and evolved human social dynamics – we propose that a combination of social niche specialisation, negative frequency-dependency, balancing selection and adaptive developmental plasticity should explain adaptation for individual differences in psychology – ‘specialised minds’ – explaining some variance in personality and psychopathology trait dimensions, which share various characteristics. We suggest that anthropological research of behavioural differences should be extended past broad demographic factors (age and sex) to include individual specialisations. As a first step towards grounding psychopathology in ancestral social structure, we propose a minimum plausible prevalence, given likely ancestral group sizes, for negatively frequency-dependent phenotypes to be maintained as specialised tails of adaptive distributions – below the calculated prevalence, specialisation is highly unlikely. For instance, chronic highly debilitating forms of autism or schizophrenia are too rare for such explanations, whereas attention-deficit-hyperactivity disorder and broad autism phenotypes are common enough to have existed in most hunter-gatherer bands, making adaptive explanations more plausible.

Evolution and dysfunction of human cognitive and social traits: A transcriptional regulation perspective

Zug, R., Uller, T.

Objective(s): Evolutionary changes in brain and craniofacial development have endowed humans with unique cognitive and social skills, but also predisposed us to debilitating disorders in which these traits are disrupted. What are the developmental genetic underpinnings that connect the adaptive evolution of our cognition and sociality with the persistence of mental disorders with severe negative fitness effects?

Methods: Based on theoretical considerations and an extensive review of the literature, we argue that loss of function of genes involved in transcriptional regulation represents a crucial link between the evolution and dysfunction of human cognitive and social traits. The argument is based on the haploinsufficiency of many transcriptional regulator genes, which makes them particularly sensitive to loss-of-function mutations.

Results: We present evidence that several human brain and craniofacial traits evolved through partial loss of function (i.e. reduced expression) of transcriptional regulator genes, and we discuss how these findings are compatible with the idea of human self-domestication. Moreover, we explain why selection against loss-of-function variants supports the view that mutation-selection-drift, rather than balancing selection, underlies the persistence of psychiatric disorders. Finally, we discuss testable predictions.

Conclusion(s): Loss of function of transcriptional regulator genes links evolution and dysfunction of human cognitive and social traits.

Social bonding boosts health behaviours and psychological wellbeing

Newson, M., Tunngene, B., van Mulukom, V.

Objective(s): At times of turmoil – such as during disasters, social upheavals, or pandemics – our social bonds can be key to receiving support and gaining certainty about the right course of action. We sought to understand how social bonds with close social circles (i.e., family, friends) and extended groups (i.e., country, government, humanity) relate to engagement in health behaviours and psychological wellbeing.

Methods: We distributed longitudinal surveys in over 120 countries during the COVID-19 pandemic, creating two global datasets (N= 13,264) pertaining to social contact, trust in science, social bonding, health behaviours, and wellbeing.

Results: People distanced most when they thought their close social circle did. Despite its legal and health threats, face-to-face contact was still positively associated with wellbeing, and use of messaging apps had a negative association. Trust in science had a small, indirect effect on adherence to the rules. Bayesian regression analyses controlling for stringency of local rules showed that following the guidelines was positively associated with wellbeing even for people in high-risk groups. Only bonding to one's family was associated with engagement in health behaviours, whereas being strongly bonded with both close circles and extended groups predicted less anxiety and depression and better wellbeing, particularly for those who were bonded with a greater number of groups.

Conclusion(s): Close and extended social bonds offer different sources of support and direction during challenging circumstances. To achieve behavioural change during crises, policymakers must emphasize shared values and harness the social influence of close friends and family.

Friday, April 21st

Health

10.45-1.15pm; Lecture Theatre 1

The relative contribution of proximity with kin and property ownership on women's health among the matrilineal Khasi (India)

Lamarque, L., Langstieb, B., Raymond, M., Alvergne, A.

Objective(s): Matriliney has been shown to benefit women's bargaining power, autonomy and health outcomes compared to patriliney. However, the mechanisms by which matrilineal kinship benefits women's health are unclear. Here we evaluate the relative contribution of proximity with kin (matrilineal women often live close to their kin) and material inheritance (property is often transmitted to females) on the health of matrilineal Khasi women living in Meghalaya (India). Although most Khasi stick to matrilineal traditions, inheritance patterns vary due to recent political attacks on female-biased inheritance, and matrilocal norms are fading away because of increasing urbanization.

Methods: This pre-registered study draws on data from the 2015 Demographic and Health Survey of Meghalaya (n = 7,000) covering both urban and rural areas. We evaluate the extent to which the gender of the household owner and the household composition (nuclear, co-living with kin or co-living with in-laws) associate with a woman's level of anaemia (severe, moderate, mild, not anaemic), a proxy for environmental and somatic stressors. Associations are estimated using mixed effects multinomial logistic models and adjusted for confounding but not mediating effects according to hypothesized directed acyclic graphs.

Results: We predict that (1) anaemia is lower among property-owning women as compared with other women, (2) anaemia is lower for women co-residing with kin, (3) the relative contribution of proximity with kin and property ownership changes along a rural-urban gradient. The analysis might reveal the cooperative and/or conflictual nature of relations with kin, who simultaneously provide help and increase competition for household resources.

Conclusion(s): The findings will help clarify the mechanisms underlying the positive association between matriliney and women's health by disentangling co-living with kin from inheritance. The results will also shed light on understudied variation in women's health in a matrilineal society.

Exploring tactile disgust in relation to text-based methods, tactile qualities, and self-reported health status

Dlouhá, D., Kaňková, Š.

Objective(s): The aim was to experimentally evoke and measure tactile disgust and compare it with disgust measured by text-based methods. Additionally, we wanted to observe possible effects of other self-reported variables regarding infection and health on the disgust sensitivity obtained through these different methods.

Methods: In a cross-sectional study, 113 participants (aged 19-50) completed the Disgust Scale-Revised (DS-R) and the Pathogen domain of the Three Domains of Disgust Scale (TDDS) along with the Perceived Vulnerability to Disease (PVD) questionnaire and questions about their health status. The participants were then given 6 disgust-evoking tactile stimuli hidden in boxes to evoke and measure disgust. The participants rated their feelings of disgust and tactile qualities regarding every object.

Results: The sum of rating how disgusted the participant was by each object and how disgusted they would be if they put each object in their mouth, significantly positively correlated

with all subscales of DS-R, the TDDS Pathogen domain and the PVD Germ Aversion subscale. People who considered themselves to be more sick reported significantly higher disgust at the thought of putting the objects in their mouth and Contamination disgust of DS-R. The object with the highest average disgust rating was rated the most “slimy” and “wet”.

Conclusion(s): The results showed that evoking tactile disgust is a reliable method of measuring disgust sensitivity. Moreover, the slimy and wet qualities, associated with pathogenic environments, are considered the most disgusting. The caution regarding interpersonal and oral transmission adaptively increases when the individual needs higher protection against pathogens.

Causal models of human growth: New tools to explore linked dynamics of culture and health

Bunce, J.A., Fernández, C.I., Revilla-Minaya, C.

Objective(s): Our objective was to develop new models of human growth that, compared to existing models, provide deeper insight into the mechanisms driving inter-individual and inter-population variation in children's growth trajectories.

Methods: Building on general theories linking growth to metabolic rates, we develop causal parametric models of height and weight growth, incorporating a novel representation of human body allometry and a phase-partitioned representation of ontogeny. These models permit separation of metabolic causes of growth variation, potentially influenced by diet and disease, from allometric factors, potentially under strong genetic control. We estimate model parameters using a Bayesian multilevel statistical design applied to temporally-dense height and weight measurements of U.S. children, and temporally-sparse measurements of Indigenous Matsigenka children from Amazonian Peru.

Results: This analysis suggests that observed differences in growth patterns between these two populations have both metabolic and allometric causes particular to infant, child, and adolescent growth phases, many of which we can tentatively interpret in light of ecological and cultural differences.

Conclusion(s): These theoretical growth models constitute an initial step toward a better understanding of the causes of growth variation in our species, while potentially guiding the development of appropriate, and desired, healthcare interventions in societies confronting growth-related health challenges. Furthermore, they constitute new tools to more rigorously explore the linked dynamics of culture and health.

Does the need to signal illness legitimacy influence patients' decision to undergo medical treatment?

De Barra, M.

While the primary goals of medical treatment are typically to shorten illness or relieve symptoms, they may also serve to communicate a need for care. Drawing on costly signalling theory, I suggest that treatments – and in particular aversive treatments – may help patients to legitimise their illness and thereby enable access to crucial support during convalescence. Our previous work demonstrated that participants are more inclined to provide care to people who undergo treatment, especially when that treatment is painful. Here I explore whether patients “choose” to undergo treatment in order to legitimise their illness and access additional care. Such a pattern would suggest that treatments act as signals rather than cues of need. A pre-registered cross-sectional study of 194 chronic pain patients shows that those who report that others doubt the legitimacy of their illness are more likely to accept aversive treatments. Furthermore, two pre-registered experiments ($n = 653$) indicate that subtle manipulations to patients' sense of social support may increase their willingness to accept treatment. These results suggest that patients' treatment decision-making is informed by the social consequences of their choices. Signalling theory may help explain the surprising longevity of some ineffective medical procedures.

Do people apply a precautionary heuristic when stating intentions to get vaccinated?

Matthijssen, M.A.M., van Leenwen, F., Cloin, J.C.L.M., van de Goor, L.A.M., Achterberg, P.

Objective(s): How do people make decisions about taking precautions such as getting vaccinated and what influences these decisions? We investigated whether people use a precautionary heuristic when stating intentions to get vaccinated. The precautionary heuristic assumes that the decision to take precautionary action is based on judgments that one is exposed to a hazard and that the action is an appropriate precaution. We addressed two limitations of previous research. First, we explicitly modeled the “if-then” structure proposed by the heuristic by including interactions between perceived hazard and perceived efficacy. Second, we tested how the heuristic relates to established predictors of vaccination intentions (i.e., individual differences in worldviews, such as religious orthodoxy, spiritual beliefs).

Methods: We tested moderation and mediation models on a Dutch representative sample collected in 2021 ($N = 2003$) for five diseases (i.e., COVID-19, Flu, HPV, Measles, Yellow fever).

Results: The analysis showed that participants applied a precautionary heuristic for all diseases. Especially, including the multiplicative interaction that reflects the “if-then” structure of the heuristic increased the explained variance compared to standard additive threat and coping models. Furthermore, the heuristic partially mediated the effects of individual differences in worldviews and ideology.

Conclusion(s): The current findings support the hypothesis that people make decisions about precautionary behaviors by applying a decision rule that integrates estimates of hazard and efficacy. Interventions to increase vaccination uptake might focus on how individual differences in worldviews influence the precautionary heuristic.

The Cost of Reproductive Effort for Female Ageing in a Contemporary Human Population

Alvergne, A., Michel, P., Argenti, A.

Objective(s): Ageing, the accumulation of molecular and cellular damage over time, is a major driver of chronic diseases, the leading causes of death and disability worldwide. However, the determinants of variation in how fast individuals age are poorly understood. Life-history theory posits that organisms face a trade-off between reproduction and the maintenance of the soma, but direct tests in contemporary populations are scarce. Here we aim to test the hypothesis that variation in the speed of ageing is due to differential investment in reproductive effort over the life-course.

Methods: We analyzed longitudinal data from women living in a post-transitional society (the UK, $n = 117\ 105$). We first used Cox-regression proportional hazard models to estimate the association between reproductive effort (number of offspring, number of sex partners) and the risk of mortality. Second, we used multi-level regression models to evaluate associations between reproductive effort and proxies of biological ageing: levels of C-Reactive Protein and leucocyte telomere length. We accounted for phenotypic correlations between reproduction and health.

Results: We found that not having offspring is associated with an increased risk of mortality, which is not due to reverse causation, but possibly to a lack of social capital brought about by children. Among those who have children, there is a positive association between the number of offspring and the risk of mortality, but only among the poorest section of the population. We also found that physiological markers of immuno-senescence (e.g., level of C-Reactive Protein, length of telomeres) increase with the number of offspring.

Conclusion(s): In this population, the relation between reproduction on mortality follows a U shape and is mediated by both socio-economic and relational resources. The cost of reproduction falls mainly on the poorest women.

Who gets contraceptive side-effects? A reproductive ecology approach to understanding variation in hormonal contraceptive side-effects among Ethiopian women

Stevens, R., Gurmu, E., Negash, A., Kebede, L., Alvergne, A.

Objective(s): A key barrier to reducing unmet need for contraception worldwide is the experience of contraceptive side-effects, yet the drivers of side-effect variation remain poorly understood. We aimed to test the reproductive ecology hypothesis that women with lifestyles associated with having lower endogenous reproductive hormone levels experience more side-effects.

Methods: We recruited two groups of new contraceptive users, 152 injectable users and 92 implant users, and a control group of 109 non-users across urban and rural locations in Ethiopia. Using a pre-tested locally specific side-effects measurement tool based on qualitative work, we measured participants' symptoms at baseline and across three months.

Sociodemographic data relevant to reproductive hormone levels, including economic status, diet, activity levels, and infection histories, were also collected. We investigated whether

factors associated with lower reproductive hormone levels in the reproductive ecology literature increased the number and severity of side-effect symptoms experienced by contraceptive users. Directed acyclic graphs were used to determine adjustment covariates to minimise confounding and obtain accurate effect estimates.

Results: Users of hormonal contraceptives experienced a greater number and severity of side-effect symptoms than non-users, adjusting for symptoms at baseline. Several factors associated with lower reproductive hormone levels increased the number of symptoms experienced among contraceptive users, but not among non-users. For instance, after three months of hormonal contraceptive use, the number of side-effect symptoms was greater among those who had poorer quality diets, greater food insecurity, or a recent infection prior to use.

Conclusion(s): Although limited by a lack of direct reproductive hormone measures, this research provides a novel understanding of side-effect symptom prevalence and aetiology in different socioecological settings. Our findings challenge current public health discourse that assumes side-effects are both equally likely for all women and mostly minor concerns or misconceptions that women can be educated out of.

Childhood physical activity in the BaYaka hunter-gatherers.

Kretschmer, L.D.W., Bann, D., Chandbary, N., Dyble, M., Salali, G.D.

Recent research in adult hunter-gatherers has revealed physical activity levels far in excess of those in post-industrial populations. While these studies have been used as evolutionary models of physical activity, an understanding of physical activity in childhood is missing. To address this gap, we collected accelerometry data from 23 BaYaka children (ages 5 to 17) over a period of 7 days. From this we examined how active BaYaka children are, how this activity is structured between individuals, and how it is structured within and between days. To understand how BaYaka children differ in their activity patterns from post-industrial populations, we compared activity profiles in BaYaka children with children in the Millennium Cohort (UK) and the National Health And Nutrition Examination Survey (NHANES). We observed BaYaka children to undertake three times the WHO recommendations of moderate to vigorous activity, with amounts of movement increasing with age but no difference between boys and girls. Volumes of activity varied across the day with individuals waking at dawn, peaking in activity in the late morning, remained high throughout the afternoon before declining after sunset. Compared to post-industrial populations, BaYaka children engaged in an additional hour of moderate-to-vigorous activity a day, showed an increase with age compared to a decrease in the US, observed greater variation between days, and more fluctuation throughout the day, despite less variation in wake and sleep times.

Cultural evolution

10.45-1.15pm; Lecture Theatre 2

Prey-producing predators 2.0: Cumulative cultural evolution and our fragile future

Efferson, C., Richerson, P.J., Schief, M., Weinberger, V.

What effect might a capacity for cumulative cultural evolution have on a sustainable human future? To address this question, we derive and analyze a model with unusual features characterizing human activities over the long-run. First, human population dynamics draw heavily on consumer-resource modeling in ecology in that humans must consume biological resources to produce new humans. Second, the model also draws heavily from economic growth theory in that humans do not simply consume biological resources; they also produce the resources they consume. Humans are, in short, prey-producing predators. Finally, humans use two types of technology. Predation technology affects the rate at which humans extract resources for consumption. Production technology controls how effectively humans convert their own labor into new resources. The dynamics of both types of technology are subject to cumulative cultural evolutionary processes. A given technology can either progress or regress depending on investments in maintaining the technology. The resulting model exhibits a wide range of dynamical regimes. That said, the system is routinely sensitive to initial conditions, with wildly different outcomes given the same parameter values. This result undermines the notion that fundamentals strongly influence how a human economy performs. Moreover, the system exhibits a basic fragility in the sense that human activities often lead to the endogenous extinction of the human species. This can happen gently, or it can happen painfully with periods of explosive human activity involving super-exponential growth then collapse. This result suggests that cumulative culture readily supports a fragile future.

Cumulative risky technological evolution with major and minor updates

Toyokawa, W.

Models of cultural evolution have overlooked the possibility that new innovations might not always be reliable, sometimes causing serious dysfunctions until they are refined by minor updates such as bug fixations. The theory of technological CCE must account for such dynamics between major updates (v1, v2 etc.) and minor updates (v1.1, v1.2 etc.). To examine the nature of major and minor innovation, I introduced a “risk” of technology into a baseline agent-based model developed by Mesoudi (2011): now the resulting utility from a given set of technological traits was subject to noise, sometimes giving high rewards and sometimes low rewards. I assumed that higher-level technologies cannot be added to the repertoire when the baseline technology gives very low reward. Agents could either search for new major innovations that may successfully improve the technological level, or search for new minor innovations by following the “minor innovation rate”, a culturally evolvable individual trait, to reduce the risk-level of technological traits. I found that agents tended to invest all the budget into major innovation, gambling to become the most successful individual among the population while free-riding others’ minor innovations. As a result, technology collapsed eventually, and the equilibrium technological level achieved after thousands of generations remained very low. However, if the “minor innovation rate” was allowed to transmit vertically, while technological traits being transmitted through payoff-biased obliquely copying, the minor and major innovation started to co-operate and the CCE emerged.

The importance of Moort (family) is highlighted through embedding Australian Aboriginal perspectives into evolutionary medicine curricula

Coall, D.A., Robertson, F., Nannup, N., McAulay, D., Nannup, D., Hill, B.

Objective(s): As our world and learning environments change at ever increasing rates, the value of diverse perspectives around any issue has never been more important. This presentation details a collaborative research relationship with staff in Kurungkurl Katitjin, the Centre for Indigenous Australian Education and Research, that has used the concept of koodjal djinang (two-way seeing) that culminated in the embedding of Aboriginal perspectives in evolutionary medicine teaching.

Methods: Over the past six years a rewarding exchange of information, ideas and perspectives between Aboriginal and non-Aboriginal researchers using the koodjal djinang (two-way seeing) concept has investigated family systems and the environment. These relationships, and the generosity of the Kurungkurl Katitjin staff and Elders, identified the value of presenting both local Nyoongar and European perspectives together to transform teaching material.

Results: Specific examples of synergies where koodjal djinang has been used include a discussion of research integrity where an example of scientific racism is critiqued, and the counterpoint provided through a Nyoongar documentary Synergies demonstrating the independence and convergence between Nyoongar and non-Aboriginal knowledge systems. The vital role of family and grandparents in Nyoongar culture and the resilience it brings now and over the past 50,000 years is highlighted through a Moort (family) research program exploring the traditional and contemporary Nyoongar family system. This is applied to the evolution of human longevity.

Conclusion(s): The embedding of Nyoongar perspectives into evolutionary medicine curricula is broadening students’ perspectives on content. What better way to understand our changing world than from one of the oldest living cultures?

Aggregate observable cultural production captures economic development in pre-industrial societies

De Dampierre, C., Thouzeau, V., Baumard, N.

The study of the dynamics of pre-modern civilizations over time presents many difficulties. Historical materials are highly heterogeneous from one period to the next, from one area to the other, making it difficult, if not impossible, to create long time series. Also, the mass of specialized academic literature is such that it is impossible for an individual or even a group of individuals to have a systematic, consensual and comprehensive approach to a civilization, let alone to all civilizations. In this paper, we take advantage of the fact that large collaborative platforms have organized, homogenized and synthesized academic knowledge at the individual level, collating sources and academic works for more than 200,000 creative individuals of the past. Collective intelligence thus allows tapping into an underused material in history, namely cultural productions (e.g. poems, treatises, musical pieces). Because cultural works require time, energy, resources, and human capital on the part of both creators and their public, they indirectly reflect a wide range of economic and social processes. And because they can be abstracted from their initial material incarnation (e.g. manuscripts), their survival rate tends to be higher. We demonstrate that production captures historical changes in economic development (urbanization, population density, GDP per capita), as well as important geographical shifts in history (e.g., East to West in Antiquity, West to East in Japan, North to South in China). We then show that this method allows us to have a much more fine-grained representation of the dynamics of pre-industrial civilizations, suggesting that economic growth before the Industrial Revolution was not limited to a few Western societies. Finally, this method provides new insights about the causes of economic development, the origins of the little and great divergences (i.e. between Southern and Northern Europe, China and Japan, the West and the Rest), and the role of culture and religion in the dynamics of civilizations.

Abortion attitudes across cultures: Exploring the role of gender inequality, abortion policy, and individual values

Adair, L., Ferenczi, N., Lozano, N.

Objective(s): Social norms act to promote cohesion, coordinated action and the achievement of complex, shared goals (Roos et al., 2015). Norms differ across cultural contexts, and are shaped by social (e.g., sanctions; Strimling et al., 2018), individual (e.g., internalisation), and environmental (e.g., pathogen prevalence; Gelfand et al., 2011) forces. From an evolutionary perspective, it is unclear why or how norms and judgements about abortion vary across cultures. When abortion judgements across cultures are explored (Adamczyk, 2022; Loll & Hall, 2019), single-item measures are typically used to assess abortion attitudes. To address this gap, survey methods were used to assess abortion attitudes in the US (N = 215), UK (N = 206), Mexico (N = 215), and India (N = 215).

Methods: Participants (Mage = 41; 45% women) reported their abortion attitudes (vignette approach; Hans & Kimberly, 2014), gender role attitudes (Garcia-Cueto et al., 2015), endorsement of the motherhood mandate (adapted from Park & Hill, 2014), sexual strategy (Penke, 2011), belief in moralising gods (adapted from Laurin et al., 2012), and guilt/shame proneness (Cohen et al., 2011).

Results: Individuals living in countries with greater gender inequality ($r(851) = -.103, p = .003$; 95% CI [-.165, -.040]) and more restrictive abortion policy ($t(849) = -4.01, p < .001$; 95% CI [-1.97, -.68]), endorsed more restrictive abortion attitudes. Individuals who endorsed more traditional gender role ideologies ($b = .11, p = .003$), reported belief in moralising gods ($b = .17, p < .001$), and reported more long-term (compared to short-term) sexual strategies ($b = .30, p < .001$), reported more restrictive abortion attitudes (R² adjusted = .16, $F(4, 845) = 41.50, p < .001$). Factors that predict abortion attitudes differ across sampled cultures.

Conclusion(s): We conclude that both contextual factors, as well as individual factors, shape individuals' attitudes towards abortion. Implications regarding the bidirectional relationship between attitudes and policy in reproductive health are discussed.

Historical myths as culturally evolved technologies for coalitional recruitment

Sjilmassi, A., Safran, L., Baumard, N.

Humans are unique in their ability to cooperate at the scale of clans, tribes, ethnic groups or entire nations. Most theories propose that large-scale cooperation should be driven by characteristics of the present—like shared norms and efficient sanctioning institutions—or in the future—like economic prospects of protection against risk. Yet, across cultures, one of the most remarkable manifestations of social cohesion in large-scale entities is the belief in a shared, distinct and immemorial past. Human communities around the world celebrate their ancestral roots, commemorate their fallen heroes and recall ancient battles. Why do humans put so much effort into celebrating a long-gone past? Integrating insights from social cognitive science, evolutionary anthropology, cultural history and political economics, we show that the cultural success of historical myths is driven by a specific adaptive challenge for humans: the need to recruit coalitional support to engage in productive collective action and prevail in conflicts. We propose that historical myths are cultural technologies whose typical narrative structure—consisting of martyrs, heroes, golden ages, shared victories and sufferings—should be conceived as a set of hyper-stimuli intended to activate specific features of their targets' coalitional psychology—with the aim to recruit coalitional support. Because incentives for coalition formation vary across individuals, social groups and environments, our theory also makes precise predictions about the variability in the prevalence, distribution, and particular content of historical myths—thus accounting for the recurrence of “narrative wars”.

Cultural Domain Evolution Among UK Migrants and Nationals

Rosun, N., Badiani, F., Hanson, J., Ferenczi, N., Willard, A.K., Kline, M.A.

Objective(s): This study investigated ongoing cultural changes resulting from intercultural contact in the UK. We targeted the largest migrant, and majority national groups within the UK to identify 1) how they learn about their heritage culture and other cultures that they are exposed to; 2) domains of the heritage culture to maintain; and 3) domains of other cultures to adopt. The aim of this study is to build a better understanding of the cultural learning pathways underlying cultural evolutionary change, as well as how these cultures are changing.

Methods: We collected 109 participants from various ethnic groups (Black, Polish, South Asian, Black) from Prolific UK. Participants completed a questionnaire containing freelist items about cultural learning (heritage and other), maintenance, and adoption. Freelists were systematically coded, then analysed for salience scores using the AnthroTools package in R.

Results: For each group, family was the most salient source for heritage cultural learning, followed by media sources. Media was also the most salient source for other cultural learning. Heritage cultural domains to maintain were food, language, and values (e.g., hospitality). Adopted cultural domains varied more. The most salient category for the British and Black groups was ‘Nothing’. South Asian and Polish groups included categories such as romantic relationships, secularism, and tolerance.

Conclusion(s): We found that all cultural learning pathways (vertical, horizontal, and oblique) are all involved in both heritage and other cultural learning. Although media falls into oblique learning pathways, we should consider how media-based learning is similar to other processes of oblique learning since they employ different modes (person-person, vs object to person). We find that maintained cultural domains align with identity markers, however participants also want to change some of the values associated with these markers by adopting them from other cultures.

Alcohol and the rise of civilizations

Hrnčič, V., Chira, A.M., Gray, R.D.

Objective(s): It is argued that alcoholic beverages played a crucial role in the rise of large-scale societies, as intoxication enhances human creativity, alleviates stress, builds mutual trust, and increases cooperation. Although many lines of evidence support this theory, it has not yet been empirically tested. Our aim is to uncover the causal relationship between alcohol consumption and social complexity using a global sample of cultures and phylogenetic methods.

Methods: We collect ethnographic data on the presence of native alcoholic beverages and different aspects of cultural complexity for 186 non-industrial societies around the world (Standard Cross-Cultural Sample, or SCCS). We use eHRAF World Cultures and D-PLACE databases as data sources. We consider several alternative metrics of cultural complexity, including cultural group size, political complexity, resource-use intensification, and technological and social differentiation. We first investigate the relationship between the presence of alcoholic beverages and our metrics of cultural complexity in Bayesian logistic models that account for several environmental and cultural cofounders, as well as for the effects of horizontal and vertical transmission. To reveal the evolutionary dynamics between alcohol and cultural complexity, we use a global phylogeny for SCCS societies and a recently developed Bayesian model of coevolution.

Results and Conclusion(s): Preliminary results show that alcohol and cultural complexity are positively linked, and both traits have a relatively strong phylogenetic signal. Subsequent co-evolutionary analyses will (1) uncover whether alcohol and cultural complexity underwent a co-evolutionary relationship, and (2) allow us to compare the strength of alcohol's influence on complexity and vice-versa.

Can phylogenetic methods accurately capture spatially explicit demographic histories?

Kriznar, M.

Objective(s): Phylogenetic methods are used for the reconstruction of historical relationships between linguistic lineages. The methods, designed for biological evolution, assume vertical transfer of variants. Little work has been done to determine how the introduction of horizontal transmission, potentially ubiquitous in cultural evolution, affects the accuracy of these methods. In this study we tested the robustness of phylogenetic methods when applied to a demographic process with horizontal transmission of traits between groups. Methods: We generated synthetic population histories represented as phylogenetic trees. We then simulated variant data on the trees, sampled the terminal nodes and used standard phylogenetic reconstruction methods to construct best fitting trees from the leaf data. We assessed the robustness of the phylogenetic reconstruction as the quartet distance between the best fitting tree and the 'true' tree representation of the demographic history.

Results: Phylogenetic methods are reliable only when the borrowing is constrained to groups with a recent common ancestor. In addition, borrowing can be constrained by either shared ancestry or spatial proximity, leading to significantly divergent inferred trees. Trees generated under a spatial borrowing model are comparable to trees where borrowing was allowed between branches with a common ancestor within 60% of the total height of the tree.

Conclusion(s): Results suggest that the introduction of even limited horizontal transmission based on spatial proximity, rather than relatedness, in a spatially explicit simulation with extinction, produced phylogenies that are potentially indistinguishable from random trees. The rate of horizontal transmission plays a comparatively minor role in the divergence between the original and the reconstructed phylogenetic topologies.

Mysteries without pedagogy. Evolved mental mechanisms explain the puzzle of apparently irrational cargo cult imitation

Umbres, R.

This paper attempts to solve a century-old puzzle: why did various Melanesian groups engage in independent and spontaneous but apparently irrational mimicry of Western behaviours and artefacts? The answer comes from the intersection of evolved mental mechanisms for representing means-goal relationships with a unique historical context of cultural contact between people separated by a vast asymmetry in technology and social organisation. Melanesians aimed to replicate the salient but unexplainable success of White people in receiving "cargo" (food, tools, clothes, etc). Evolved dispositions for teleological reasoning (Gergely and Csibra 2006) and epistemic vigilance (Sperber et al 2020) drove the apparently irrational - but psychologically and contextually reasonable - copying of ineffective practices like military parades, radio communication, or keeping flowers in vases. These ostensive but apparently goalless behaviours were inferentially interpreted as mysterious but relevant means to obtain economic and social prosperity. A qualitative review of a large archive of books and articles written by lay observers and anthropologists confirms that cargo cultists selectively imitated cognitively-opaque behaviours but not intelligible ones. A comparative analysis of the adoption/adaptation of Christian rituals and theology further elaborates on the causal role played by folk epistemologies of secret knowledge and the reciprocal mistrust between natives and colonisers. Further inspection of the ethnographic material suggests that competition between "big men" in acephalous societies may have also encouraged the spread and intensity of mimicking cults during World War cornucopia. Engaging with the growing conversation around cultural imitation (Whitehouse 2021, Jagiello et al 2022), this study offers support for three main claims: Ostensive-referential communication (or natural pedagogy - Csibra and Gergely 2009) is crucial for efficient cultural replication, learning rituals and technology uses similar mental inclinations, and researchers should pay more attention to the natural context of cultural transmission.

The phylogenetic ancestry of witchcraft beliefs

Wu, B.H., Peacey, S., Grollemund, R., Mace, R.

Objective(s): Witchcraft belief, the idea that individuals harm others using supernatural means, is found on every continent. The witchcraft phenotype has traits that are recurrent and vary cross-culturally, including the association of sorcery with cannibalism, the sex most likely to be accused, and the concept of familiars (magical helpers). We examined the vertical transmission of such beliefs in the population dispersal of Bantoid and Bantu societies across sub-Saharan Africa which began several thousand years ago.

Methods: We coded 8 traits using materials sourced from the Ethnographic Atlas and eHRAF, databases with historic ethnographic material from small-scale societies. We matched societies to a phylogeny of Bantoid-Bantu languages (N = 79). We tested phylogenetic signal and reconstructed ancestral states using BayesTraits.

Results: The ideas that witchcraft is a physical substance and witches' have familiars showed significant phylogenetic patterns and were inferred to have been present at the tree root. Others, such types of familiar, were distinct in certain clades, while the sex of 'witches' did not show phylogenetic signal.

Conclusion(s): Results indicate that traits associated with witchcraft beliefs vary in their vertical transmission: some are locally distinct while others are present throughout the tree and appear to be highly ancestral. They evolve at different rates: the lack of phylogenetic signal for the sex most accused of witchcraft may indicate this can alter rapidly in response to changing socio-ecology. Diverse patterns in these traits may elucidate the evolution of witchcraft beliefs over several thousand years.

Exploring the social clustering of female genital mutilation/cutting (FGMC) attitudes among Arsi Oromo agropastoralists in Ethiopia

Myers, S., Gurmu, E., Alvergne, A., Gibson, M.

Objective(s): Over 25 million (or two thirds) of Ethiopian women are thought to have undergone FGMC, the second highest national number in Africa. While establishing why and how FGMC persists is a priority for global policy-makers, how FGMC norms are socially transmitted and maintained remains unclear. Here we combine social network data with cultural evolutionary theory to explore the social clustering of FGMC preference among Arsi Oromo agropastoralists in Ethiopia, who practice cutting shortly before marriage.

Methods: FGMC preference, demographics, and socio-centric social network data assessing 4 relationship types (marriage preparation advice, chatting, financial support, respect) were collected from 2549 adults (50.5% women) in 2021. Using combined social relations and stochastic block modelling we will test whether FGMC preference predicts social ties reflecting advice transmission or chatting. With additional demographic and network data we will further explore whether such ties are prestige-, resource-, kin-, or homophily-biased.

Results: Six and a half percent self-reported being pro-FGMC. Exploratory analysis suggests shared FGMC preference clusters in both advice and chatting networks: advice – 29.1% of pro-FGMC egos named at least one pro-FGMC alter, while 14.8% of those anti-FGMC did; chatting – 33.3% of pro-FGMC egos named at least one pro-FGMC alter, while 13.9% of those anti-FGMC did. Advice and chatting networks show moderate overlap with each other and financial support networks. Advice networks are heavily kin-biased and chatting networks highly homophilic for gender. Neither network overlapped strongly with respect networks.

Conclusion(s): Self-reported pro-FGMC preference was rare. Preliminary findings indicate attitudes cluster both among those sharing advice regarding the marital preparation of daughters and socializing, suggesting social reinforcement may help maintain pro-FGMC attitudes at low levels. We will discuss findings from the next steps of our analysis, which includes exploring the predictors of FGMC preference homophily, and how our results can inform more effective anti-FGMC interventions.

Cognition

10.45-1.15pm; Jeremy Bentham Room

Early life adversity jointly regulates Body-Mass Index and Working memory development

Jacquet, P.O., Furkas, B., Bouton, S.

Objective(s): We aimed to characterize whether and how individuals trade off cognitive development against somatic growth by studying the interrelationship between individuals trajectories of Working memory (WM) capacities and Body-Mass Index (BMI), so as their associations with multiple dimensions of early life adversity.

Methods: We fit a series of multi-group Latent Growth Curve models to data drawn from the Early Childhood Longitudinal Study, Kindergarten Class of 2010-11 (ECLS-K:2011), a large, longitudinal and representative US cohort [final N = 15 257 (7 448 female)]. Longitudinal data at 1-year intervals, collected between ages 6 and 11 (6 timepoints in total), were utilized. BMI was calculated from direct measures of weight and height; WM was assessed using standardized scores on the Numbers Reversed test. Early life adversity dimensions of Deprivation, Threat and Unpredictability were modelled as sums of z-scores from a variety of available indicators. Growth curves were modelled separately for boys and girls, and involved an intercept, a linear slope, and a quadratic slope parameters.

Results: WM growth was characterized by a concave function, whereas BMI growth was characterized by a convex function. The WM and BMI intercepts were negatively related in both sexes. Furthermore, in boys the BMI linear slope was negatively predicted by the WM intercept, whereas in girls the BMI quadratic slope was negatively predicted by both the WM intercept and the WM linear slope. However, the inclusion of early life adversity factors somewhat attenuated these relationships. Adversity factors were generally related negatively to WM, and positively to BMI, with Deprivation showing the strongest effects.

Conclusion(s): Somatic and cognitive growth showed opposing trends in the studied period, with some evidence supporting a direct negative relationship between the two. Dimensions of early life adversity – especially Deprivation – were related to the development of both, with generally opposite signs.

Modelling Emergent Communication with Deep Learning Networks

Konwenhoven, T., Kleijn de, R., Raaijmakers, S., Verhoef, T.

Objective(s): Although natural language processing recently improved dramatically, large language models still lack an understanding of how language relates to the real world [1], known as the grounding problem. One solution may be to let grounded vocabularies emerge from interactions between humans and machines [2,3]. We studied human behaviours in an emergent communication paradigm by integrating cognitive, evolutionary, and computational approaches.

Methods: We use the Embodied Communication Game (ECG) [4] to investigate the emergence of shared signal—meaning mappings. Here, participant pairs solve a shared task without access to conventional means of communication, enforcing the emergence of a new communication system. Typically, this happens by negotiating a set of sequential signals that acquire meaning through interactions. Individual differences in Personal Need for Structure (PNS) influence how this process develops [5]. We trained deep learning models on captured data to mimic human behaviours and used hyperparameter optimization to approximate latent human cognitive variables (learning rate (lr) and exploration (τ)) to explain human behaviour.

Results: Out of 23 pairs, 14 were able to play the ECG successfully. Bidirectional LSTM networks capture human behaviours better than unidirectional LSTM networks (BF10=6.63e+11, d=1.66). For both model types, there is a positive relation between lr and τ , and for biLSTM models, we find a positive relationship between lr and high score. In the current data, lr or τ cannot explain differences in PNS.

Conclusion(s): The models can capture behaviours associated with the emergence of new communicative systems and suggest that participants who adopt new behaviours faster are more successful in creating these systems. Our results indicate that human sequence processing in the ECG is influenced by expected future states [6] and that random and uncertainty-directed exploration strategies are combined to develop optimal behaviour [7-8]. Future work should investigate the emergence of signals between humans and machines.

The effect of emotional recognition and theory of mind on relationship quality

Lindová, J., Tomková, V., Příkrýlová, K.

It is generally assumed that the adaptive function of human theory of mind and emotional recognition is effective and complex cooperation. The human romantic pair bond - a complex cooperative unit with division of labour and resource sharing - could be a context in which these social-cognitive skills are specifically advantageous. We studied theory of mind, emotional recognition, conflict communication and relationship adjustment in 147 long-term heterosexual romantic couples. Both partners filled in the Dyadic Adjustment Scale to measure relationship quality and the Romantic Partner Conflict Scale to assess communication styles. Emotional recognition was measured using the Geneva Emotion Recognition Test and the Reading the Mind in the Eyes Test, and the Faux Pas test was used to measure theory of mind. Statistical analyses were performed in APIM. Our results showed that men's, but not women's emotional recognition had a positive effect on their own (beta = .29, $p < .001$) and their female partners' (beta = .25, $p = .009$) compromising communication in relationship. Theory of mind did not affect relationship communication. In addition, compromising communication significantly predicted dyadic adjustment in women (beta = .34, $p < .001$). Taken together, our results indicated that couple's compromising communication mediates the relationship between men's emotional recognition and women's dyadic adjustment. We conclude that human monogamous bonds might represent an important social context where the human ability of reading emotions from interpersonal signals and cues is of adaptive value.

Cognitive foundations of ideological orthodoxy and free speech repression

Marie, A., Petersen, M.B.

Political and religious movements often bind around shared ideological narratives denouncing an evil or villains encroaching on a sacred value, such as national grandeur, the faith, or political, racial, or gender equality. In their most devoted activists, this triggers moral motivations to affirm and protect the narrative from being argumentatively challenged (i.e., orthodox mindsets), with free expression and nuance as the primary casualties. Repressive reactions range from expressions of outrage or public shaming on social media to the “deplatforming” of controversial speakers to censorship and imprisonment of dissidents. Orthodox mindsets are puzzling because of the often disproportionate righteousness with which they try to protect cherished narratives. We suspect that orthodox mindsets may derive from three main evolved cognitive foundations. First, over-sensitive dispositions to detect outgroup threat and exploitation. Second, motivations to highlight universally relevant threats to mobilize ingroup members for cooperative benefits and against rival groups. Third, to accrue prestige within the ingroup by signaling personal devotion to the causes one's allies value. In line with arguments about self-deception, those behaviors are often displayed by activists sincerely committed to the ideological movement's tenets.

The origin and structure of the intentional level of analysis: Humans' intuitive psychology as the output of difference detecting mechanisms.

Pietraszewski, D.

Our ability to predict and understand human behaviour intentionally—in terms of beliefs, desires, and goals—is generally acknowledged to be underwritten by cognitive adaptations. However, the full entailment of this acknowledgement has yet to be appreciated. In principle, from an evolutionary perspective, the content of our intuitive psychology—mentalistic concepts such as a “self”, “attention”, “memory”, “intelligence” and so on—reflects the output of difference detecting mechanisms (Lukaszewski et al., 2020): Each mentalistic concept picks out a set of conditions that varied within the lifespan of each individual organism, and yet were also enduring enough over multi-generational phylogenetic time to build a conceptual entry within the design of the mind. Crucially, such mentalistic constructs are based around those features of the world that varied and that were benefit to keep track of. They do not, even in principle, describe how any of mental processes actually work. This point has far-reaching and deflationary consequences for the behavioral and psychological sciences, as it suggests the conceptual bedrock of these sciences—constructs such as “attention”, “memory”, “intelligence”, etc.—do not carve the mind up by the joints, nor do they even pick out processes, per se. How to integrate such concepts into a mature evolutionarily-informed cognitive science will be discussed.

The belief type continuum: An evolutionary, dual-inheritance explanation of the acquisition and transmission of conspiracy beliefs in light of paranormal and religious beliefs

Van Mulukom, V.

Objective(s): The objective is to present a comprehensive framework that explains the acquisition and transmission of beliefs using evolutionary frameworks, and the differences between paranormal, conspiratorial, and religious beliefs. The aim herein is to explain the evolutionary puzzle of conspiracy beliefs: are they like paranormal beliefs – phantoms of the mind – or like religious beliefs – important group identity markers?

Methods: The research is based on a literature review of the cognitive and evolutionary science of religious and paranormal beliefs, combined with the social psychology of conspiracy beliefs.

Results: Using dual-inheritance theories, I argue that the acquisition and transmission of beliefs can be explained through a combination of cognitive and cultural factors. While the human cognitive apparatus makes beliefs possible, it is the cultural context that gives rise to specific beliefs. What is more, certain beliefs are closer related to cognitive biases (e.g., paranormal beliefs), whereas others are more closely related to cultural context (e.g., religious beliefs). The background and implications of this proposal will be discussed. I argue furthermore that the puzzle of conspiracy beliefs can be explained by the fact that they are in between paranormal and religious beliefs on this spectrum: They are strongly influenced by individual cognitive biases and cultural contexts at the same time. This also explains why conspiracy beliefs increase in challenging times: It is then that the cultural contexts change radically (e.g., increased uncertainty and threat), and when the cognitive resources to rely on are low (e.g., information and trust) as a result of certain cognitive predispositions, epistemological and existential needs are fulfilled through the endorsement of conspiracy theories.

Conclusion(s): By assessing their reliance on cognitive and culture factors during acquisition and transmission, beliefs can be meaningfully distinguished and understood. This in turn can contribute to predictions of where and when such beliefs occur.

The psychological and cultural consequences of two centuries of global economic growth

Boon-Falleur, M., Baumard, N.

Living standards have improved dramatically over the past 200 years : life expectancy has risen from 29 years to about 70 years, infant mortality before 5 years old has fallen from 49% to 4%, and per capita energy consumption went from 20 gigajoules to 80 gigajoules per year. Today, most of the world's population can fulfill basic needs (housing, sanitation), has access to modern medicine (e.g. vaccines, antibiotics) and to primary and secondary education. This increase in living standards has triggered important phenotypic changes in people's physiology through adaptive plasticity. Humans in modern ecologies are taller, have lower body temperatures, lower testosterone levels, earlier menarche, more brittle cranial bones, and a greater investment in neural development. Similarly, their psychological traits have adapted to a more resource abundant world. There is evidence that increased resources lead to changes in Big Five personality traits, with humans in modern ecologies being more open, more agreeable, more extroverted and conscientious, and less neurotic. Increased resources also lead to less time discounting and a greater sense of agency. Such changes in psychological traits and preferences at the individual level have an impact on culture. We review causal evidence that changes in people's preferences lead to more extensive cooperation, greater tolerance, more exploratory preferences such as scientific exploration or fantasy worlds, and a greater emphasis on personal development. Our results contrast with the prevailing view that cultural differences between countries cause individual level differences in preferences.

Cognitive Ingredients in Fictional Storytelling: A Comprehensive Framework

Dubourg, E., Thouzeau, V., Benchoit, T., Boon-Falleur, M., Fiorio, G., Fitouchi, L., Mercier, M., Mercier, H., Sijlmasi, A., Zhong, Y., Baumard, N.

Objective(s): Narrative fictions are highly successful in human cultures, notably because they trigger psychological mechanisms, for instance for finding mates (e.g., in romance fictions), exploring the world (e.g., in adventure and speculative fictions), or avoiding predators (e.g., in horror fictions). In this presentation, we aim at putting forward a general framework to study the associations between specialized psychological mechanisms (e.g., for mating, exploring) and specific features in fictions (e.g., love at first sight, imaginary worlds).

Methods: We review the relevant literature in psychological and evolutionary sciences, and (1) list more than seventy adaptive challenges, (2) identify the adaptive psychological mechanisms that evolved in response to such challenges, (3) specify the sources of their adaptive variability (personality traits, biological sex, age, and ecological conditions), and (4) link them to fictional features that resemble and sometimes exaggerate real-life cues (that such mechanisms are designed to detect and respond to). During this iterative process, we took feedback from more than 30 evolutionary and cognitive scientists interested in fiction.

Results: Focusing on the adaptive variability of such specialized psychological preferences, this framework addresses both the question of the great diversity of narrative fictions and the question of their structure: It explains and generates a wide range of predictions about the non-random combinations of content features in fictions, according to the way the mechanisms they tap into adaptively vary.

Conclusion(s): This comprehensive framework lays the ground for a theory-driven research program for the study of narrative fictions, their content, their distribution, their structure, and their cultural evolution.

Self-bias for own artworks reveals art as extended self

Straffon, L., Agnew, G., Desch-Bailey, C., van Berlo, E., Gocłowska, M., Kret, M.

Objective(s): Self-relevant information is prioritized in cognitive processing. It is generally assumed that art is highly self-relevant and that artworks carry information about the personal attributes of their creators. In a recent study, we tested the role of self-relevance in attentional preference for visual artworks.

Methods: We designed experimental conditions focusing on three mechanisms of self-relevance: endowment, effort, and familiarity. We had 70 participants make abstract paintings, observe the creation of artworks by others, and copy other people's artworks. These were then incorporated into a dot-probe task. Participants also ranked their own and others' artworks.

Results: In agreement with our hypothesis, Studies 1 and 2 found a consistent attentional bias towards self-made artworks. This was found even when subjects had been exposed to other-made artworks that were highly familiar (Study 2). Study 3 failed to find a visual attention self-bias for copies of others' artworks. However, participants ranked their own artworks higher in all conditions.

Conclusion(s): Our results confirmed that own artworks get preferential visual attention and are judged more favourably than other-made artworks. Our results support the hypothesis that self-relevance is a key component of aesthetic preference and suggest that artworks are perceived as extensions of the self. This is consistent with archaeological evidence that the earliest aesthetic practices, in the form of body ornamentation, were produced to signal social identity. The implications of our findings for the evolutionary function of visual art and the role of art in the construction of identity are discussed.

Sketching out the mind: A Mathematical and AI approach to understanding drawing behaviour

Sueur, C., Beltzung, B., Pelé, M.

For more than 73000 years, drawing behaviour has been present in Homo sapiens. This form of expression is the premise of writing and allows for the transmission of more complex ideas or emotions than through verbal communication. Thus, this behaviour has been used as a window to the mind, especially in children, both at the psychological level, helping for example in the diagnosis of depression, and at the physical level, by analysing the development of fine motor skills. Biases such as adultcentrism make it difficult to understand and extract all the information hidden in the drawings in humans and anthropomorphism for other animals. One possible approach to overcome these difficulties is to analyse these drawings in a purely objective way, through the definition of new mathematical indices. A first option that will be presented is the use of fractals, which have revealed the evolution of the efficiency of spatial use of drawing between chimpanzees and humans, as well as differences related to the temporal intermittency of drawing between young children and adults. Then, different approaches using deep learning will be explained, demonstrating how convolutional neural networks can be used to analyse the drawing behaviour. The first proposed methodology will consider nonfigurative orangutans' drawings to assess whether deep learning can refine previous findings discovered through traditional analyses, by using transfer learning and feature extraction. The second methodology consists in using transfer learning and predictions to study the development of such behaviour, by considering children's and adults' drawings. The results bring a new light on the evolution of drawing in Hominidae and conduct to great applications in terms of human behaviour and psychology.

Unravelling Children's Language and Theory of Mind Competence in Freely-Told Narratives using Computational Linguistics

Van Dijk, B.M.A. Duijn, M.J.

Objective(s): We compiled ChiSCor, a new language resource consisting of 619 stories told freely by Dutch children aged 4-12, to examine the relation between children's linguistic, narrative and socio-cognitive competencies. In this contribution we focus on mental complexity of characters children created, a proxy for Theory of Mind (ToM) [1] competence used in developmental research [2], which we predict as a function of computationally extracted linguistic features from the stories.

Methods: Stories were collected in classroom environments, where children were invited to individually tell a fantasy story to class peers. Experts labelled the stories by indicating characters' mental complexity ranging from flat 'Actors', to 'Agents' having basic perceptive, emotional, and intentional capacities, to fully-blown 'Persons' with complex inner lives. We built a classifier encoding linguistic information to predict the occurrence of each character type in a story.

Results: Our classifier obtained good explainable performance (F1-macro = .80). 'Actor'-stories were overall syntactically and lexically less complex; 'Agent'-stories involved more use of deictic terms and higher syntactic complexity; Person stories involved clausal complementation, higher lexical diversity and complexity, and higher syntactic complexity. Our classifier drew purely on text-internal features that were not strongly correlated with age, on which existing research has often focussed [3].

Conclusion(s): Our results are in line with theoretical and experimental work [4-5] suggesting that language builds crucial scaffolds for a child's ability to understand and reason about the social world. Additionally, our results provide insight into the exact features involved in such scaffolding, and emphasise that language abilities make a contribution independent of age, which is informative for education and pedagogy. Our corpus, protocols, and code will be made freely available to the community and have wide potential for further research on the links between language, storytelling, and cognitive abilities.

Theory of Mind: Children aged 5-12 vs. a Large Language Model (GPT3)

Van Duijn, M.J., de Valk, W.M., van Dijk, B.M.A., van der Putten, P.W.H.

Objective(s): Large Language Models (LLMs) such as OpenAI's ChatGPT have recently shown striking abilities to produce natural language, performing at or beyond human level on many tasks including conversation and creative writing. Here we feed language-based Theory of Mind (ToM) tests to GPT3 [1] and compare its performance to that of children aged 5-12. Our aim is not to test whether LLMs like GPT3 possess ToM. Rather, comparing the model output with that of children of different ages allows us to map out parallels in commonly made mistakes, and to identify conditions that influence performance.

Methods: As part of a larger study* we ran a test battery including the Strange Stories Task [2] for children aged 5-7 (n=44) and a test of recursive ToM [3] for children aged 7-12 (n=42). Tests were presented verbally (voice-over) and in written form, accompanied by illustrations, and questions were multiple choice and/or open text fields eliciting brief motivations. We wrote a Python script feeding the same questions to the text-davinci-003 model via the OpenAI API, with minimal adaptations to prompt the desired output format.

Results: Parallel to children in our sample, GPT3 performs quite well overall, but makes more errors on more complex questions, particularly those involving irony, sarcasm, and recursive ToM. Also like humans [4], the model is highly sensitive to how information regarding these more challenging ToM questions is being presented.

Conclusion(s): Our results are in line with existing findings on ToM in children [5] and adults [3], and deepen current understanding of the limitations of LLMs [6-8]: they falter when a grounded understanding of the (in this case social) world is required. We discuss the implications and future potential for research into the mechanisms that support ToM, and into what is needed for a ToM capacity to evolve.

The BIASR Model: Confirmation Bias Emerges from an Approximation to Bayesian Reasoning

Pilgrim, C., Malthouse, E., Sanborn, A., Hills, T.T.

Objective(s): Confirmation bias is the tendency for people to search for and assimilate information in a way that supports their existing beliefs. Despite explanations in the literature, the question of why we have confirmation bias does not have a satisfactory answer. We provide an explanation based on a normative model of information processing with cognitive constraints. We consider 5 confirmation bias behaviours: biased evaluation, biased assimilation, belief perseverance, attitude polarisation, and confirmation bias in the selection of sources.

Methods: We ask how a rational agent should update their beliefs in a world with uncertain information and uncertain reliabilities in sources of information. We model this as a Bayesian network in the BIASR model (Bayesian updating with an Independence Assumption and Source Reliabilities). Rational updating introduces dependencies between beliefs, which scale exponentially with the number of beliefs being tracked. Realistic agents would therefore not be able to stay rational due to memory constraints, and would instead need to approximate rationality. Confirmation bias emerges from this approximation.

Results: We replicate the 5 forms of confirmation bias listed above through simulation. We show that our model has greater explanatory power than other information processing models in the literature. We also simulate specific experimental results from existing studies.

Conclusion(s): In the BIASR model confirmation bias emerges from evolutionary pressures to maintain large world models while minimising cognitive resources. This complements other explanations including motivated reasoning and arguments related to social cognition.

Costly signalling and cooperation

2.15-3.45pm; Lecture Theatre 1

Applying the smoke detector principle to moral categorization

Rottman, J., Foster-Hanson, E., Bellersen, S.

When people who are generally loyal or generally fair have a momentary lapse in their moral behavior, how does that impact whether we continue to think of them as loyal people or fair people—and how is this shaped by our own relative valuation of loyalty and fairness? From the perspective of research on conceptions of the “true self”, one possibility is that we reliably project our most strongly held moral values onto others, even after these people lapse. In other words, people who highly value generosity should consistently expect others to be generous, even after they act frugally in a particular instance. However, reasoning from an error-management perspective predicts that we should be especially prone to revoke membership from our most deeply cherished moral categories upon witnessing moral lapses, given the potential costs of affiliating with people who do not reliably adhere to our core moral values. In other words, people who most highly value generosity should be quickest to stop considering others to be generous if they act frugally in a particular instance. Across two studies conducted on Prolific (N = 720), we found consistent evidence indicating that people are most likely to heavily weight moral lapses when categorizing others in highly cherished moral categories, indicating that error management explains our moral character assessments more accurately than beliefs about good true selves. This tendency toward moral stringency carries significant practical and theoretical implications.

Honesty in signalling games is maintained by trade-offs rather than costs

Számádó, S., Zachar, I., Czégel, D., Penn, D.J.

Objective(s): Signal reliability poses a central problem for explaining the evolution of communication. According to Zahavi's Handicap Principle, signals are honest only if they are costly at the evolutionary equilibrium; otherwise, deception becomes common and communication breaks down. Theoretical evaluations of the Handicap Principle are difficult, however, because finding the equilibrium cost function in such signalling games is notoriously complicated. Here, we provide a general solution to this problem and show how cost functions can be calculated for any arbitrary, pairwise asymmetric signalling game at the evolutionary equilibrium.

Methods: We investigate the conditions of honesty in a general asymmetric signalling game. We reverse engineer the general solution by means of a Taylor series decomposition.

Results: Our model clarifies the relationship between signalling costs at equilibrium and the conditions for reliable signalling. It shows that these two terms are independent in both

additive and multiplicative models, and that the cost of signalling at honest equilibrium has no effect on the stability of communication. Moreover, it demonstrates that honest signals at the equilibrium can have any cost value, even negative, being beneficial for the signaller independently of the receiver's response at equilibrium and without requiring further constraints. Our results are general and we show how they apply to seminal signalling models.

Conclusion(s): Our results refute the claim that signals must be costly at the evolutionary equilibrium to be reliable, as predicted by the Handicap Principle and so-called 'costly signalling' theory. We argue that the evolution of reliable signalling is better understood within a Darwinian life-history framework, and that the conditions for honest signalling are more clearly stated and understood by evaluating their trade-offs rather than their costs per se. Last but not least, our results highlight why signals are expected to be efficient rather than wasteful.

The weak effect of social information on costly social behaviour: a preponderance of altruism over spite in one-shot anonymous interactions.

Watson, R., Morgan T.J.H., Kendal R.L., Van-de-Vyver J., Kendal J.

Objective(s): Mutually costly ("spiteful") behaviour, while rare in the animal world, is surprisingly common in humans, despite our cooperative tendencies. Yet, spiteful behaviour remains vastly understudied and little is known about its proximal causes. The principle goal of our study was to investigate the role of social learning in the incidence of spite. Methods: We ran an online experiment. In part 1, participants earned points (worth real money) in a game against a bot. In part 2, participants could engage in one-shot costly social behaviour either positively (altruism) or negatively (spite) to affect the points of another player. Before choosing their costly social behaviour, participants received social information indicating that previous participant(s) had either been altruistic, neutral, or spiteful. We varied the source of the social information as being either the majority of participants (conformity) or the individual who had thus far been most successful at earning points (success bias).

Results: We found: 1) Relative to an asocial control, exposure to altruistic social information slightly increased altruism, while exposure to neutral or spiteful social information slightly decreased altruism. There was no effect of conformity or success bias. 2) Participants were generally altruistic and spite occurred at very low rates. 3) High scoring participants were less altruistic than lower scoring participants.

Conclusion(s): Our results suggest that social information has only a small effect on social behaviour, and that, contrary to expectations from some social media platforms, participants can be altruistic in one-shot online interactions with an anonymous partner.

Sex Differences in Costly Signaling in Rural Western China

Dongzhi, C., Ge, E., Du, J., Mace, R.

Objective(s): Costly rituals convey the commitment to communities and advertise trustworthiness and cooperativeness to peers, which might explain why humans perform costly religious rituals. Here, we compare the efficacy of occasional public displays versus regular but less public acts for prestige enhancement.

Methods: We collected data on religious practices ranging from daily routine practices to infrequently elaborate distant pilgrimages among residents of an agricultural Tibetan village, as well as their reputational standings.

Results: We find that religious practices are mediated by demographic factors such as wealth, age and gender. Women are more inclined to daily religious activities, but men are more predisposed to distant pilgrimages.

Conclusion(s): Distant pilgrimages increase the perception of all prosocial characteristics. In contrast, daily practices are positively associated with nominations of devoutness but not with other qualities. Devoutness sometimes negatively relates to other reputational qualities, limiting the interpretation of religiosity as only about signaling prosociality.

Religious signalling and trust-based cooperation among mothers in the USA and Gambia

Chuvp, R.

Objective(s): Signalling theories of religion propose that public religious acts honestly communicate signallers' trustworthiness. Previous experiments suggest that those who engage in costlier rituals are trusted more. Field work suggests that participation in costly rituals is associated with more cooperation overall. We combined approaches from both to investigate trust-based cooperative interactions and publicly expressed religiosity among 1,100 mothers in the US and Gambia.

Methods: Trust-based cooperation: Mothers listed people from their cooperative social networks (N~11,000), and indicated who provided financial help (given or loaned) during the last year and how frequently these gifts occurred. Public religiosity: We created an ethnographically informed scale of publicly expressed religiosity sensitive to the religious contexts of the individual sites. Analysis: Hierarchical models were used to predict the probability of financial help, and among those who helped, the frequency of help. Covariates included kinship and support provided by mothers to their social networks' members.

Results: We found a positive association between women's public religiosity and financial help. However, the result was driven by the Gambian site and social network members' gender. Specifically, men were more likely to provide financial help and were more sensitive to mother's public religiosity than were other women.

Conclusion(s): In a real-world setting, religious signalling is positively associated with trust-based cooperation. This relationship goes beyond reciprocity and kinship, but the association is contingent upon culture and the gender structure of signallers' social networks.

Religiosity and gender bias structure social networks in a Tibetan population

Ge, E., Cairang, D., Mace, R.

Many have attempted to explain the evolutionary origins of religion and some suggest that religiosity promotes cooperation, but the empirical works evaluating the links between religious practices and cooperative social networks have been surprisingly few. Whether religious celibacy helps structure local social support remains to explore. Here, we draw on the religiosity and social support network data among residents of an agricultural Tibetan village to evaluate whether people are more likely to establish supportive relationships with religious individuals and consanguineous kin of celibate monks. We also examine the gender-specific correlations between religiosity and personal network characteristics. We found that religious practices foster supporting social relationships overall. Kin of celibate monks enjoys more social acceptance not only by the enhanced probability of having a supportive relationship but also by denser connections among them. Engagement in pilgrimage is associated with larger networks for males but not for females, whereas partaking in daily practice correlates with denser networks for both males and females. Particular religious acts may help individuals gain particular types of social network benefits, but benefits are gender dependent.

Sensitivity analyses to explore causal assumptions from observational data: An example assessing whether religiosity promotes cooperation in UK parents

Major-Smith, D.

Objective(s): Causal inference from observational data is notoriously difficult, and relies upon many unverifiable assumptions, such as no confounding or selection bias. Despite this, researchers often must rely on less-than-ideal observational data to infer causality. Here, I will demonstrate how a range of sensitivity analyses can be applied to explore whether a causal interpretation from observational data may be justified. I aim to answer the causal question "Does religiosity promote cooperative behaviour?" as a motivating example applying these methods.

Methods: I will use data from the parental generation of a large-scale (n = approx. 14,000) prospective UK birth cohort (the Avon Longitudinal Study of Parents and Children; ALSPAC). ALSPAC has detailed information on religiosity and potential confounding variables, while cooperation will be measured via self-reported history of voluntary blood donation. The basic analysis will be a logistic regression with religiosity as the exposure and blood donation as the outcome, adjusting for a range of relevant potential confounders. Sensitivity analyses will include: i) comparing models with different confounding structures (as the assumed confounding model may be incorrect); ii) quantitative bias analyses to assess the extent of unmeasured confounding necessary to alter conclusions; and iii) multiple imputation to examine potential selection bias due to missing data. Analyses will be repeated in the mother and partner cohorts.

Results: This project is a Registered Report with the analysis plan recently accepted (December 2022) in Evolutionary Human Sciences (<https://osf.io/z5gcm/> (<https://osf.io/z5gcm/>)). Analyses will be conducted by the EHBEA conference in April.

Conclusion(s): These analyses are intended to illustrate how sensitivity analyses can be used to aid research using observational data with the specific aim of causal inference. These approaches are frequently employed in disciplines such as Epidemiology, and it is hoped that they will become increasingly common in evolutionary-informed research to help improve our understanding of causality.

Kinship

2.15-3.45pm; Lecture Theatre 2

What does bride price do?

Brandl, E., Collieran, H.

Many societies practice bride price, where the family of the groom makes a payment to the family of the bride to seal the marriage. International organizations believe that bride price undermines gender equality because it establishes the husband's rights over the wife's reproductive capacities. But does bride price really harm women? To answer that question, we conducted a qualitative review of recent ethnographic publications from Melanesia. Some argue that bride price promotes gender-based violence: it emboldens the husband and his relatives to control the wife's labour and fertility while the wife's relatives monitor her premarital conduct, exercise pressure on partner choice, and extract resources from the husband. Moreover, the associated repayment obligations (in the case of divorce), interdependence between families, and custody rights can trap women in unwanted marriages. Others argue that bride price values women's contributions, secures women's and children's access to resources, and raises their standing in the husband's family. An inclusive fitness perspective may explain both: the woman's kin use bride price to consolidate ties, secure resources, and lay the groundwork for further marriages. As the woman's marriage prospects are tied to her reputation and influence the wellbeing of the wider family, her relatives have an incentive to restrict her choices. Paternity uncertainty motivates the man's side to demand restrictions on the woman's conduct before and during the marriage. In turn, the woman's side uses bride price as an honest signal of the groom's family's wealth, social support, and willingness to invest in the wife and children. The material needs of the wider family (and lower relatedness to affines compared to natal kin) incentivize both sides to extract labour and resources from the other. In sum, bride price may be a double-edged sword, with outcomes shaped by the competing strategies of different relatives.

Kin selection favours religious traditions: Ancestor worship as a cultural descendant-leaving strategy

Stucky, K., Gardner, A.

There has been a long tradition of research on the unusual extent of cooperation found in human societies, with researchers investigating a variety of general explanatory mechanisms and more recently the role of religious systems in some of these in particular. In this context, it has been suggested that cultural traditions such as ancestor worship might have evolved as a descendant-leaving strategy of ancestors by encouraging increased altruism specifically between distant kin. More precisely, Coe et al. (2010) have presented a mechanism of cultural transmission exploiting social learning biases, by which ancestors might have been able to establish parental manipulation of kin recognition and perceived relatedness as a traditional behaviour, eventually leading to increased altruism among co-descendants and thereby maximising the respective ancestor's inclusive fitness. This manipulation attempt might have given rise to what has been termed as "ancestor-descendant conflict", however. Here, we develop a demographically explicit model based on Coe et al.'s assumptions, quantifying the proposed ancestor-descendant conflict and investigating the evolutionary feasibility of religiously motivated cultural norms promoting increased altruism among co-descendants. Our analysis reveals that such norms could indeed lead to an increase in overall altruism with potential for ancestor-descendant conflict as anticipated by the authors. More importantly, we found that natural selection could favour traditions encouraging increased altruism towards co-descendants in a range of conditions, given a demographically conditional variability of parental manipulation and given that information about kin relations would be tied to the respective norm. This would allow individuals to direct their altruistic behaviour towards co-descendants as opposed to non-kin, thereby offsetting some of the inclusive fitness costs incurred by the increase in expressed altruism. Ancestors who introduced cultural traditions such as ancestor worship might therefore have been more successful in leaving descendants, potentially resulting in the spread of such cultural traditions.

Network ecology of marriage

David-Barrett, T.

Objective(s): The institution of marriage is an understudied phenomenon in behavioural sciences. All patriarchal systems invented a form of formal marriage institution. People living in foraging cultures and in matriarchy tend to form long-lasting romantic primary relationships with substantially weaker social control.

Methods: 1. Mathematical simulations using the modelling framework of the Microfoundations theory generated the empirical hypotheses; 2. Seven-language, cross cultural survey and experiment tested the hypotheses.

Results: Mathematical simulations show that replacing distant direct kin with in-laws increases the interconnectedness of the family social network graph, especially for mid-sized groups. Thus, if a formal institution increases the stability of these graphs (and because integrated social networks have higher trust) recognising in-laws reduces free riding within the group. The derived empirical hypothesis is that people test network compatibility of the two families during mate choice. The empirical results showed that people (i) assess potential long-term partners for affinal compatibility, (ii) test potential long-term partners for direct affinal compatibility, and (iii) assess potential partners for indirect affinal compatibility. As predicted by the Microfoundations theory, all these effects are driven by the fertility-urbanisation dynamics.

Conclusion(s): The recognition of in-laws increases the cooperativeness of medium sized groups, especially in high fertility, rural populations. Formal marriage institution increases the stability of such cooperation. This is especially important in cultures with individual rather than collective property concepts. This may explain why all and only patriarchies invented this tradition, and why it has been waning the past decades in most cultures around the world.

Evolution of Kinship Structures Driven by Marriage Tie and Competition

Itao, K., Kaneko, K.

In many traditional societies, people are categorized into several cultural groups, or clans, within which they believe they share ancestors. Clan attributions provide certain rules for marriage and descent. Such rules between clans constitute kinship structures. Anthropologists have revealed several types of kinship structures, e.g., generalized exchange characterized by matrilineal (or patrilineal) cross-cousin marriage or generalized exchange characterized by bilateral cross-cousin marriage. Here, we propose an agent-based model of traditional societies to reveal the evolution of kinship structures. In the model, several societies compete. Societies themselves comprise multiple families with parameters for cultural traits and mate preferences. These values determine with whom each family gets married, cooperates, and competes. They are transmitted to a new generation with slight mutation. Families can cooperate with those having similar traits to their own or mates, whereas those with similar preferences will have more mating conflicts. The growth rate of each family is determined by the number of cooperators and competitors within a society. Evolutionary simulation demonstrates that family traits and preferences diverge to form several clusters that can be regarded as clans. Subsequently, various kinship structures emerge, including clan endogamy, dual organization, and generalized or restricted exchange. These structures emerge depending on the necessity of cooperation and the strength of mating competition. Finally, we conducted the statistical analysis using the Standard Cross-Cultural Sample, a global ethnographic database. It empirically verified the theoretical results on the environmental dependence of kinship structures. Such collaboration between theoretical and empirical approaches will unveil universal features in anthropology.

Cross-cousin marriage among Tsimane forager-horticulturalists during demographic transition and market integration

Dakgero, A., Beheim, B. A., Gurven, M., Kaplan, H., Lukas, D.

We examine how the practice of cousin marriage is affected by recent dynamics of mate availability, fertility and child survival in a small-scale population experiencing rapid market integration and acculturation, the Tsimane foragers-horticulturalists of the Bolivian Amazon. Specifically, we examine recent changes in (i) the frequency of cross-cousin marriage (ii) the composition of the pool of eligible partners and its influence on marriage decisions and (iii) the fitness consequences of cross-cousin marriage for both men and women. Using genealogical data collected by the Tsimane Health and Life History Project over the last two decades (28000 individuals, observed years from 1876 to 2015), we estimate the prevalence of cross-cousin marriage with respect to the availability of eligible partners in the population, and age-specific fertility and survival curves by marriage type. We also measure the rates of change in each of these calculations longitudinally. We find that cross-cousin marriage is associated with earlier age at first reproduction and higher fertility, but lower offspring survival, in some of the recorded time intervals. We observe these effects to all have decreased over the last several decades. Although Tsimane marry cross-cousins more than expected by chance, we also observe a recent decline in the frequency of cross-cousin-marriage. Cultural changes in marriage practices might reflect the underlying societal changes that shape partner availability and fitness returns of different marriage choices.

Group relatedness and wealth consolidation under different cousin marriage and inheritance rules

Perez Vekilla, A., Campbell, O.L.K.

Many human societies vary their marriage practices, such as whether endogamy or cousin marriage is normative, and this can greatly affect social structure, particularly group relatedness. Furthermore, some groups have a preference for a parallel cousin marriage, where an individual marries a cousin to whom they are related through their parent's same sex sibling, whereas others prefer cross-cousins (related through their parent's opposite sex sibling). Female inheritance, high levels of warfare, whether wealth is movable or immovable, and overall wealth consolidation have all been raised as reasons as to why individuals may practice cousin marriage. Additionally, much of the literature implicitly assumes that parallel cousin marriage should produce more highly related groups and more intensive kinship than cross-cousin marriage, and this assumption forms the basis of many verbal models. We build an agent-based model comparing different types of cousin marriage and different inheritance rules to test some of these assumptions. Specifically, we compare differences in group relatedness and wealth consolidation under different combinations of cousin marriage and inheritance. By growing a network of inheritance, we track both economic resource and gene flows from parents to their children under these diverse kinship regimes. This allows us to compare cousin marriage and inheritance rules on a quantitative basis, using measures of average relatedness, wealth distribution and intergenerational wealth escape of kinship groups.

Genetic markers of a history of cousin marriage and justification of honour-based violence across 50 ethnic groups.

Campbell, O.L.K., Padilla-Iglesias, C., Mace, R.

Objective(s): Cousin marriage is common cross-culturally and societies vary in regard to which type of cousin is preferential to marry. Anthropological literature has long posited that cousin marriage may be associated with the emergence of honour cultures – cultures characterised by strong retributive norms and violence in response to threats of dishonour. Methods: We match average genomic inbreeding coefficients of ethnic groups – a measure of the historical practise of cousin marriage – with survey data on justification of honour killings against both men and women across 50 cultures. Secondly, we test whether our associations hold between regions within countries, mitigating doubts that results are due to omitted country level variables. Results: We show some of the first empirical evidence that consanguineous marriage is associated with honour-cultures cross culturally and that this holds for honour-based violence directed against both men and women, something that has been little discussed in previous literature. By using multiple data sets at different levels, we also demonstrate that this holds at the ethnicity level, the country level and between regions within countries. Lastly, we provide tentative evidence that female inheritance and pastoralism, both of which have been raised as possible causal factors, cannot explain the association between cousin marriage and honour.

Beyond humans

2.15-3.45pm; Jeremy Bentham Room

Associations between Intentional gestural communication and sociality in wild East African chimpanzees (*Pan troglodytes schweinfurthii*): A social network approach

Roberts, S.G.B., Roberts, A.I.

Objective(s): The selection pressures arising from primate sociality are proposed to have played a major role in shaping both communication and cognitive skills, but whether primates can adjust their use of communication to the social environment is still unclear. The objective of this study is to examine how the use of intentional gestural communication is associated with social context and strength of social bonds. Methods: We used focal scan samples to collect data from 12 chimpanzees from the Budongo Forest, Uganda. We used social network analysis and multi-level modelling to examine the associations between use of intentional communication, the social context and strength of social bonds. Results: Visual gestures were used across a greater variety of social contexts (e.g., grooming, mating) than auditory or tactile gesture. Further, chimpanzees with strong social bonds had higher rates of visual gestures. Finally, more complex gestural communication (e.g. gestures accompanied by calls, use of persistence in gestural communication) was associated with a longer amount of time spent in close proximity, and stronger social bonds. Conclusion(s): These results show that chimpanzees use intentional gestural communication flexibly according to the social context and use different types of gestural communication with different social partners. This suggests that gestural communication may play an important role in helping chimpanzees meet the challenges of living in large social groups.

Between-group competition and cognitive evolution in primates

Majolo, B., Wakes, S.J., Ruta, M., Willems, E.P.

The social intelligence hypothesis predicts a positive relationship between social complexity and brain evolution. Recently, Ashton and colleagues (2020) proposed the 'Napoleonic intelligence' hypothesis. They argue that the nature (friendly or violent) and frequency of interactions between groups can drive brain evolution. Moreover, Cowl & Shultz (2017) found a positive association between brain size and intensity of agonism in female primates. Our study aims to test whether between-group competition shapes brain evolution in primates. We extracted data from the literature, in 86 primate species, on five measures of between-group competition: frequency (events/day) of between-group encounters, proportion of aggressive between-group encounters, occurrence of multicide during encounters, home-range overlap and degree of territoriality (D-index). We used brain weight (controlling for body weight) as a proxy of cognitive complexity. We used comparative phylogenetic analyses to estimate the relationship between cognitive complexity and between-group competition whilst controlling for socio-ecology (diet, group size). We found weak support for the 'Napoleonic intelligence' hypothesis. Our results suggest that between-group competition does not have a significant effect on brain evolution. Alternatively, brain evolution may have been affected by the need to maintain social cohesion within the group, which in turn encapsulates various measures of social complexity, including the quality and frequency of between-group interactions.

What can we learn from studying curiosity in animals? Insights from primatology

Forss, S., Willems, E., Motes-Rodrigo, A., Dongre, P., Mabr, T., van de Waal, E.

Curiosity is a core mechanism of intelligence, indispensable for driving life-long learning. In a broad and catch-all sense, curiosity is described as an intrinsic motivation to acquire novel information in the absence of any immediate extrinsic reward. Empirical studies on curiosity have been dominated by the fields of psychology and neuroscience, and with a few exceptions the prevailing data sets use methods relying on linguistic skills. Thus, existing knowledge and current transdisciplinary debates focus on humans and exclude other animal species. Here, I draw from studies on non-human primates (great apes and vervet monkeys) using novel object paradigms on what we can learn regarding the evolutionary roots of curiosity using behavioural measures. Results from four ape species (chimpanzees, bonobos, Sumatran- & Bornean orangutans, N=101) revealed that curiosity followed a linear

gradient in accordance with these species sociality. We propose the social curiosity hypothesis to explain the observed pattern and that curiosity may have evolved in interaction with sociality. Results from vervet monkeys (N=167) suggest that intraspecifically curiosity varies depending on environment and habituation level. Wild individuals show less curiosity than captive conspecifics and unhabituated monkeys are less curious than habituated individuals. We conclude that both social and ecological factors will play a role in the evolution of curiosity.

Phonetic invariance in chimpanzee calls: Toward an evolutionary theory of speech motor control

Ekström, A.G.

Objective(s): Human speech motor control allows speakers to achieve predictable resonances values for vowel phonemes. The evolution of this capacity, however, has yet to be explored. Here, I explore vocal tract resonances in Chimpanzee (Pan troglodytes) hoots, with the goal of determining apparent articulatory targets in vocal production.

Methods: Resonances were estimated from chimpanzee barks (N=79, 8 individuals) and pant hoots (N=36, 4 individuals). Because articulation appears largely constant across early-bout pant hoots, only the first voiced segment was used for this call type. To control for potential repositioning effects from somatosensory or auditory feedback, resonances were estimated only from the first glottal pulse observed for each call. Further, because resonances likely differ significantly between individuals (as in humans), comparisons were not made between individuals, but only within individuals.

Results: Early observations suggest that chimpanzees call types do not differ significantly within individuals, but that for each individual, a target articulatory target is reached for an executed in call production.

Conclusion(s): This effort represents tentative steps toward an evolutionary theory of speech motor control. The subject of non-human primates call “innateness” will also be discussed.

What is it like to be an octopus? Anthropocentrism, embodiment and the limits of cognitive convergence

Barton, R.

What is it that evolved during cognitive evolution? The dominant paradigm in the field of cognitive evolution assumes, implicitly or explicitly, that it was some generalised property or properties supporting ‘intelligent behaviour’. As a result, attention focuses primarily on how ‘intelligent’ (flexible, innovative etc) different species are, rather than on what qualitative differences there might be that are associated with different lifestyles and mediated by different neural systems. This focus is in large part because humans see ourselves as ‘intelligent’, and because we do love to rank things. It drives researchers to emphasise cognitive evolutionary convergence, yet the comparative method can also be used to study divergence. I propose that the field has the balance of emphasis wrong in this respect: we should not be asking whether or to what extent an octopus has a “Theory of Mind”, but what sort of cognitive processes are likely mediated by its radically different (to vertebrates), independently evolved CNS associated with an invertebrate body and phylogenetic history. Along the way I suggest a rethink of several commonly employed concepts, including distinctions between domain generality versus specificity and associative learning versus complex cognition.

Posters

5.45-8.00pm; North Cloisters

Physical Dominance Perception in the Human Visual Periphery from Neutral and Intimidating Male Gait

Anguera, A., Leslie, C., McCarthy, K.

The perception of physical dominance and threat has been recently discussed in the evolutionary and cognitive psychology domains of the literature. Biological motion provides some insight into the different traits that allow an individual to accurately and quickly assess the physical formidability of another, avoiding the unnecessary use of force when fighting prospects are largely unbalanced. Selective pressures towards the accurate and quick detection of fearful stimuli in the visual periphery have also been recently suggested. This project aims to explore the perceptions of dominance and threat of a moving target (i.e., walker) presented in the visual periphery of a rater. This project contained two discrete parts; the first informed the collection of biological motion, and creation of visual stimuli from 34 men ‘walkers’ in different walking conditions (baseline / intimidating) recorded through a high-end motion capture system. Confounding variables in the walkers were removed with the creation of a standardised virtual avatar. The second part followed up by presenting 33 ‘raters’ with animated videos of all of the walkers in their visual periphery. This was ensured by utilising eye-capture technology, and a gaze-contingent computer experiment to prevent the participants’ gaze from fixating directly on the stimuli. This study found that raters are able to accurately ascertain the physical dominance of walkers in their visual periphery. Another finding of this study is that when shown an intimidating walk, raters are more likely to both rate it as more physically dominant, and have a higher intent of crossing the street to avoid a physical confrontation. This study suggests the existence of a link between the previously unexplored areas of visual periphery fearful stimuli detection, and physical dominance in men.

“It’s my decision because it’s my body”: Exploring the role of social support in abortion decisions

Baker, J., Kayser, S., Sifaki, K., Adair, L., Lozano, N.

Objective(s): An evolutionary perspective highlights how abortion fits within a diverse repertoire of strategies that have been used by humans for thousands of years (Nurge, 2011) to control their family size, reproductive timing, and ultimately enhance their survival and success. This perspective captures the duality of abortion decisions, as both highly individual (Anglin et al., 2010) and influenced by social and environmental factors (Virgo & Sear, 2016). Research tends to focus on demographic patterns (rather than psychological processes) or emphasize the role of partners and parents (to the exclusion of others in a social network). To address these gaps, we examine abortion decision-making processes, with an emphasis on social and relational factors.

Methods: Semi-structured interviews were conducted via Zoom/phone with women (ages 18 - 73; 50% White British) in the UK both with (N =14) and without (N =14) abortion experience. Each interview transcript was coded by two researchers and thematic analysis (Braun & Clarke, 2012) was used to identify key themes.

Results: We identified five themes: (1) certainty reflects feelings of “rightness” and confidence in the decision to terminate, often associated with the careful and reflective approach applied to the decision-making process, (2) independence captures emphasis on decisional autonomy, including assertions that a pregnant person can and “should” hold ultimate decision-making power, (3) secrecy as stigma management highlights disclosure strategies as abortion decisions are navigated, (4) practical concerns reflect factors (e.g., health, financial resources) that were often considered “better” reasons to terminate than other relevant concerns, and (5) relational concerns capture the role that social support, overwhelmingly focused on the role of a romantic partner, plays in shaping abortion decisions.

Conclusion(s): While evolutionary models underscore the importance of community support in reproductive decisions, our findings suggest that abortion stigma may be isolating women from these support mechanisms.

Examining the influence of contexts related to mating and threat on women’s makeup usage.

Biesiadecka, M., Baryla W., Szymkow A.

The main function of cosmetics is to improve the physical appearance of the face (Bates & Russell, 2018). Makeup can be used to enhance evolutionarily important facial indicators of high mate value. Studies (e.g., Jacob et al., 2009) have indicated that makeup increases attractiveness and likelihood of courtship behaviors. For example, makeup is perceived by men as a signal of a woman’s willingness to engage in sexual activity (Mafra & Varella, 2020). Vast majority of makeup research has been limited to everyday makeup use, while the amount of makeup can vary depending on the situation. For instance, in the context of courtship behaviors, women can feel a stronger motivation to increase their attractiveness and might be more inclined toward signaling their availability. On the other hand, as suggested by Valentova et al. (2021), there are situations in which an individual does not want to stand out from the crowd, seeking to avoid attention instead. It is possible that in a situation of danger, a woman may want to make her appearance worse, or at least avoid enhancing it.

Two studies were designed to test whether the amount and diligence of makeup application are higher in a mating context compared to no-mating contexts, and lower in a threatening context. Additionally, the mediating function of motivation and moderating role of sociosexuality and appearance orientation was examined. Results show that the relationship between context and both quantity and diligence of makeup is fully mediated by motivation, but not moderated by sociosexuality and appearance orientation.

Why do Grandparents Care?

Bullingham, R., Karthigesu, S.P., Coall, D.A.

Objective(s): Living longer than previous generations, grandparents typically have active lives and their role as a grandparent is likely one of many. Although there is a consensus that grandparents who provide childcare experience health benefits, some recent studies show inconsistent findings. Variation in frequency and intensity of grandchild care may explain these inconsistencies. Another, currently unexplored explanation, is that grandparents have different motivations (e.g., autonomy versus obligation) and it is these motivations that moderate the association between grandparenting and health.

Methods: Employing a qualitative dominant convergent mixed methods design, focus groups were used to explore grandparents' experiences of providing grandchild care. Participant surveys gathered demographics, lineage, and childcare frequency.

Results: Grandparents reported enjoying providing grandchild care and were autonomously motivated to help but acknowledged that the role comes with challenges regardless of the frequency of grandchild care. Grandparents discussed feeling appreciated and having control over the amount of care they provide. Overall, grandparent health and wellbeing was good, however, a common physical challenge for grandmothers was feeling tired. Maintaining paid employment and grandchild caregiving was a challenge for grandmothers with many reducing their hours or retiring. There was high cooperation and low conflict between parents and grandparents.

Conclusion(s): Exploring grandparent motivations provides insight into how grandparents balance healthy ageing with the challenges of providing grandchild care. It also helps us understand why grandparents care for their grandchildren and the impact their motivations may have on health. This research highlights grandparents' motivations to provide grandchild care and how they negotiate the role as they are likely moderators of the relationship between caregiving and health.

Does the menstrual cycle affect vaccine response?

Cooper, A., Faurie, C., Alvergne, A.

Objective(s): Does the menstrual cycle affect how vaccines work? Building on existing knowledge that biological sex and the menstrual cycle modulate the immune system through reproductive hormones, our objective was to determine if timing in menstrual cycle when vaccinated affected presence, severity, and number of side effects experienced. Drawing on perspectives from Evolutionary Medicine, we consider menstrual cycles as a trade-off between fertility and immunity.

Methods: Data was collated from the Clue Period Tracker App on 1,735 women who received a first Covid-19 vaccine and 1,603 who received a second Covid-19 vaccine, to examine whether vaccine side effects depend on the phase of the menstrual cycle. We also collected data on 292 adults visiting vaccination centres to analyse the effect of biological sex. We use DAG-informed regression models to test the following hypotheses (1) women will be more likely to experience side effects and at a greater severity and number than men and (2) ovulation, menstruation and the follicular stage will result in a higher probability of presence, severity and number of side effects owing to increased inflammation during these periods of the menstrual cycle.

Results: We found women experience stronger and more side effects than men. Women vaccinated in ovulation experienced more severe side effects from a first vaccine, and women vaccinated in their follicular phase more commonly experienced side effects from the first vaccine. No clear association between menstruation and side effects was found.

Conclusion(s): Our results further evidence biological sex as a significant variable in medical interventions. The trade-off between reproduction and disease prevention within the menstrual cycle is complex and requires further research with biological sampling and detailed ecological data.

Social distance as a strategy of pathogen avoidance by women in the 1st trimester of pregnancy - mediating role of disgust sensitivity not confirmed

Frankowska, N., Szymkom, A., Tolopilo, A.

The behavioral immune system (BIS) represents a set of actions to avoid contracting infectious diseases (Murray & Schaller, 2012). Since having contact with out-group members entails a heightened risk of infection (Schaller & Neuberg, 2012), pathogen avoidance motives are linked with negativity toward foreign individuals and a preference to keep a greater social distance from them (e.g., Faulkner et al., 2004; Millar et al., 2020; Szymkom et al., 2021). Importantly, these effects were shown to be mediated by disgust propensity and germ aversion (Szymkom et al., 2021). Pregnant women (especially in the 1st trimester) experience progesterone-linked immunosuppression – a reduction in the effectiveness of the immune system. The compensatory behavioral prophylaxis hypothesis (CPH) assumes that during this period, women are motivated to avoid a potential threat of infection, and disgust plays a compensatory function in this process (Fessler & Navarrete, 2003; Żelazniiewicz & Pawłowski, 2015; Kankova et al., 2022). At the same time, women express elevated ethnocentrism in the 1st trimester of pregnancy (Navarrete et al., 2007). Thus, according to CPH both disgust sensitivity and progesterone should be the highest at the beginning of pregnancy than in later trimesters. Yet, progesterone is usually the highest in the 3rd pregnancy trimester, thus the mechanisms of these effects are not conclusive (Timmers et al., 2018; Jones et al., 2018). Here, we tested the role of pathogen disgust and germ aversion as potential mediators between gestational age (trimester) of pregnant women and preferred social distance toward in-groups and out-groups presenting signs of infection or not. Our results do not confirm the significant mediating role of germ aversion or pathogen disgust. While we have observed a predicted decrease in declared nausea frequency, we did not observe a decrease in disgust sensitivity between trimesters within participants.

Conformity bias and task demonstrability in information searching

Fujikawa, M., Yokota, K., Tokuoka, M., Nakanishi, D.

Objective(s): Boyd & Richerson argue that conformity bias can characterize cultural evolution and be an adaptive strategy for obtaining accurate information in uncertain situations. Nonetheless, the evidence on conformity bias is inconclusive, with some studies showing evidence of this bias and others not. Especially, Eriksson & Coultas (2009) show that conformity bias does not occur in information-seeking tasks with questions that have no correct answer (e.g., norms.) However, conformity bias may be an adaptive strategy for answering questions with high demonstrability, such as objective questions with correct answers. This study aimed to investigate the role of task demonstrability in conformity bias by replicating the experiment in the study by Eriksson & Coultas (2009) using subjective and objective questions. The hypothesis was that conformity bias would occur when the correct response rate for objective questions slightly exceeded 50%.

Methods: In a pilot survey with 68 Japanese undergraduates, nine and five objective and subjective questions, respectively, that showed a correct response rate exceeding 50% were selected. In Experiment 1, 120 undergraduates answered to the 14 yes-or-no questions, were presented with the responses of nine, six, three, or zero other participants ("Yes" responses, within-participant design), and completed the same questions again. Experiment 2 replicated Experiment 1 with 183 crowd-workers. Results and

Conclusion(s): Both experiments showed evidence of conformity bias in both the subjective and objective questions, suggesting that conformity bias is an adaptive strategy for information searching. We discuss the differences between the experimental design of the prior study and our study.

Creativity and sexual selection: Enhanced creative thinking in the context of an attractive mate.

Galasinska-Grygorczyk, K., Szymkom, A.

Previous research suggests that creativity emerged as a result of sexual selection. If so, men and women should manifest their creativity in the context of an attractive potential partner, especially when they are single. We tested this hypothesis by arranging a dating portal context. Male (n = 241) and female (n = 242) participants, single (n = 204) and in a relationship (n = 279), viewed photos of attractive and unattractive opposite-sex mates who were told they were dating site users looking for a partner. We asked the participants to rate their attractiveness and create a self-presentation note to attract them, presenting a variety of qualities to be perceived as an original and creative person. We monitored their motivation to perform the task well. The notes were rated for fluency, flexibility, originality and overall creativity. We also counted participants' mentions of their own creativity, which they used for self-advertisement. The results show that regardless of relationship status, men were more fluent and flexible in their self-presentation, as their motivation and mood

valence increased after seeing attractive mates. Women, on the other hand, had a more original self-presentation when their mood valence increased after viewing attractive mates, but especially when they declared themselves to be in a relationship. Differences in the mating strategies of the two sexes are discussed in the context of the evolutionary aspect of creativity.

Want it all? - The (In)Dependence of Mate Preferences

Hartung, A.

Objective(s): Deciding with whom to mate is one of the most important decisions from an evolutionary perspective. Consequently, evolutionary psychology is concerned with what is attractive to potential mates. Since there is no agreement on which and how many characteristics are relevant, dimensionality reduction methods were used. Here, some researchers identified independent preference dimensions. This conflicts with the assumption that mate choice preferences are psychological adaptations, evolved to maximize an individual's fitness. If this is true, there should be an overall preference for fitness and its indicators.

Methods: In order to find out if the independence of the dimensions is a methodological artifact, the independence of the dimensions "personal/parenting qualities" and "attractiveness/social visibility" were reexamined. For this purpose, a conjoint analysis was used. Furthermore, the additive conjoint measurement was used to approach the question how the two dimensions interact. 175 persons (60.57 % female) aged 18 - 25 years were surveyed.

Results: The results showed that 81.5 % of the respondents prefer above-average levels on both dimensions. 37.15 % of the participants violated at least one necessary and testable axiom of the additive conjoint measurement. Remarkably, 25.71 % of the participants violated transitivity. Their data cannot be assumed to be ordinal scaled.

Discussion: This suggests that the dimensions are not independent and the preference for these dimensions are not additive. Additionally, these results add to the criticism of the common practice of assuming data as interval scaled.

Disgust sensitivity depending on phases of the menstrual cycle and early pregnancy

Hlaváčková, J., Dlouhá, D., Hrbáčková, H., Nouzová, K., Šeda, J., Kaňková, Š.

Objective(s): The aim was to compare the disgust sensitivity between pregnant and non-pregnant women, and for the latter, also between the phases of the menstrual cycle.

Methods: In a prospective cross-sectional study conducted between 10/2020 – 10/2022, 157 women in the first trimester of pregnancy and 335 non-pregnant women without children (124 in the follicular and 211 in the luteal phase of the menstrual cycle, based on ultrasound examination) completed the Disgust Scale-Revised (DS-R; includes Core, Animal Reminder, and Contamination subscales) and the Three Domains of Disgust Scale (TDDS; includes Pathogen, Sexual, and Moral domains).

Results: Pregnant women reported higher overall DS-R score ($t=-4.47$, $p<0.001$, $d=0.43$) Core ($t=-3.71$, $p<0.001$, $d=0.36$), Animal Reminder ($t=-3.41$, $p<0.001$, $d=0.33$), Contamination ($t=-4.06$, $p<0.001$, $d=0.39$), Pathogen ($t=-2.18$, $p=0.030$, $d=0.21$) and Sexual disgust score ($t=-3.00$, $p=0.003$, $d=0.29$) than controls, but not overall TDDS score and Moral disgust score. In controls, disgust sensitivity did not differ between women in the follicular and luteal phases of the menstrual cycle.

Conclusion(s): Our findings that pregnant women had higher disgust sensitivity related to pathogens than non-pregnant controls are consistent with the Compensatory Prophylaxis Hypothesis and also support the hypothesis of a functional behavioral immune system operating through the emotion of disgust when one needs to be more protected against infectious diseases. No significant difference between menstrual cycle phases could be explained by the fact that changes in disgust sensitivity between these phases are generally small and tend to be detected when other factors, such as increased pathogen exposure, are at play.

Preferences for facial sexual dimorphism across culture and the sexual orientation spectrum

Hokxleimer, I.J., Bjornsdottir, R.T., Ishii, K.

Objective(s): Judgments of attractiveness have many important social outcomes, highlighting the need to understand how people form these judgments. One aspect of appearance that impacts perceptions of attractiveness is facial sexual dimorphism (femininity/masculinity). However, extant research has focused primarily on White, Western, heterosexual participants' preferences for sexual dimorphism, limiting generalisability. Indeed, recent research indicates that these preferences vary by culture, and other work finds differences between gay/lesbian and heterosexual individuals. Yet, aspects of identity such as culture and sexual orientation do not exist in isolation from one another; rather they intersect, leaving a critical gap in current understanding. Our research aims to bridge across these hitherto separate areas of inquiry to provide a comprehensive and generalisable understanding of sexual dimorphism preferences.

Methods: We are currently in the process of recruiting 1,110 British and 1,110 Japanese participants, evenly split by gender and sexual orientation categories (bisexual, gay/lesbian, heterosexual; see <https://osf.io/7gmnf> (<https://osf.io/7gmnf>) for pre-registration). Since recent evidence suggests that attraction to men and women may be better conceptualised as two independent axes, we are also collecting a second, continuous measure of sexual orientation. Participants are judging White and East Asian women's and men's faces in two different experimental tasks: a two-alternative forced choice task assessing preferences for feminised versus masculinised morphs of faces; and a second task, in which participants interactively manipulate faces' sexual dimorphism to optimise facial attractiveness.

Results and Conclusion(s): This project will allow us to test how perceivers' culture, gender, and sexual orientation as well as faces' gender and ethnic background predict preferences for sexual dimorphism. This will provide a more comprehensive understanding of both the relationship between sexual dimorphism and attractiveness judgments and how perceivers' and faces' intersecting identities predict an important social judgment.

Some like it hot (but mostly they don't): Genital and subjective sexual arousal in response to the copulatory movements of different animal species

Hála, M., Krejčová, L., Norák, O., Polysková, K., Bártoná, K.

Studies have reported sex differences in human genital response and subjective rating of arousal while exposed to erotic video. Men show category-specific sexual responses, while women do not. A possible explanation is that for men, the key features eliciting arousal are cues relating to the preferred sex, whereas for women the crucial cues are sexual activities such as copulatory movements (CM). Although women's subjective arousal is category-specific, they react also to stimuli of non-preferred sex and even to other animal species (Chivers & Bailey, 2005). This study aims to test the level of sexual arousal in response to the CM of nonhuman species differing in phylogenetic distance to humans. We use vaginal and penile plethysmography to test the level of genital arousal in response to the CM of different nonhuman species, and a 9-point Likert scale to assess the subjective arousal. The heterosexual participants ($n=42$, 21F, mean age = 27.59 ± 7.91) were randomly presented with eleven 1-minute soundless videos depicting copulation (human female-male, human female-female, bonobos, gorillas, lions, zebras, rabbits, guinea pigs, budgerigars, crickets, and lizards). Averaged genital responses to each stimulus were standardized to allow for comparison between sexes. Mixed-design ANOVA revealed the effect of stimulus type ($\omega^2 = 0.35$). The largest responses were for human female-male and human female-female stimuli. They differed from other stimuli but not from each other. All other stimuli were indistinguishable from each other. There was no difference in genital responses between men and women ($\omega^2 = 0$). For subjective arousal, we found the effect of stimulus type ($\omega^2 = 0.72$). Again, human female-male and human female-female stimuli were the most arousing. Men rated human stimuli as more sexually arousing than women. Our results suggest that CM displayed in non-human species is not a sufficient cue for eliciting genital or subjective sexual arousal.

The influence of the Dark Triad of personality on altruistic behaviour

Jaszczyńska, I.

Objective(s): From an evolutionary point of view, altruistic behaviour towards relatives has often been explained by kin selection while altruistic behaviour toward non-kin by reciprocal altruism. The aim of the study is to assess the influence of the Dark Triad of personality on altruistic behaviour. Traits of the Dark Triad include Machiavellianism, narcissism, and psychopathy. Altruistic behaviours in this study were defined as being or willing to become a blood donor, an organ donor and a bone marrow donor. In addition, the study aims to find out why people do not become donors.

Methods: The study uses the Polish version of the Dirty Dozen measure, the Generative Altruism Scale (GAIS) and additional questions about the willingness to donate blood,

organs and bone marrow. The survey is addressed to people aged 18 – 65 living in Poland.

Results: Based on previous studies, I predict that the Dark Triad of personality will correlate negatively with non-kin altruism. I also expect differences in the answers depending on the age of the respondents. Young people may have different approach to helping others, while older people may have more health contraindications.

Conclusion(s): People who help are often guided by a selfless desire to help others, but the amount of help is also influenced by other factors. I hope that my research will contribute to raising awareness about the sources of human motivation to engage in altruistic behaviour. I also hope that thanks to them we will understand what factors influence the resignation from helping others.

Who changes more? Fertility plans before and during COVID-19 pandemic

Jurczak, A., Marcinkowska Trimboli, U., Mijas, M., Nenke, I., Kossakowska, I.

Objective(s): Limited access to prenatal care, perceived financial insecurity and decreased mental well-being were shown to affect reproductive plans of Poles. All these factors have been also linked in research with socioeconomic changes following the COVID-19 pandemic. The aim of the study was to investigate changes in reproductive plans in response to COVID pandemic among cisgender men and women.

Methods: Study sample consisted of 2074 adult women and 155 adult men with no diagnosed fertility issues. Participants filled an online questionnaire which included questions on their reproductive plans, COVID-related stress, place of residence, education, economic and relationship status, and parity. Multivariate logistic regression models were conducted to explore predictors of reproductive plans in men and women separately.

Results: In the studied group 32.5% of women (28.9% revealed decreased and 3.6% increased intentions to have children) and 23.0% of men (20.7% revealed decreased and 2.3% increased intention to have children) declared that the pandemic influenced their reproductive plans. In both genders' COVID-related stress was related to decreased intention to have children (women: OR=1.13, 95% CI: 1.11-1.15; men: OR=1.19, 95% CI: 1.08-1.31) adjusted for age, sexual orientation, and professional situation. Increase in financial situation negatively impacted women' reproductive plans (OR=0.74, 95% CI: 0.65 - 0.86).

Conclusion(s): Our results indicate that the socio-economic changes following COVID-19 pandemic may have an unprecedented effect on shaping the current society and future generations.

Associations between nausea and vomiting in pregnancy, disgust sensitivity, and first-trimester maternal serum free β -hCG and PAPP-A

Kaňková, Š., Hlaváčová, J., Roberts, K., Benešová, J., Havlíček, J., Calda, P., Dlonhá, D., Roberts, S.C.

Objective(s): This study aimed to assess the associations between disgust sensitivity and nausea and vomiting (NVP) during the first trimester of pregnancy and to examine the predictive roles of the two biochemical markers measured during first-trimester screening (free β -human chorionic gonadotropin (hCG) and pregnancy-associated plasma protein A (PAPP-A)) on both disgust sensitivity and NVP symptom severity.

Methods: We prospectively collected two independent samples (S1: n1 = 202, S2: n2 = 397) of women in the first trimester of pregnancy who, in both samples, completed the Index of Nausea, Vomiting, and Retching and Disgust Scale-Revised, and additionally (in S2 the Pathogen domain of the Three Domains of Disgust Scale. We also measured hCG and PAPP-A in maternal serum.

Results: Our results did not confirm any association between NVP and disgust. In both samples, disgust sensitivity was significantly negatively correlated with free β -hCG and (only in S1) with PAPP-A. In contrast, NVP was positively associated with free β -hCG and (only in S1) with PAPP-A.

Conclusion(s): We have shown that NVP and disgust may likely have different proximal causes. While low hCG levels seem to be an important indicator for activation of the behavioral immune system in the first trimester, increased hCG levels play a role in stronger symptoms of NVP, a result consistent with previous studies. Levels of PAPP-A are likely part of a larger network of immunological and endocrine responses and do not appear to provide sufficient information for predicting women's NVP and disgust sensitivity.

Observers' abilities to produce future benefits influence decisions to signal generosity

Karabegović, M., Rotella, A., Barclay, P.

Objective(s): Audience (observer) effects are well-documented in the cooperation domain: making the public nature of one's behaviour salient boosts prosociality, reflecting the evolutionary importance of reputation in a partner choice ecology. The sensitivity of the underlying evolved cognitive mechanisms are, however, less clear – especially with regard to audience features. In this study, we focus on an observer's ability to produce future benefits for the actor, and ask whether this ability influences prosocial signaling. Additionally, we explore whether observers adjust their trust according to the plausibility of a generous partner's strategic motivations.

Methods: The study was conducted online with two groups of participants, Dictators (N=341) and Observers (N=284), who played two economic games: (1) a Dictator game where Dictators made the choice of either keeping or splitting 1\$; and (2) a Trust Game with an Observer as the Trustor and the Dictator as Trustee. The experiment had four conditions, depending on the information Dictators had about their audience (none, observer-only, low-ability, high-ability). The difference between low- and high- ability was in whether cooperating with the observer could offset the initial cost of the generous signal.

Results: Sharing \$1 was more likely in the high-ability and observer-only conditions than the no observer condition. Generous Dictators were trusted more; this trust did not differ across conditions. However, in a forced partner-choice paradigm, observers expressed a preference for cooperating with hypothetical non-strategic, generous Dictators (with one, interesting exception).

Conclusion(s): Reputation management mechanisms are sensitive to observers' ability to generate future benefits, compared to the cost of the prosocial signal. Furthermore, people seem to err on the side of caution when no information about observers is available. Observers' transfers do not show the expected skepticism, but partner-choice decisions do show preferences for generous actors who are less likely to be strategic.

Grandparental influence on parents' decision to vaccinate children

Karthigesu, S.P., Chisholm, J.S., Coall, D.A.

Objective(s): Where research informed arguments are not persuasive and anti-vaccine propaganda is pervasive, grandparents, through their experience of the diseases may positively influence parents' decision to vaccinate their children. This study explored the influence of grandparents on paediatric vaccine uptake.

Methods: Using a mixed-method research design, this study investigated the perceived influence of grandparents on parents' decision to vaccinate their children in Perth, Western Australia. Qualitative data generated through focus group discussions with mothers, fathers, grandmothers and grandfathers revealed parents are indirectly influenced by grandparents. The qualitative data informed the development and distribution of a questionnaire to explore the effects of beliefs, attitudes and perceived sources of influence on paediatric vaccine uptake.

Results: Focus group data (N=73) revealed parents are influenced by grandparents' attitudes. The survey (N=278) showed that grandparents generally recognised the benefits of paediatric vaccines, reported positive social group influences and low vaccine anxiety. This was reflected by high confidence levels on the information available to them and the protection conferred to their children by vaccines. Grandparents with high scores on knowledge and positive group influence reported they would advise parents on vaccinations for their grandchildren.

Conclusion(s): The findings suggest that grandparents could have a positive impact on parents' decision to vaccinate their children if parents welcome or ask for their input. Future studies must explore whether grandparents could act as proxy public health agents in alleviating the concerns of vaccine hesitant parents.

Gene-culture coevolution in the Pacific islands

Kasser, S.M., Lala, K.N., Fortunato, L., Blanco-Portillo, J., Feldman, M.W.

Objective(s): Gene-culture coevolution (GCC) theory routinely predicts broad interactions between cultural and genetic variation that should emerge through empirical analyses, yet evidence for such associations remains sparse. By adapting tools that quantify ecological drivers of population genetic structure and local adaptation, we explore potential cultural drivers of (adaptive) human genetic variation across Oceania to evaluate the extent of GCC.

Methods: To quantify genetic variation, we use high-density SNP data from 682 Oceanian individuals across 48 island populations from a new, soon-to-be-published dataset (Oceanian Genome Variation Project). We use local ancestry inference to extract relevant genetic variation, estimating allele frequencies in the ancestral Polynesian component of these modern admixed populations. We then analyse these frequencies with two methods adapted from landscape and adaptation genomics: GESTE, a Bayesian procedure to estimate the influence of environmental factors on population genetic structure, and Bayenv 2, a robust method for correlating population allele frequencies with environmental variables to identify candidate loci for local adaptation. For the present analysis, the relevant cultural and ecological variables relating to kinship practices, subsistence, island type and climate are drawn from relevant cross-cultural datasets, namely Pulotu and the Ethnographic Atlas.

Results: We will present results on associations between cultural factors and genetic structure, as well as report on candidate loci for adaptation to dietary shifts and subsistence systems within the Pacific islands.

Discussion: We will discuss the results in the context of recent work on how culture may shape genetic variation beyond classic examples of GCC, and whether landscape ecology represents a useful analogy for quantifying GCC.

Kinship and Socioeconomic Status: Social Gradients in Frequencies of Kin Across the Life Course in Sweden

Andersson, L., Kolk, M.

Objective(s): Socioeconomic status (SES) influences demographic behavior. Moreover, SES correlates across generations. Kinship structures, therefore, likely display social stratification. Despite growing interest in the functions and the everyday experience of kinship across social strata, the prevalence and antecedents of kinship stratification are rarely studied empirically.

Methods: We have estimated socioeconomic differences in kinship using Swedish register data. We created kinship networks for the entire 1973 birth cohort and tracked kin development from birth to age 45 of this birth cohort, analyzing consanguineous kin, spouses, reproductive partners, parents-in-law, and siblings-in-law.

Results: We calculated the difference in total kinship size across the index cohort's earnings quartile and educational level. We analyzed the relative contributions of specific kin relations, demographic behaviors, and generations to this difference. The results revealed a moderate negative gradient in kinship frequency and illuminated how life-course phases and gender condition these disparities. Higher fertility of earlier generations of the low SES index cohort led to a moderate negative SES gradient in kinship size. Among low SES men of the index cohort, low fertility and union instability narrowed SES differences in the number of kin.

Conclusion(s): There are socioeconomic differences in kinship in Sweden. Such differences originate both from own behavior (e.g. number of children), and as SES is associated with kin demographic behavior (e.g. grandparents number of children). SES gradients in kinship have implications for a large number of demographic and socioeconomic outcomes.

Group success over time is associated with group size and bonding – A study of real-life adult friendship groups

Lehtonen, N., Laakasuo, M., van Duijn, M., David-Barrett, T., Dunbar, R.I.M., Rotkirch, A.

Objective(s): Friendship can be defined as long-term social bonds between non-kin, characterized by mutual affection and support. Friendship is often studied in dyadic interactions or in ego-networks, but less is known about the group level processes of friendship. Here, we investigate how group size and bonding relate to group success in real-life male and female friendship groups over several years.

Methods: We use retrospective data from the Fraternity Friendship Study, collected from fraternity alumni members (N = 272). The friendship groups were between 5 and 18 members in size and were formed 12-24 years ago. Respondents were asked about group activities both during university studies and at present. Group bonding was measured with the Inclusion-of-Other-in-Self scale and group success as meeting frequency at present.

Results: Results show that group success was related to group size. A group size of around 10 members or more appeared optimal for group success in the study context. Group bonding was also associated with group success. Furthermore, group bonding moderated the effect of group size on success, so that the association was stronger in smaller groups. There were mostly no gender differences in group size or success. However, group size had a slightly larger effect for the maintenance of female friendships, compared to male groups: especially small female groups were more likely to lose contact.

Conclusion(s): We conclude that both group bonding and group size play a role in the maintenance of friendship group ties. Altogether, these findings are valuable in sketching out the dynamics in friendship group behaviour and maintenance.

Integrating Community Perception with Social Baseline Theory, Generalised Unsafety Theory of Stress and Stigmergy: An Argument for Kith Perception

McAleavey, D.

Objective(s): The term community perception was coined by David Sloan Wilson and Daniel O'Brien to define the psychological construct that they proposed to account for a particular socially oriented cognitive process (2011). They hypothesised that in a group-living species, which must navigate not only their own but occasionally unfamiliar social environments, it would be adaptive to be able to quickly infer the kinds of social encounters that might occur there. In a series of studies conducted in the US city of Binghamton, they found that participants were able to generate accurate assessments of the social quality of unfamiliar neighbourhoods using only photographs of the physical context of the built environment. Participants were less likely to cooperate in prisoner's dilemma games when paired with a person reported to be from a neighbourhood with low social quality, which O'Brien and Wilson interpreted as indicating an adaptive social response to a potentially threatening context. The objective of this poster is to reconsider this interpretation in light of recent theoretical work on social baseline theory (Coan), generalised unsafety theory of stress (Brosschot), and stigmergy (Heylighen).

Methods: This poster comprises brief theoretical introductions to the four constructs followed by an examination of the implications for O'Brien and Wilson's community perception afforded by recent advances in the other three approaches.

Results: We find the exercise rewarding; reorienting community perception from a construct where disorder in the built environment functions as an index of ambient threat, broadly defined, to one where disorder in the built environment functions primarily as an index of the availability of reliable social partners.

Conclusion(s): Active researchers within all four approaches would benefit from an integration of their respective theoretical frameworks; a proposal for kith perception.

Partners, friends or something in between? - A systematic analysis of relationship models

Misch, K.

Objective(s): For a long time, serious monogamous relationships were considered the only object of investigation in the research of interpersonal relationships. Later, short, more sexually oriented relationships were also included. It is becoming evident that couple relationships can be very diverse and differently structured. For this reason, it will be investigated in two ways how existing and known relationship models are characterized on the one hand, and which mechanisms can offer an explanation for this diversity on the other.

Methods: This work will follow an evolutionary and behavioural-ecological approach that identifies as a basic mechanism different levels of shared resources and resulting contingency structures effecting the shown behaviour in relationships. In order to gain an overview of the phenomenon of couple relationships, qualitative interviews with grounded theory were conducted and a questionnaire was developed and evaluated. The theoretical and qualitative results were compared.

Results: The description of the relationships models by criteria, as well as the classification of these models in a ranking according to the criterion of involvement are compatible with the theoretical considerations.

Conclusion(s): It seems possible that the categorical structure of the models is a representation of the different degrees of shared resources.

Can the behavioural equilibrium follow the 80/20 rule in human beings?

Natsumeda, M., Yokota, K., Nakanishi, D.

Objective(s): The 80/20 principle is an empirical rule whereby a group or an organization are composed of both, workers and non-workers. Equilibrium based on the 80/20 rule has been observed in ant colonies, which can be explained by the kin selection theory. However, humans cooperate with non-kin, even with strangers. The question is whether the 80/20 rule can be observed among human beings. Previous research argued, theoretically, that stable behavioural equilibrium following the 80/20 rule, occurred in a situation with the payoff structure of diminishing returns (e.g., Producer-Scrounger equilibrium). As there is little empirical evidence of the 80/20 rule phenomenon among humans, we conducted an experiment using two types of group tasks (disjunctive/additive). We hypothesized that equilibrium as per the 80/20 rule, should be observed in the disjunctive task because it theoretically included the payoff structure of diminishing returns.

Methods: Seventy-five undergraduates participated in the experiment and performed five trials of a calculation task in groups of four or five. Experimental rewards were determined either by the outcome of one group member who solved the most answers (disjunctive task condition) or by the average of all members' outcomes (additive task condition).

Results: In disjunctive tasks, there was a stable difference between the performance of the higher and lower rank members across all trials. In contrast, in additive tasks, the behavioural variance within a group converged over trials.

Conclusion(s): Equilibrium based on the 80/20 rule was observed in disjunctive, but not in, additive tasks.

The effect of winning and losing in virtual reality game on the perception of male dominance

Pátková, Ž., Třebický, V., Havlíček, J., Třebická Fialová, J.

Previous research indicates that men's perception of dominance is related to individual differences in the perceiver's dominance, affected by traits such as height, self-perceived dominance, and imagining winning or losing in a competition. The ability to modulate dominance perceptions should allow for a decrease of potential costs in contests (e.g., injury, loss of resources), especially in less dominant men who may suffer more substantial losses in case of incorrect judgements. Our study aimed to test the effect of changes in dominance (winning/losing in a virtual reality (VR) game) on the perception of male dominance.

Forty-six men ($M = 24.4$ ys, $SD = 4.30$ ys, range = 18-36 ys) participated in 2 sessions a week apart. To manipulate their dominance status, participants played an immersive first-person shooter game in VR against a trained confederate who controlled the outcome (participants won and lost once, order randomized). After each game, participants rated 45 male facial and full-body photographs for dominance. They completed the Positive and Negative Affect Schedule questionnaire before and after each game to test the effect of winning/losing on the affective state.

Using a mixed-effects model, we found no significant effect of winning/losing on the dominance ratings. The ANOVA test showed a significant positive effect of winning on positive affect, while losing did not influence positive affect in the first session. Further, in the first session, the losing participants had a significantly higher negative affect score after losing than the winners.

Even though winning/losing in a VR game was associated with changes in positive and negative affects, our results show no difference in perceived dominance and are not in line with the previous studies showing that changes in dominance modulate men's perception of other men's dominance.

An index of her own: An investigation of the proportion of women indexed in evolutionary psychology textbooks.

Pollet, T.V., Kovárová, K., Boret J.

Objective(s): Research has consistently documented that women receive fewer citations than men do in the scientific literature. In disciplines such as ecology and geology women also receive less coverage than men do in textbooks. Here, we examined the proportion of women indexed in leading textbooks in evolutionary psychology.

Methods: The (perceived) gender of the authors listed in the indices of four major recent textbooks in evolutionary psychology was coded. The textbooks were: Buss (2019) *Evolutionary Psychology - New Science of the Mind* (6th ed.), Welling & Shackelford (2020) *The Oxford Handbook of Evolutionary Psychology and Behavioral Endocrinology*, Workman & Reader (2021) *Evolutionary Psychology: an introduction* (4th ed) and Workman, Reader & Barkow (2020) *The Cambridge Handbook of Evolutionary Perspectives on Human Behavior*. We did not pre-register any hypothesis as we simply intended to describe how many were included in each. We report descriptive statistics and a Random Effect meta-analysis on the proportions (using ML estimation and the Agresti-Coull method for the estimation of the confidence) via the 'meta' package in R.

Results: With the exception of Welling & Shackelford (2020), where 43.00% of entries were women, fewer than 1 in 4 individuals indexed were women in the other textbooks (Buss (2019): 23.87%; Workman & Reader (2021): 16.00%; Workman, Reader & Barkow (2020): 20.42%). Across all four books, 280 out of 950 entries were women (24.95%, 95%CI: 16.47% to 35.93%).

Conclusion(s): One textbook provided a sizable representation of women, whereas the other three textbooks featured women substantially less than men - with men outnumbering women 3 to 1. We discuss the strengths and limitations of our approach. We further discuss which women were included consistently and implications for gender representation in evolutionary psychology.

Children Residence Patterns and Labor Allocation: a pre-pre-registration

Pretelli, I.

Objective(s): Children and teenagers represent both a cost and a source of labor for households in subsistence societies. Substantial work has been done to estimate production and consumption along the lifetime in a number of ethnic groups, in order to understand how families keep their economies in balance. However, much of this work did not keep into account residential mobility of children and teenagers, who can move between often related households and lighten the burden of growing families by either removing a cost or providing labor. Because individuals often engage in age- and sex-specific activities and have equally specific needs, certain household compositions are better at distributing the necessary tasks across its members: for example, in an household full of toddlers, a teenage girl could be a precious supplier of childcare, while a young child could perform menial tasks such as bringing objects for an ailing grandmother. Here I propose to test the hypothesis that families strategically manipulate household composition to optimize the workloads of individuals by relocating their younger members, so that the costs and benefits provided by individuals of different age and class categories can be efficiently combined.

Proposed Methods, Results and Conclusion(s): I present a workflow that starts by describing the causal factors involved in the decision making of extended families concerning the residence of children. Based on the assumed causal relations between the factors to be analyzed, I propose an analysis tested on simulated data, as well as the data collection plan. Finally, I discuss the implications for cooperative breeding of the specifically human ability to strategize household composition based on the labor needs of family nuclei.

Who cares after mom dies?

Schacht, R., Starkweather, K.E., Macfarlan, S.J., Meeks, H., Frasier, A., Smith, K.R

Rapid rates of birth paired with altricial and slow to develop young are characteristic of our species. This pattern, though, puts mothers into an investment dilemma by way of needing to provide for multiple dependent young simultaneously. Traditional models of the family center on nuclear households as a solution to this problem whereby help, primarily through resource provisioning, comes from the father. It is increasingly well-documented, however, that others, beyond the pair-bond, are necessary for household functioning and child provisioning. What is less well known, though, are their relative roles for child outcomes. Here we look to over 400,000 births across a ninety-year, natural fertility period in the US state of Utah and evaluate how the presence (or absence) of available caregivers (fathers, grandmothers, aunts, cousins, and siblings) impacts both child survival and fertility outcomes after maternal death. Our analyses yield two sets of results. 1) Children with an older sister or grandmother (maternal or paternal) living in the same household experience significantly lower mortality risk. 2) The death of a father results in depressed fertility outcomes (but not survival), as does the presence of an older sister or grandmother (maternal or paternal). Additionally, father presence results in later ages at first and last birth. In sum, we find that several household members play significant roles, however their effects vary by whether we look to survival or fertility. Thus, our findings add to a growing literature, highlighting how diverse sources of household investment variably impact child outcomes across the lifecourse.

Properties underlying infants' reluctance to touch plants: A re-analysis of seven studies

Schlegelmilch, K., Rioux, C., Wertz, A.E.

Objective(s): Plants have played an important role in human life and evolution. Previous studies have found that infants and toddlers exhibit selective behavioral strategies towards plants. However, it is not yet known which features underlie infants' responses to plants. To investigate this, we are re-analyzing data from seven studies that assessed infants' or toddlers' reaching behaviors towards plants compared to other kinds of objects (i.e., novel artifacts, familiar artifacts, natural objects). Consistent with a plant-selective behavioral avoidance strategy, infants and toddlers were more reluctant to touch plants compared to the other object categories in these studies.

Methods: With the combined data of the seven studies (total N = 368; age range = 5.16–47.9 months), we explore which visual properties drive infants' and toddlers' reactions to plants. We ask: (A) Can touch-behavior be explained by plant-typical properties? (B) Is touch probability or latency predicted by other visual properties of the objects? (C) How do exemplars of the object categories vary in touch probability or latency?

Results: Thus far, we have found (A) that plant-typical properties (i.e., green, leafy, branchy) predicted neither touch probability nor latency beyond the objects' identity (all p 's > .26) within the novel-artifacts category, where plant-typical properties varied most. (B) We then computationally assessed low-level visual properties (e.g., fractality, contrast) of the objects. These rarely predicted infants' and toddlers' touch probability as much as object category itself. (C) However, the predictive value of individual object exemplars varied within the object categories. To ascertain visual properties that could explain this variance, we are currently compiling additional object properties that will be assessed by adult raters (e.g., symmetry, curvature, complexity).

Conclusion(s): With the resulting catalog of properties, we hope to identify visual properties that predict touch behaviors—indicating which features infants use to distinguish plants from other entities.

Repercussions of patrilocal residence on mothers' social support networks among Tsimane forager-farmers

Seabright, E., Alami, S., Kraft, T.S., Davis, H.E., Caldwell, A., Hooper, P., McAllister, L., Mulville, S., Veile, A., von Rueden, C., Trumble, B., Stieglitz, J., Gurven, M., Kaplan, H.

While it is commonly thought that patrilocal residence is associated with worse outcomes for women and their children due to lower social support, few studies have examined whether the structure of female social networks covaries with post-marital residence. Here, we analyse scan sample data collected among Tsimane forager-farmers. We compare the social groups and activity partners of 181 women residing in the same community as their parents, their husband's parents, both or neither. Relative to women living closer to their in-laws, women living closer to their parents are less likely to be alone or solely in the company of their nuclear family (odds ratio (OR): 0.6, 95% CI: 0.3–0.9), and more likely to be observed with others when engaging in food processing and manufacturing of market or household goods, but not other activities. Women are slightly more likely to receive childcare support from outside the nuclear family when they live closer to their parents (OR = 1.8, 95% CI 0.8–3.9). Their social group size and their children's probability of receiving allocare decrease significantly with distance from their parents, but not their in-laws. Our findings highlight the importance of women's proximity to kin, but also indicate that patrilocal residence is not costly to Tsimane women.

The number of sons born may increase the risk of metabolic syndrome in post-reproductive women

Sekula, K., Galbarczyk, A., Klimek, M., Nenko, I., Jasienka, G.

Objective(s): Reproduction is costly, thus it is expected that women with high parity, especially those having sons, will have poor health in older age. In fact, increased risks of some diseases has been shown in women with high costs of reproduction. The aim of this study was to explore the relationship between the total number of children, and the number of daughters and number of sons, and the risk of metabolic syndrome (MetS) among post-reproductive women. Metabolic syndrome is diagnosed when a person has at least three of five medical conditions: abdominal obesity, hypertension, hypertriglyceridemia, low concentrations of HDL cholesterol and insulin resistance or type 2 diabetes.

Methods: We analyzed data for 425 women from a rural Polish population at the Mogielica Human Ecology Study Site. Women had 3.9 (SD 2.14) children, including 2.1 (SD 1.50), sons and 1.8 (SD 1.43) daughters on average. 219 participants (51.5%) were classified as having metabolic syndrome.

Results: Logistic regression analysis showed a positive relationship between the overall number of children and the likelihood of being diagnosed with MetS (OR=1.20; 95% CI: 1.08–1.33, p

Prestige and dominance amongst children in Colombia, Finland and the USA: An experimental and ethnographic study

Sequeira, M.E., Afshordi, N., Kajanus, A.

Objective(s): Research suggests that individuals can achieve high social status through two distinct pathways; dominance (rooted in force or threat of force) or prestige (rooted in merit and admiration from others). There is cross-cultural variation in how children reason about prestige and dominance, and we hypothesise that one socio-cultural factor that might shape their reasoning is the level of societal equality and stability. Our objective is therefore to explore how preferences for, and expectations of, leaders (i) develop between the ages of 4–12 years, and (ii) differ across three distinct socio-cultural contexts.

Methods: We selected three novel and diverse socio-cultural contexts with varying degrees of socio-political and economic stability and equality; Colombia, Finland, and the USA. We conducted an experiment with children aged 4–12 years to determine whether they: (i) discriminated between dominance and prestige, (ii) preferred to learn from a dominant or prestigious individual, (iii) associated dominant or prestigious individuals with leadership, and (iv) self-identified with a dominant or subordinate character. The interpretation of results was supported by ethnographic fieldwork conducted in Colombia and Finland.

Results: Analyses will be completed in January 2023. Preliminary analyses suggest a developmental trajectory whereby younger children (4–7 years) have less consistent preferences and expectations than older children (8–11 years). A strong preference for prestige is evident amongst children in Finland from age 5 years onwards, and a weak preference is observed in the USA, whereas children in Colombia have no clear preferences for either dominant or prestigious leadership.

Conclusion(s): Results suggest cross-cultural variation in children's preferences for and expectations of leaders, with stronger preferences for prestigious leadership being found in a more equal and stable society (Finland). Ethnographic data suggests that in Colombia, dominant leadership may be considered prestigious, therefore challenging the dichotomy of dominance vs prestige.

Are sex differences in religious celibacy driven by same-sex sibling competition?

Sherbaji, S., Micheletti, A.J.C., Mace, R.

Objective(s): Religious celibacy is found in many world religions. Studies show that life-long celibacy is associated with inclusive fitness benefits, under conditions where the decision is within parental control and celibacy would alleviate competition between same-sex siblings over parental resources (Micheletti et al, 2022; Zhou et al, 2022). However, the causes of the variation in the frequency of female and male celibacy across time and space have not been explored. The purpose of this research is to identify the factors which influence this variation.

Methods: Using eHRAF, we search for mentions of religious celibacy using keywords that are person-based (e.g., monk, nun) and institution-based (e.g., monastery, nunnery). Then, using D-PLACE, we determine whether differences in the relative frequency of male and female celibacy can be predicted by factors which drive same-sex sibling competition such as the presence of dowry or bride-price.

Results: We expect female celibacy to be more common in societies where dowry is more widely practiced, as such societies typically have greater female-female competition.

Likewise, we predict male celibacy to be more common in societies where bride-price is more widely practiced, as such societies typically have greater male-male competition.

Conclusion(s): Building on models in behavioural ecology, we hope to show why cultural phenotypes such as religious celibacy emerge and why their features vary across populations.

Does the daily number of steps affect the level of sex hormones?

Słojewska, K., Galbarczyk, A., Klimek, M., Tubek-Krokosz, A., Ścibor, M., Jasienska, G.

Objective(s): Physical activity is one of the most important behaviors that improve health. Sex steroid hormones play an important role in physiological processes. Levels of these hormones are associated with risks of many diseases, and high lifetime exposure increases breast cancer risk. Sex hormones produced in menstrual cycles can be modified by lifestyle, mostly by factors related to energetics. Accordingly, physical activity has been shown to influence levels of sex hormones in women. The aim of this study was to investigate whether recommendations of taking 10,000 steps a day are sufficient to reduce the sex hormone levels in premenopausal women.

Methods: The participants were 90 healthy women aged 21-36 who performed at least 180 minutes/week of moderate physical activity for 2 complete menstrual cycles. Physical activity was measured by Fitbit wristband accelerometers. Estradiol and progesterone concentrations were measured in daily-collected saliva samples in the second menstrual cycle. Multiple regression models were used to test the effects of exercise expressed as number of steps on estradiol and progesterone levels.

Results: There was a significant negative association between average number of steps taken daily and salivary progesterone levels after adjusting for potential confounding factors (age, education, BMI). Women who took more than 10,000 steps a day had significantly lower (19%) progesterone levels compared to women who took less than 10,000 steps. The association between physical activity and estradiol levels was statistically insignificant.

Conclusion(s): Our results indicate that taking at least 10,000 steps a day reduces levels of progesterone, but this intensity of physical activity may not be high enough to influence levels of estradiol. Potential health implications of exercise-induced changes in levels of sex hormones observed in this study are not well understood. However, currently recommended levels of physical activity may not be sufficiently high for premenopausal healthy women to significantly reduce their risk of postmenopausal breast cancer.

All I want to give you for Christmas: Mate value discrepancy in romantic relationships as a predictor of Christmas gifts' value.

Szymkow, A., Frankowska, N., Tolopilo, A., Galasinska, K.

Mate value refers to an individual's relative desirability on the "mating market" (Sugiyama, 2005). Individuals with higher mate value are more attractive and have greater access to high-value mates compared to those of lower mate value (Starratt et al., 2016). How an individual perceives his or her own mate value, as well as how one perceives the mate value of one's partner, have substantial influence on relationship satisfaction (Conroy-Beam et al., 2016), jealousy (Sidelinger & Booth-Butterfield, 2007), and mate retention behaviors (Buss & Shackelford, 1997). And as various research suggests, of greatest importance here is the mate value discrepancy – that is, the difference between an individual's mate value and his or her partner's mate value (Conroy-Beam et al., 2016; Buss & Shackelford, 1997; Danel et al., 2017; Sela et al., 2017). Those who perceive that they have lower mate value than their partner are usually more satisfied with the relationship and at greater risk of losing their partner to a rival. As a consequence, they should be more motivated than their partner to maintain the relationship. In one study we test (study in progress) whether mate value discrepancy significantly predicts the amount of money partners spend on buying a Christmas gift for their partners. We predict that the lower the mate value of a partner comparing to his/her partner, the more money one spends on the Christmas gift. The mediating role of relationship satisfaction and commitment will be tested.

Step-gap in adult children's investment towards older parents: financial, practical and emotional support

Hämäläinen, H., Tanskanen, A.O., Pettay, J., Danielsbacka, M.

Objective(s): According to inclusive fitness theory individuals are predicted to invest more resources (e.g., time, money, materials) in their closely related relatives than more distant ones. Investments between step-relatives indeed tend to suffer from a 'step-gap', meaning that step-relatives invest less in each other compared to their genetically related counterparts. Here, we examine investments toward stepparents by adult children. We argue that if children and stepparents have lived together in the same household during children's childhood, they may form a 'kin like' -bond and perceive themselves as emotional kin, leading adult children to invest more in their stepparents.

Methods: Using data from the German Family Panel (pairfam) we utilize multiple regression analyses to examine the provision of financial, practical and emotional support to (step)parents by their adult children. We investigate (i) do children provide more support to their biological parents than stepparents, (ii) is the duration of childhood co-residence associated with the support provided to (step)parents, (iii) does the duration of childhood co-residence reduce the step-gap in the provided support toward parents.

Results: More investments are channeled towards biological parents than stepparents. Moreover, the length of childhood co-residence is positively associated with the level of provided support toward (step)parents, and the length of co-residence reduces the step-gap in the provision of support toward (step)parents.

Conclusion(s): Although stepparents receive less support from adult children compared to genetically related parents and children, the length of shared time during childhood diminishes the severity of step-gap.

A Review work: examining the breadth and limits of conditions under which human pair bonds can be established -using the comparative population's method among human case studies-

Teramoto, R., Takada, A.

Recent promiscuous primate studies have shown relatively stable female-male pair bonds were not limited to among pair species. At the same time, these studies have questioned the strict pathways used in previous studies to predict the evolutionary background of extended human pair bonds. To better understand human pair bonding, it would be beneficial for future research to examine a wide range of possible scenarios beyond conventional assumptions and to conduct human studies that can be compared with the circumstances surrounding these promiscuous species pairs. Therefore, this study used The Human Relations Area Files (HRAF) to distinguish the detailed nuances of human pair bond definitions (pair-living, pair-breeding, or sexual pair bond) that have been confounded among studies and a review of case studies, focusing on inter-group pair variations and their conditional limitations. In particular, this study focused on the sub-components of the marriage pairs, assuming that the below indicators could generate pair variation.: (1) the tolerance for divorce (which could measure whether pair maintenance was forced or not), (2) the tolerance for sexual activity outside the pair and within the pair, and (3) the marriage system. The most important finding of this review was to show the cases in which marital status did not necessarily guarantee actual sexual activity. We have also suggested that more cross-cultural case studies are needed to redefine the sexual pair bond, focusing on the period of non-sexual pairing as the uniqueness of the human pair.

Does daily habitual activity affect immune system functioning?

Tubek-Krokosz, A., Galbarczyk, A., Klimek, M., Słojewska, K., Ścibor, M., Baran, J., Jasienska, G.

Objective(s): Physical activity is one of the most important factors having a significant impact on health and life expectancy. Physical activity could affect the activation of immune processes, enhancing the functioning of the immune system. It is recommended that people take at least 10,000 steps a day. The aim of the study was to investigate whether daily habitual activity affects the biomarkers of immune system of young women.

Methods: This study was conducted among 84 urban Polish women aged 21-36 (mean=27.3, SD=4.21). Daily, habitual physical activity data was collected using Fitbit wristband accelerometers for one week. On the last day of observation blood sample was then taken for immunological analyses. Linear regression models were used to test relationships between daily physical activity expressed as number of steps and selected biomarkers of the immune system. Student's t-test was used to determine the difference between women who took at least 10,000 steps per day versus less active women.

Results: Women took 9589 (SD=3011) steps per day on the average. Only 33 (30.8%) of them took at least 10,000 steps per day. There was no significant linear relationship between the number of steps and the white blood cells (WBC) count ($p=0.159$), the percentage of total lymphocytes ($p=0.400$), T lymphocytes ($p=0.670$), T helper/inducer cells ($p=0.069$), T cytotoxic/suppressor cells ($p=0.117$), B lymphocytes ($p=0.526$), and natural killer (NK) cells ($p=0.286$). No significant differences were detected in immune biomarkers between women who took at least 10,000 steps per day versus less active women (p value varied from 0.483 to 0.921).

Conclusion(s): These results suggest that moderate physical activity does not affect the immune biomarkers assessed in this study.

Mothers of small-bodied children and fathers of vigorous sons live longer

Valge, M., Meitern, R., Hõrak, P.

Objective(s): According to the theory of life-history evolution, the key traits that contribute to Darwinian fitness, i.e., growth, survival and reproduction, evolve in a coordinated manner. Life-history traits (traits directly related to survival and reproduction) materialise through physiology and behaviour. This means that in addition to basic components of fitness, anatomical, physiological, behavioural and psychosocial variation too accumulates non-randomly among the individuals within populations. This premise is well supported by genome-wide association studies demonstrating moderate to strong genetic correlations between life-history and associated physiological, behavioural and health-related traits. We investigate lifespan as a potentially informative marker for life-history speed, as it subsumes the impact of a diverse set of morphometric and behavioural traits.

Methods: We examined associations between parental longevity and various anthropometric traits in a sample of 4000-11 000 Estonian children in the middle of the 20th century. The offspring phenotype was used as a proxy measure of parental genotype, so that covariation between offspring traits and parental longevity (defined as belonging to the 90th percentile of lifespan) could be used to characterise the microevolutionary aggregation of longevity and anthropometric traits. We predicted that larger linear dimensions of offspring are associated with increased parental longevity and that testosterone-dependent traits are associated with reduced paternal longevity.

Results: 12 of 16 offspring traits were associated with mothers' longevity, while three traits (rate of sexual maturation of daughters and grip strength and lung capacity of sons) robustly predicted fathers' longevity. Contrary to predictions, mothers of children with small bodily dimensions lived longer, and paternal longevity was not linearly associated with their children's body size (or testosterone-related traits).

Conclusion(s): Our study thus failed to find evidence that high somatic investment into brain and body growth clusters with a long lifespan across generations, and/or that such associations can be detected on the basis of inter-generational phenotypic correlations.

Disgust sensitivity relates to attitudes toward gay men and lesbian women across 31 nations

Van Leeuwen, F., Inbar, Y., Petersen, M.B., Aaroe, L., Barclay, P., Barlow, F.K., de Barra, M., Becker, D.V., Borovoi, L., Choi, J.A., Consedine, N.S., Conway, J.R., Conway, P., Cubela Adoric, V., Demiri, E., Fernández, A.M., Ferreira, D.C.S., Ishii, K., Jakšić, I., Ji, T., Jonaityte, L., Lewis, D.M.G., Li, N.P., McIntyre, J.C., Mukherjee, S., Park, J., Pawlowski, B., Pizarro, D., Prodromitis, G., Prokop, P., Rantala, M.J., Reynolds, L.M., Sandin, B., Sevi, B., Srinivasan, N., Tewari, S., Yong, J.C., Žeželj, I., Tybur, J.M.

Objective(s): Previous work has reported a relation between pathogen-avoidance motivations and prejudice toward various social groups, including gay men and lesbian women. It is currently unknown whether this association is present across cultures, or specific to North America. The aims of the current study were twofold: (1) to assess the cross-cultural variability of the association between individual differences in pathogen-avoidance motivations and antigay attitudes, and (2) to assess the specificity of this relationship (i.e., does it generalize to other outgroups associated with norm violations?).

Methods: Analyses of survey data from adult heterosexuals ($N = 11,200$) from 31 countries that included measures of pathogen disgust sensitivity (an individual-difference measure of pathogen-avoidance motivations) and antigay attitudes.

Results: Analysis showed a small relation between pathogen disgust sensitivity and antigay attitudes. Analyses also showed that pathogen disgust sensitivity relates not only to antipathy toward gay men and lesbians, but also to negativity toward other groups, in particular those associated with violations of traditional sexual norms (e.g., prostitutes).

Conclusion(s): These results suggest that the association between pathogen-avoidance motivations and antigay attitudes is relatively stable across cultures and is a manifestation of a more general relation between pathogen-avoidance motivations and prejudice towards groups associated with sexual norm violations.

Testing the functional flexibility of the behavioral immune system: The influence of the harshness of the environment on perceived disgust

Váňová, B., Kuba, R., Třebická Fialová, J., Schnumbergová, D., Havlíček, J.

Objective(s): According to Compensatory Prophylactic Hypothesis, behavioral immune system is expected to be functionally flexible and various factors can affect disgust sensitivity, including the perceived harshness of the environment. Such example of a harsh situation is first aid where disgust may reduce the willingness to provide first aid or significantly slow the performance itself. The aim of the project is to test the effect of completing a first aid course on perceived disgust sensitivity and willingness to provide first aid. We expect that an increase in environmental harshness will be associated with a decrease in pathogen and sexual disgust; furthermore, disgust levels will increase one month after exposure to the harsh environment.

Methods: We collect data at the first aid courses and other courses similar in structure at the Faculty of Science, Charles University. Based on G*Power ver 3.1.9.7 (Erdfeider et al., 2009), and on the experience with average attendance at the selected courses, the expected sample size is 150 participants in the experimental (currently 189 before, 133 right after, 75 month after course; aged 18-43; mean 21.4; SD 2.52) and 150 in the control group (currently 215, 121, 64; aged 18-41; mean 20.06; SD 1.96). Participants are asked to complete three sets of questionnaires – before, right after and month after completing the course. It consists of purpose-built questionnaire regarding the participant's first aid experience/participation in the course and standardized questionnaires - The Three-Domain Disgust Scale (Tybur, 2009) evaluating pathogen, sexual and moral disgust, the Injury Disgusting Questionnaire (Kupfer, 2018), The Culpepper Disgust Image Set (Culpepper, 2018) consisting of pathogen-salient images with their matching pathogen-free counterparts, the Ten Item Personality Inventory (Gosling, 2003) assessing the personality characteristics, and the 20 item Trait Anxiety Inventory (Spielberger, 2010) assessing the stable traits of anxiety.

Results: Data collection is still ongoing and the project results will be presented at the conference and discussed in the Compensatory Prophylactic Hypothesis framework.

The trade-off between childcare and foraging work on BaYaka nursing mothers in the Republic of Congo

Visine, A., Boyette, A., Lew-Lery, S., Ouamba, Y.R., Sarma, M., Jang, H

Nursing mothers face an energetic trade-off between childcare and work. In hunter-gatherer societies, this trade-off occurs when mothers go on out-of-village expeditions for food acquisition. Mothers either leave infants with allomaternal caregivers in the village or take infants on expeditions. Leaving infants in the village may allow mothers to increase work efficiency, but mothers may not travel far and come back to the village quickly for breastfeeding. Alternatively, mothers may not travel far when they took infants with them, as carrying infants drains maternal energy and taking infants on long trips can harm them. Moreover, looking after infants may reduce work efficiency, resulting in lower food returns. Despite these controversial predictions, the trade-off on forager mothers is not investigated in detail. Here we aimed to investigate how taking infants on work expeditions or leaving them in the village affect nursing mothers' travel duration, distance, energy expenditure and foraging productivity, and how group composition may mitigate this trade-off. During 7 weeks in 2022, we distributed 20 GPS and heart rate measuring devices to 20 BaYaka mothers on a daily basis from morning to evening. We recorded the time when the mothers left and returned to the village, work type, the presence of infants in groups, group composition, and measured the weight of food collected. In total, we collected 650 person-days GPS and heart rate measures and 598 food returns from 20 mothers during 985 expeditions. Mothers took infants with them for 53.3 % of the 985 expeditions. Using Bayesian multilevel modelling, we will test how the presence of infants and the presence of possible helpers in groups affect mothers' travel duration, distance, energy expenditure and food returns. This study will provide an insight into mothers' decision making associated with childcare and food acquisition.

Does maternal anxiety during pregnancy relate to infant sleep behaviour?

Wachowicz, A., Galbarczyk, A., Danel, D., Apanasiewicz, A., Klimek, M., Ciochoń, A., Ziomkiewicz, A., Marcinkowska, U. M.

Objective(s): Anxiety affects our quality of life. Especially, pregnant women might be predisposed to increased anxiety levels during pregnancy. The prenatal period is an especially important one because it prepares child's metabolism and physiologically for the post-natal life. The aim of this study was to explore the relationship between maternal anxiety level during pregnancy and sleep behaviour of her child as a possible maker of the prenatal environment.

Methods: This prospective study was conducted among 162 Polish mothers -who had given live birth during the COVID-19 pandemic. Logistic regression was used to expose whether the infant had problems with falling asleep (yes/no) during the day, evening and at night depending on the level of maternal anxiety.

Results: We found no significant association between maternal anxiety during pregnancy and the problems with falling asleep of her baby during the day ($OR=0.98$, 95%CI=0.95-1.01), evening ($OR=0.99$, 95%CI=0.97-1.02), and at night ($OR=1.01$, 95%CI=0.98-1.05).

Conclusion(s): We did not find any significant relationship between maternal anxiety level during pregnancy and infant's sleep behaviour. However it should be noted that maternal anxiety were made during pregnancy might change after the birth of the child. Other, more dominant prenatal factors, maternal postnatal characteristics (i.e.. post-natal stress) as well as paternal influence could also play a role in infants sleep behaviour.

Revisiting Hockett's Design Features from an Evolutionary and Neurocognitive Perspective: The Case of Discreteness

Zhang, E.Q., Pleyer, M.

Objective(s): Hockett's Design features of language (e.g. Hockett, 1960) have proven highly influential within the field of linguistics in describing the differences between human language and animal communication. However, they have been criticized for not paying enough attention to their underlying cognitive abilities (Waciewicz & Żywiec, 2015). Here, we revisit Hockett's Design Features of Language from an evolutionary and neurocognitive perspective. In doing so, we demonstrate that the neurocognitive foundation for these features should be explored, so that the evolutionary trajectory of language will be better understood.

Methods: We take discreteness, the fact that language is built up from discrete units such as phonemes, as a case study. Categorical perception (CP) has been proposed to be the precursor of discreteness, as it is a process by which continuous inputs are perceived discretely and as distinct categories across modalities (Zhang & Shi, 2022). We review cognitive and neurological studies on categorical perception in nonhuman animals to illustrate the underlying cognitive and neural basis for the discreteness of language.

Results: CP has been observed in a wide number of animals, including crickets (Wytenbach et al., 1996), frogs (Baugh, Akre & Ryan, 2008), birds (Dooling et al., 1987), rodents (Kuhl & Miller 1975) and primates (Sinnott & Brown 1997). In both birds and mammals, CP is supported by homologous subcortical structures, such as the basal ganglia. Furthermore, similar neural structures in birds and mammals respond robustly to auditory (Prather et al., 2009) and rhythmic (Lampen et al. 2017) categorical changes. Similarities have also been found between the neural implementation of CP of sounds in human and non-human primates and birds at the cortical level (Tsunada et al., 2011).

Conclusion(s): As the case of discreteness shows, a neurocognitively informed re-evaluation of Hockett's design features offers the potential of turning them into a useful analytical device for evolutionary comparisons of human language and animal communication.